1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
<REV>	<ECN>	<ECO_DESCRIPTION>	<ECODATE>

J44 MLB SCHEMATIC - PVT
08/20/2013

ALIASES RESOLVED

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-9879	1	SCHEM,MLB,J44	SCH	CRITICAL	
820-3476	1	PCBF,MLB,J44	PCB	CRITICAL	

DRAWING TITLE
<PART_DESCRIPTION>

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BOM Groups

BOM GROUP	BOM OPTIONS
J44_COMMON	ALTERNATE,COMMON,J44_COMMON1,J44_COMMON2,J44_COMMON3,J44_COMMON4,J44_PROGPARTS
J44_COMMON1	TBTHV:P15V,SKIP_5V3V3:AUDIBLE,SPI:DUAL_IO
J44_COMMON2	EDP,EDP_LS_CAP,CAMERA_3V3:S0,CAM_WAKE:NO,CAM_XTAL:NO,MEM_ODT:PU,VCORE_FETS
J44_COMMON3	XDP,LPCPLUS,BKLT:PROD,CPUTHRM:ALRT,LOADRC:NO,OTHERRC:NO,DDRRC:NO,TBTRC:NO,BMONRC:NO
J44_PROGPARTS	SMC_PROG:PVT,BOOTROM:PVT,TBTROM:PVT,TPAD_PSOC:PROG
ENGISNS	LOADISNS,OTHERISNS,DDRISNS,TBTISNS,BMONISNS

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4596	1	HSWULT,SR18A,PRQ,C0,2.4,28W,2+3,3M,BGA	U0500	CRITICAL	CPU_HSW:2.4G
337S4597	1	HSWULT,SR189,PRQ,C0,2.6,28W,2+3,3M,BGA	U0500	CRITICAL	CPU_HSW:2.6G
337S4598	1	HSWULT,SR188,PRQ,C0,2.8,28W,2+3,4M,BGA	U0500	CRITICAL	CPU_HSW:2.8G
338S1247	1	IC,TBT,FR-4C,A0,PRQ,CIO,SR1JC,FCBGA288	U2800	CRITICAL	
338S1186	1	IC,BCM15700A2,S2 PCIE CAMERA PROCESSOR	U3900	CRITICAL	
376S1194	2	MOSFET,N-CH,30V,15.3A,12M,8P 3.3X3.3 DFN	Q7310,Q7320	CRITICAL	VCORE_FET:VSHY
376S1193	2	MOSFET,N-CH,30V,22A,6.0M,8P 3.3X3.3 DFN	Q7311,Q7321	CRITICAL	VCORE_FET:VSHY
376S0964	2	MOSFET,N-CH,25V,30A,9.6M,8P 3.3X3.3 DFN	Q7310,Q7320	CRITICAL	VCORE_FET:REN
376S1104	2	MOSFET,N-CH,25V,30A,6.1M,8P 3.3X3.3 DFN	Q7311,Q7321	CRITICAL	VCORE_FET:REN

Programmables (All Builds)

TBT

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341S3918	1	EPROM,FALCON RIDGE (V13.7) J44	U2890	CRITICAL	TBTROM:PVT

SMC

341S3922	1	IC,SMC-B1,EXT(V2.16F39),PVT,J44	U5000	CRITICAL	SMC_PROG:PVT

EFI ROM

341S3924	1	IC,EFI ROM (V0116),PVT,J44	U6100	CRITICAL	BOOTROM:PVT

PSOC

341S3862	1	IC,TRKPD/KYBD PSOC,CU ONLY(V224) J44	U4801	CRITICAL	TPAD_PSOC:PROG

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1053	376S0604		ALL	Diodes alt to Fairchild
128S0311	128S0329		ALL	NEC alt to Sanyo
138S0739	138S0706		ALL	Samsung alt to Murata
197S0481	197S0480		ALL	Epson alt to NDK
152S0461	152S1645		ALL	Cyntec alt to Vishay
376S1080	376S0820		ALL	Diodes alt to On Semi
155S0667	155S0583		ALL	Panasonic alt to TDK
138S0725	138S0724		ALL	Samsung alt to Murata
376S1032	376S0855		ALL	Toshiba alt for Diodes Dual
376S1129	376S0855		ALL	NXP Alt for Diodes Dual
376S1089	376S1128		ALL	NXP Alt for Diodes Single
353S3452	353S1286		ALL	Maxim alt to Microchip
376S1180	376S0761		ALL	Renesas alt to Vishay
128S0364	128S0264		ALL	Sanyo 2nd Factory alt
107S0254	107S0241		ALL	Cyntec alt to TFT
138S0843	138S0674		ALL	Samsung alt to Murata (BKLT)
138S0803	138S0639		ALL	Samsung alt to Murata (BKLT)
138S0846	138S0811		ALL	Samsung alt to Murata (BKLT)
197S0542	197S0544		ALL	NDK alt to TXC
197S0545	197S0544		ALL	Epson alt to TXC
152S1876	152S1804		ALL	TDK alt to Toko
107S0255	107S0240		ALL	Cyntec alt to TFT
107S0250	107S0248		ALL	Cyntec alt to TFT
127S0164	127S0162		ALL	Rohm alt to Vishay
353S4070	353S4069		ALL	Pericom alt to TI DP Mux U9750
353S4068	353S4069		ALL	NXP alt to TI DP Mux U9750
353S3814	353S3812		ALL	TI alt to NXP
311S0649	311S0541		ALL	ONsemi alt to Toshiba
128S0436	128S0392		ALL	Kemet alt to Sanyo

SYNC_MASTER=SHART_J44 SYNC_DATE=11/27/2012

PAGE TITLE
BOM Configuration

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
685-0037	COMMON,MLB,J44	J44_COMMON
985-0030	DEV,MLB,J44	XDP_CONN
639-4881	PCBA,MLB,2.4G,8GB HYNIX,J44	BASE_BOM,CPU_HSW:2.4G,RAM_8G_HYNIX
639-4882	PCBA,MLB,2.4G,8GB ELPIDA,J44	BASE_BOM,CPU_HSW:2.4G,RAM_8G_ELPIDA
639-4883	PCBA,MLB,2.4G,8GB MICRON,J44	BASE_BOM,CPU_HSW:2.4G,RAM_8G_MICRON
639-4964	PCBA,MLB,2.6G,8GB HYNIX,J44	BASE_BOM,CPU_HSW:2.6G,RAM_8G_HYNIX
639-4965	PCBA,MLB,2.6G,8GB ELPIDA,J44	BASE_BOM,CPU_HSW:2.6G,RAM_8G_ELPIDA
639-4966	PCBA,MLB,2.6G,8GB MICRON,J44	BASE_BOM,CPU_HSW:2.6G,RAM_8G_MICRON
639-4884	PCBA,MLB,2.8G,8GB HYNIX,J44	BASE_BOM,CPU_HSW:2.8G,RAM_8G_HYNIX
639-4885	PCBA,MLB,2.8G,8GB ELPIDA,J44	BASE_BOM,CPU_HSW:2.8G,RAM_8G_ELPIDA
639-4886	PCBA,MLB,2.8G,8GB MICRON,J44	BASE_BOM,CPU_HSW:2.8G,RAM_8G_MICRON
639-4740	PCBA,MLB,2.4G,16GB HYNIX,J44	BASE_BOM,CPU_HSW:2.4G,RAM_16G_HYNIX
639-4741	PCBA,MLB,2.4G,16GB MICRON,J44	BASE_BOM,CPU_HSW:2.4G,RAM_16G_MICRON
639-5192	PCBA,MLB,2.6G,16GB HYNIX,J44	BASE_BOM,CPU_HSW:2.6G,RAM_16G_HYNIX
639-5194	PCBA,MLB,2.6G,16GB MICRON,J44	BASE_BOM,CPU_HSW:2.6G,RAM_16G_MICRON
639-4887	PCBA,MLB,2.8G,16GB HYNIX,J44	BASE_BOM,CPU_HSW:2.8G,RAM_16G_HYNIX
639-4888	PCBA,MLB,2.8G,16GB MICRON,J44	BASE_BOM,CPU_HSW:2.8G,RAM_16G_MICRON
685-0073	VCORE FET,REN,J41	VCORE_FET:REN
685-0072	VCORE FET,VSHY,J41	VCORE_FET:VSHY

Main DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM_8G_ELPIDA	8G_ELPIDA,RAMCFG3:L,RAMCFG2:H,RAMCFG1:L,RAMCFG0:L,PPDDR:1V35
RAM_8G_HYNIX	8G_HYNIX,RAMCFG3:L,RAMCFG2:H,RAMCFG1:L,RAMCFG0:H,PPDDR:1V35
RAM_8G_MICRON	8G_MICRON,RAMCFG3:L,RAMCFG2:H,RAMCFG1:H,RAMCFG0:L,PPDDR:1V35
RAM_16G_HYNIX	16G_HYNIX,RAMCFG3:H,RAMCFG2:L,RAMCFG1:L,RAMCFG0:H,PPDDR:1V35
RAM_16G_MICRON	16G_MICRON,RAMCFG3:H,RAMCFG2:L,RAMCFG1:H,RAMCFG0:L,PPDDR:1V35

Main DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	
333S0703	16	IC,SDRAM,4GBIT,512MX8,DDR3L-1600,REV B,78FBGA		CRITICAL	8G_ELPIDA	U2300,U2310,U2320,U2330,U2340,U2350,U2360,U2370,U2500,U2510,U2520,U2530,U2540,U2550,U2560,U2570
333S0667	16	IC,SDRAM,4GBIT,512MX8,DDR3L-1600,GEMMA,78FBGA		CRITICAL	8G_HYNIX	U2300,U2310,U2320,U2330,U2340,U2350,U2360,U2370,U2500,U2510,U2520,U2530,U2540,U2550,U2560,U2570
333S0660	16	IC,SDRAM,4GBIT,512MX8,DDR3L-1600,V80A,78FBGA		CRITICAL	8G_MICRON	U2300,U2310,U2320,U2330,U2340,U2350,U2360,U2370,U2500,U2510,U2520,U2530,U2540,U2550,U2560,U2570
333S0711	16	IC,SDRAM,8GBIT,DDR3L-1600,HUMA,96P FBGA		CRITICAL	16G_HYNIX	U2300,U2310,U2320,U2330,U2340,U2350,U2360,U2370,U2500,U2510,U2520,U2530,U2540,U2550,U2560,U2570
333S0709	16	IC,SDRAM,DDR3L-1600,DUAL_512MX8,78FBGA		CRITICAL	16G_MICRON	U2300,U2310,U2320,U2330,U2340,U2350,U2360,U2370,U2500,U2510,U2520,U2530,U2540,U2550,U2560,U2570
333S0714	16	IC,SDRAM,4GbX8,DDR3-1866,FDIE,78B,FBGA		CRITICAL	8G_ELPIDA_1866	U2300,U2310,U2320,U2330,U2340,U2350,U2360,U2370,U2500,U2510,U2520,U2530,U2540,U2550,U2560,U2570
333S0716	16	IC,SDRAM,4GbX8,DDR3-1866,HUMA,78B,FBGA		CRITICAL	8G_HYNIX_1866	U2300,U2310,U2320,U2330,U2340,U2350,U2360,U2370,U2500,U2510,U2520,U2530,U2540,U2550,U2560,U2570
333S0719	16	IC,SDRAM,4GbX8,DDR3-1866,V80A,78B,FBGA		CRITICAL	8G_MICRON_1866	U2300,U2310,U2320,U2330,U2340,U2350,U2360,U2370,U2500,U2510,U2520,U2530,U2540,U2550,U2560,U2570

S2 DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S0700	1	IC,SDRAM,4GBIT.DDR3L-1600,HUMA,96B BGA	U4000		

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	
685-0073	685-0072		ALL		Renesas alt to Vishay for CPU Core Mosfets
333S0704	333S0700		ALL		Elpida alt to Hynix for S2 Camera DDR3 Memory

Variable BOM Groups

BOM GROUP	BOM OPTIONS
J44_COMMON4	SMCBOARDID:16

Development/Base BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-0037	1	J44 MLB COMMON BOM	BASE	CRITICAL	BASE_BOM
985-0030	1	J44 MLB DEVEL BOM	DEVEL	CRITICAL	DEVEL_BOM

Sub-BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-0072	1	VCORE FET,VSHY,J41	VCOREFETS	CRITICAL	VCORE_FETS

SYNC_MASTER=J14 SYNC_DATE=09/04/2012

PAGE TITLE
BOM Configuration

Apple Inc.

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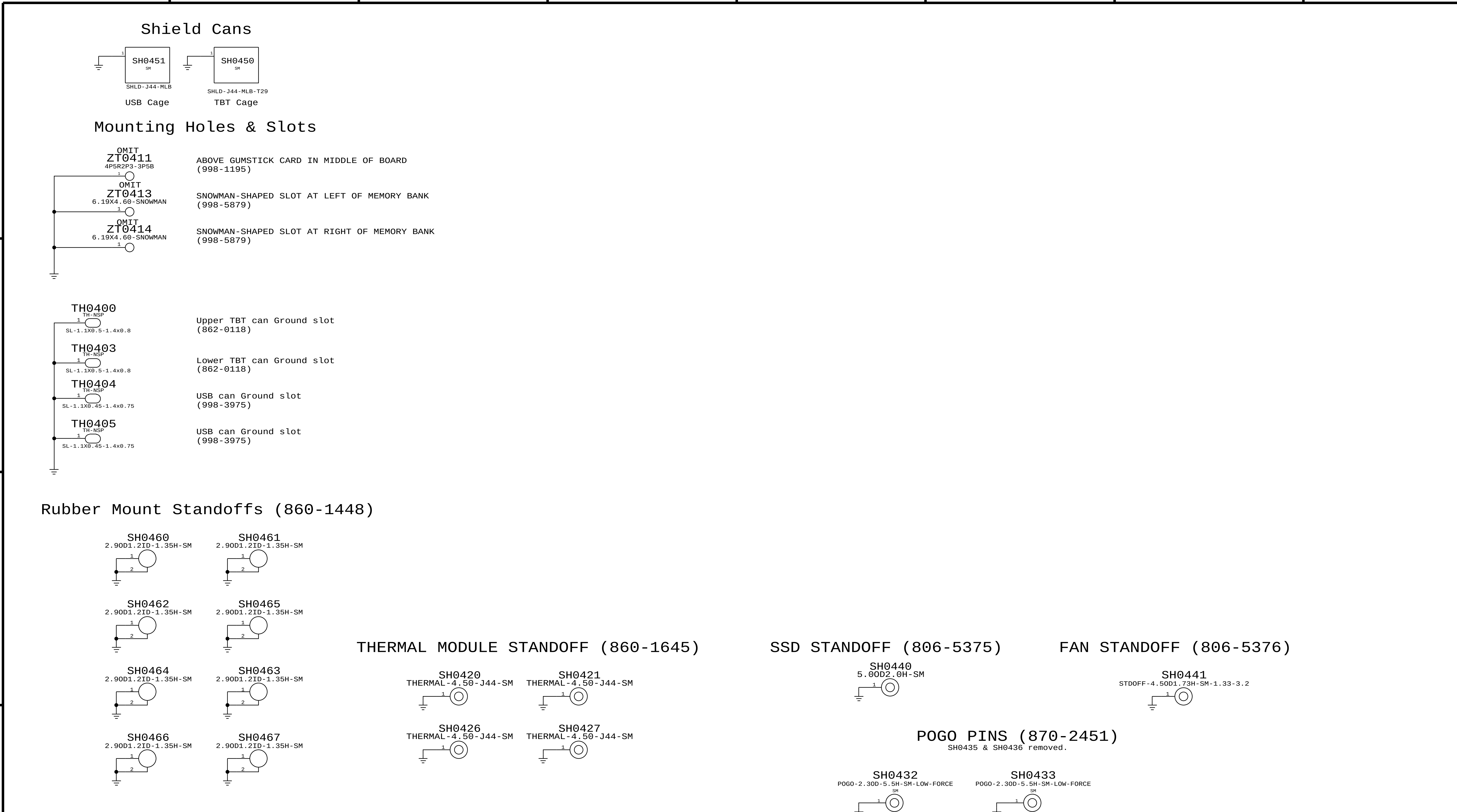

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PAGE TITLE: PD Parts

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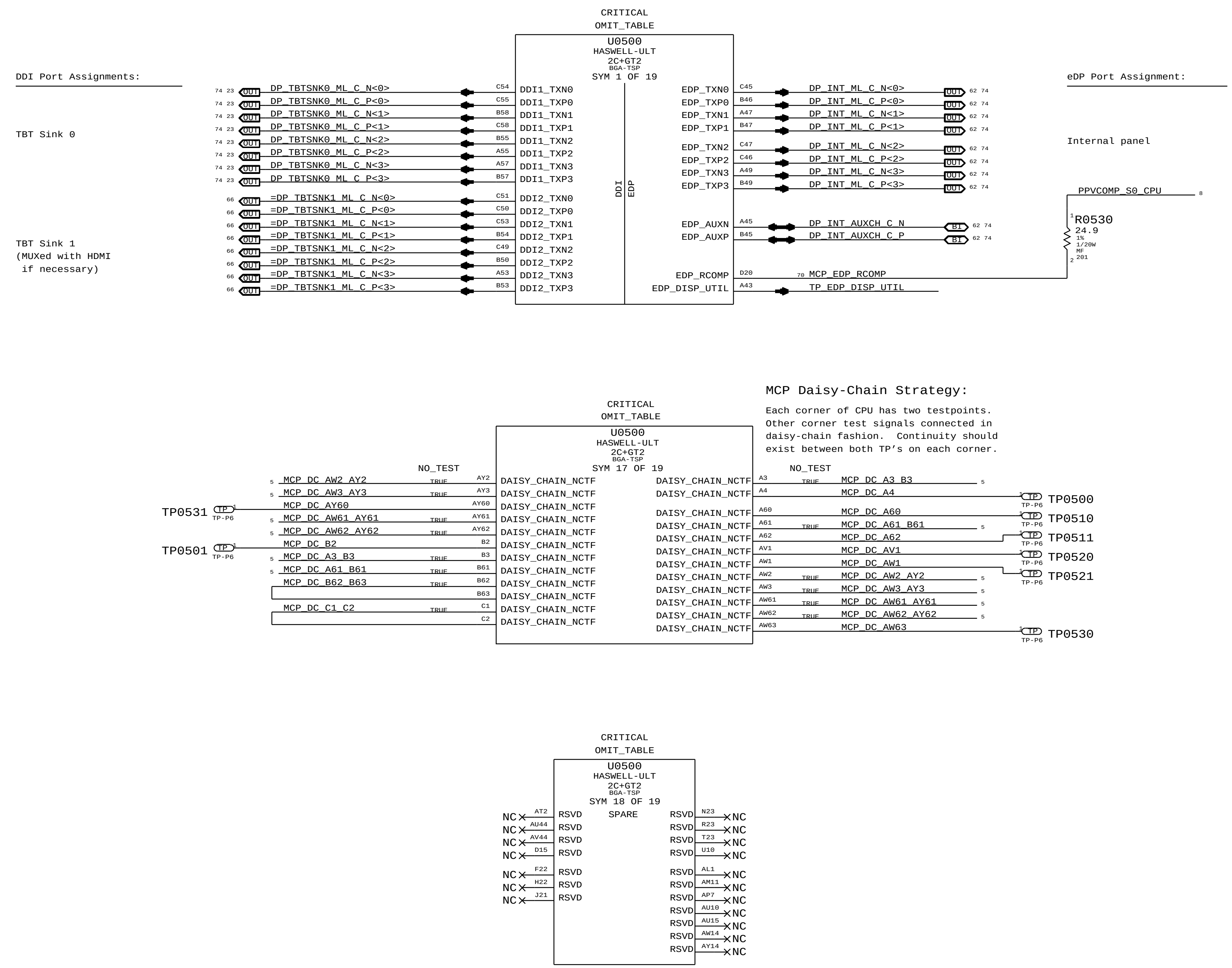

SYNC_MASTER=J41 SYNC_DATE=10/23/2012

PAGE TITLE

CPU GFX/NCTF/RSVD

Apple Inc.

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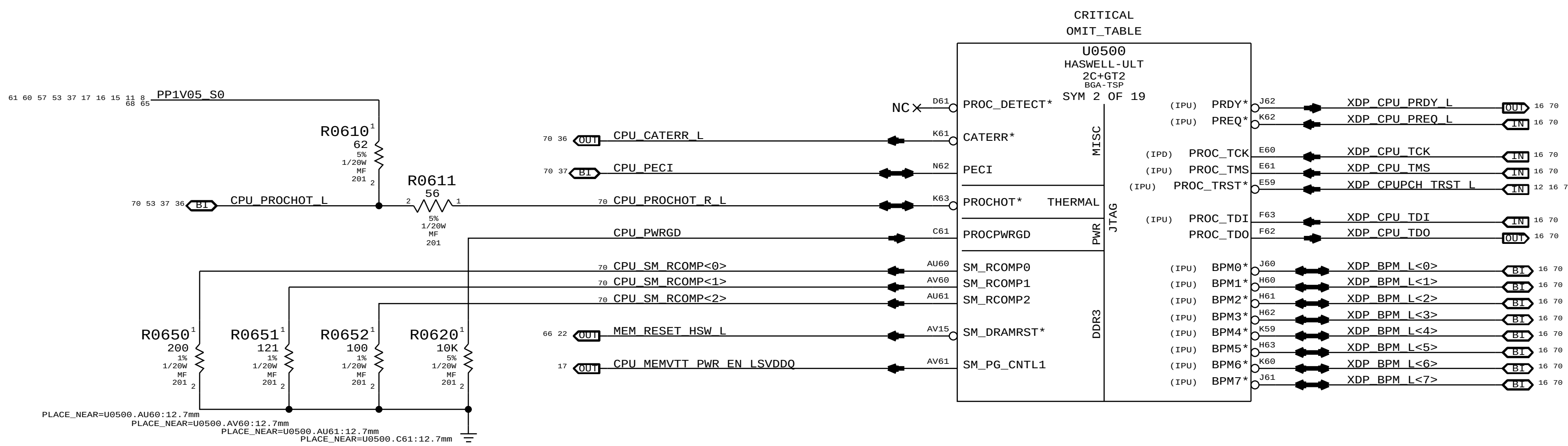
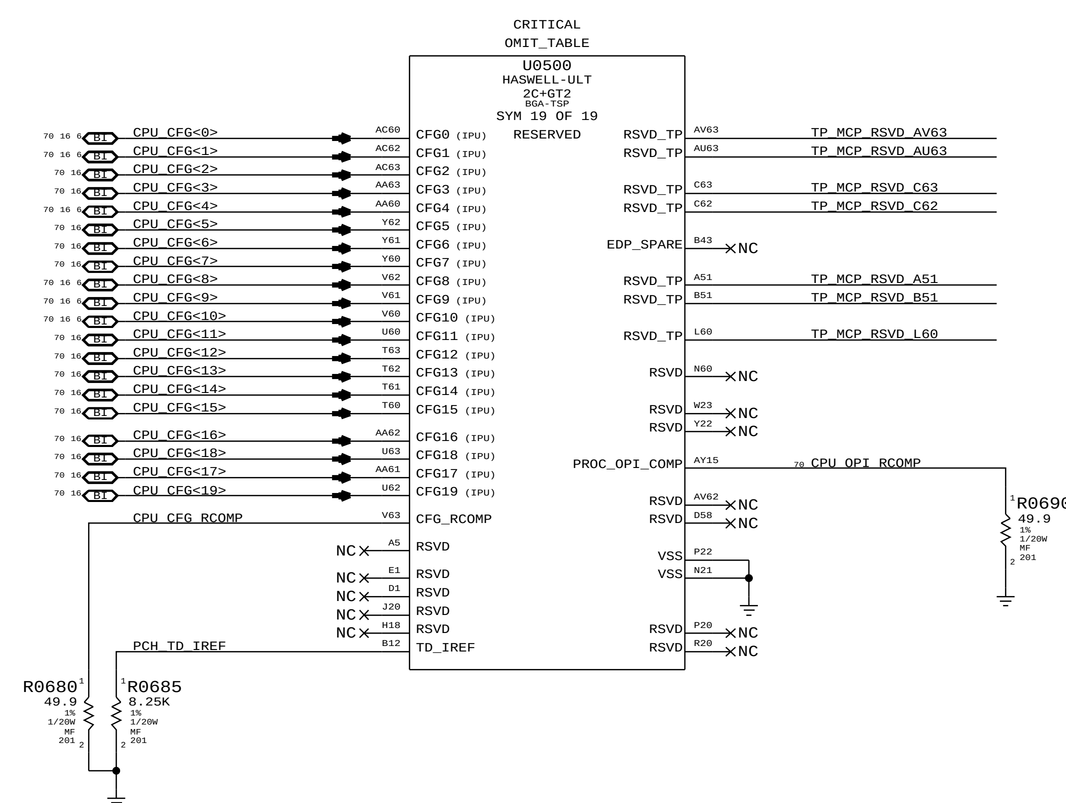
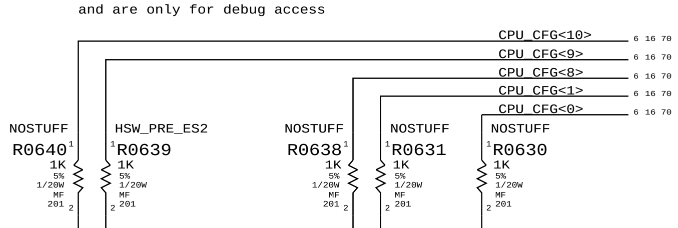


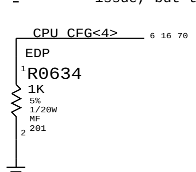


CFG<10>:SAFE MODE BOOT	1 = NORMAL OPERATION	0 = POWER FEATURES NOT ACTIVE
CFG<9> :NO SVID-CAPABLE VR	1 = VR SUPPORTS SVID	0 = VR DOES NOT SUPPORT SVID
CFG<8> :ALLOW NOA ON LOCKED UNITS	1 = NORMAL OPERATION	0 = NOA ALWAYS UNLOCKED
CFG<4> :eDP ENABLE/DISABLE	1 = DISABLED	0 = ENABLED
CFG<1> :PCH-LESS MODE	1 = NORMAL OPERATION	0 = PCH-LESS MODE
CFG<0> :RESET SEQUENCE STALL	1 = NORMAL OPERATION	0 = STALL AFTER PCU PLL LOCK

These can be placed close to J1800
and are only for debug access



NOTE: Pre-ES2 CPUs have issue with Sx cycling, must set CFG<9> low to avoid issue, but this locks CPU VR at 1.7V Vboot (CPU Sighting #4391569).



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PAGE TITLE

CPU Misc/JTAG/CFG/RSVD

Apple Inc.

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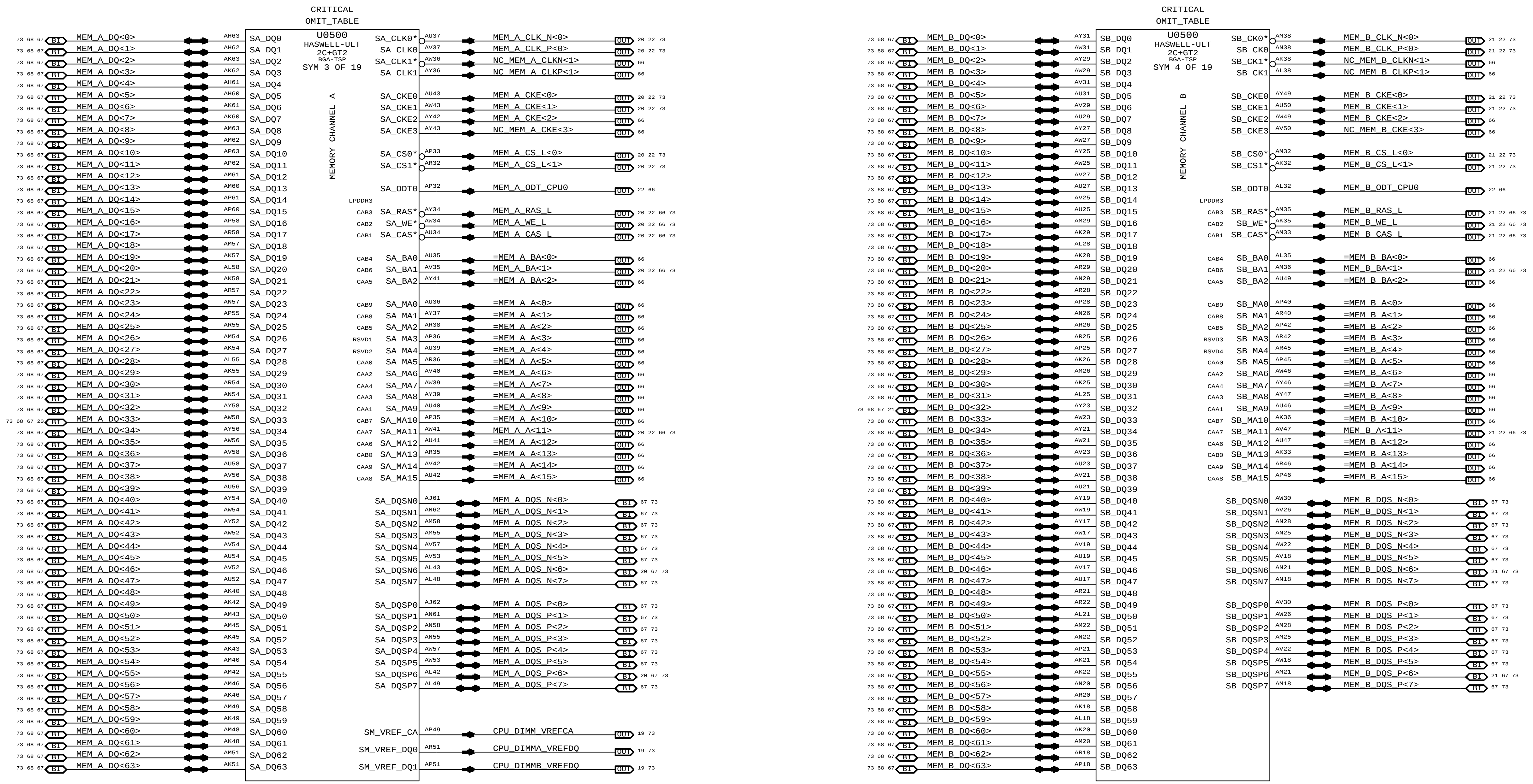

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PAGE TITLE

CPU DDR3/LPDDR3 Interfaces

Apple Inc.

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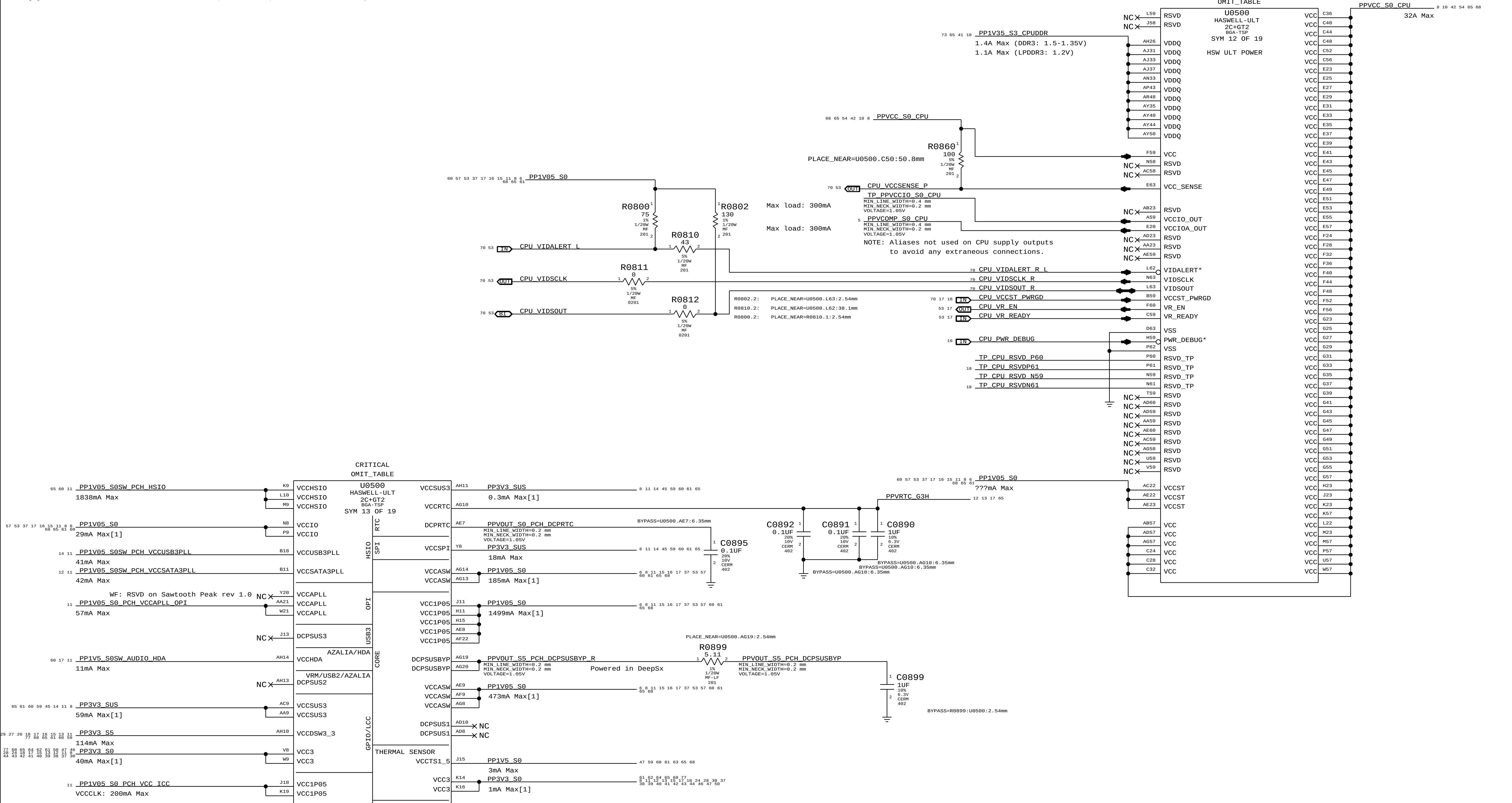
HSW-ULT current estimates from Haswell Mobile ULT Processor EDS vol 1, doc #502406, v0.9.
LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0.
Note [1] current numbers from clarification email, from Srini, dated 9/10/2012 2:11pm.



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CPU/PCH POWER

Apple Inc.

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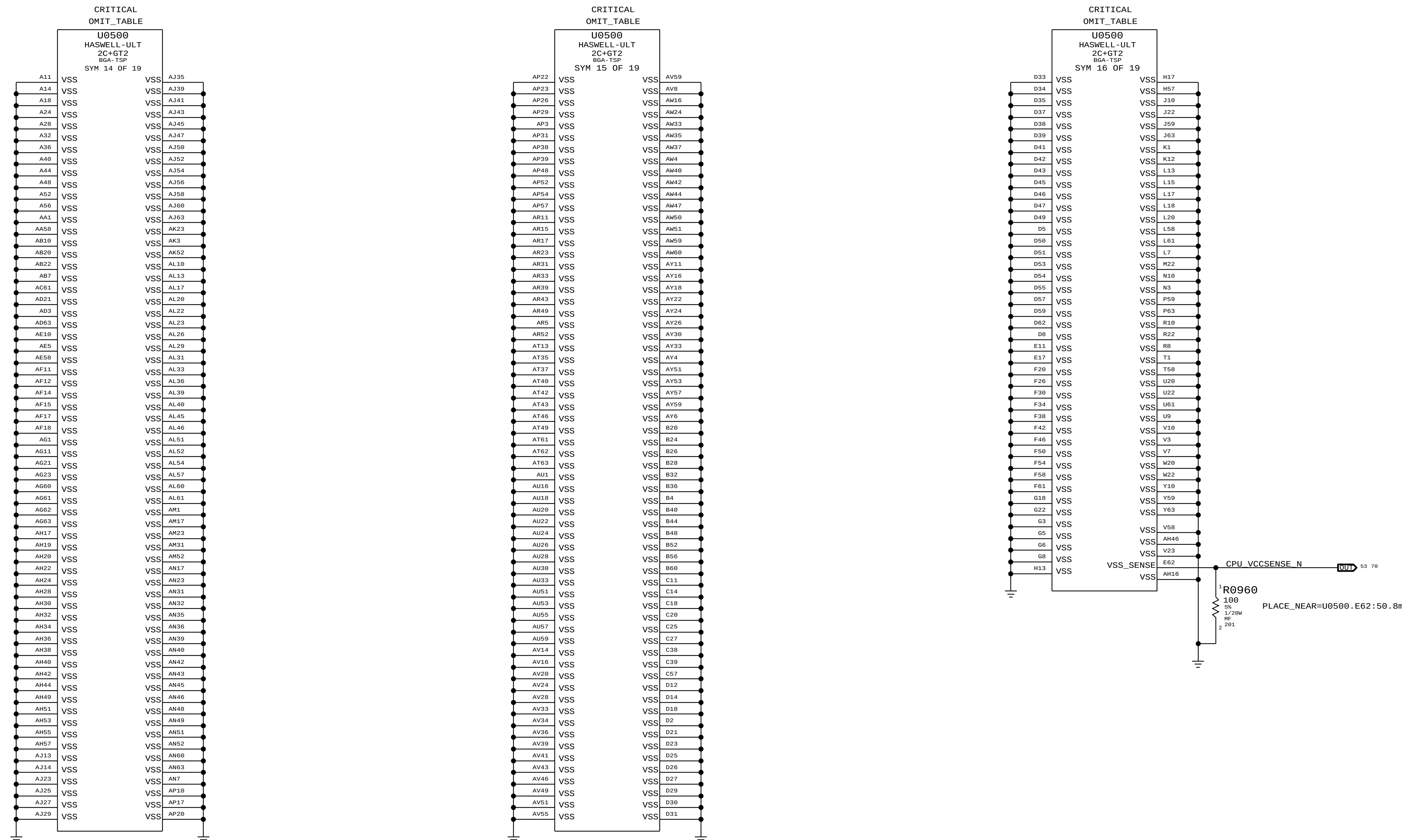

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PAGE TITLE

CPU/PCH GROUNDS

Apple Inc.

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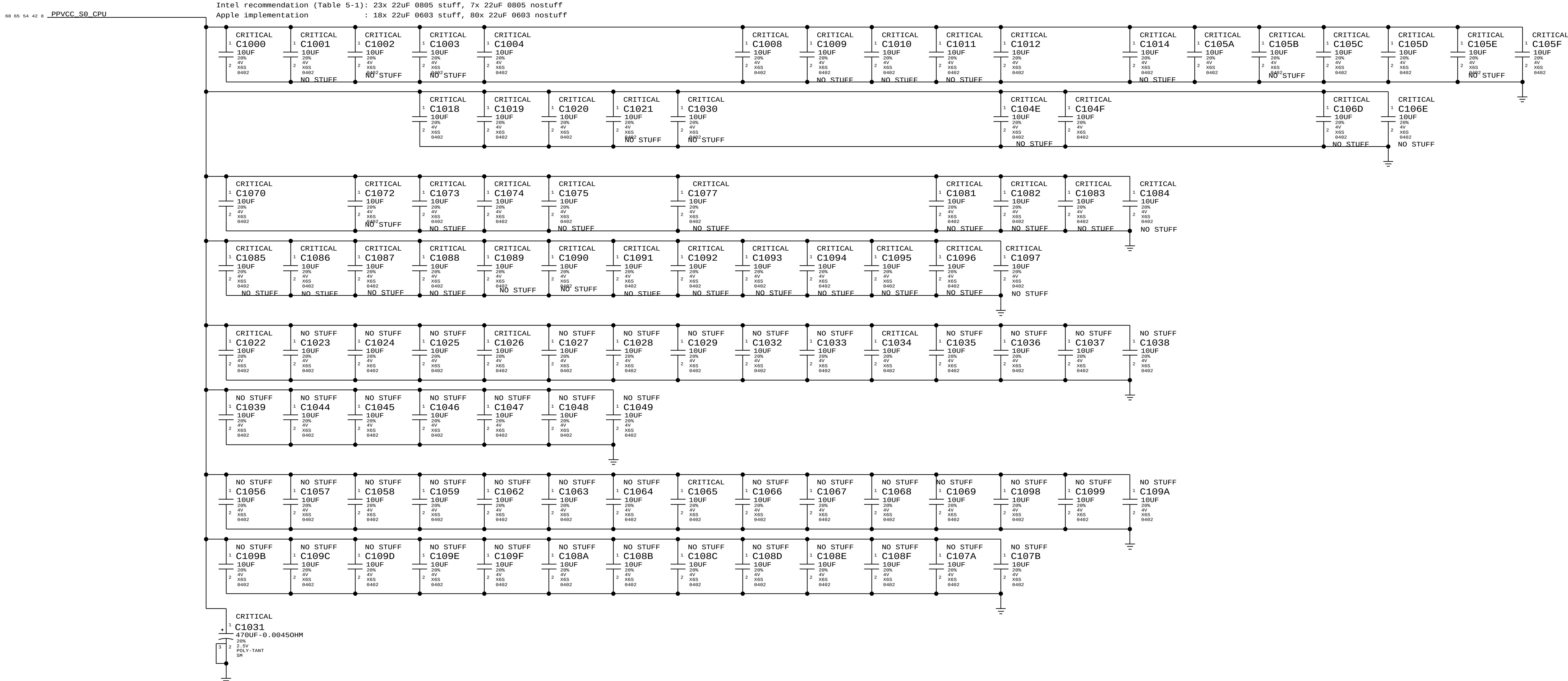
All Intel recommendations from Intel doc #503160 Shark Bay Ultrabook Platform Power Delivery Design Guide Rev 0.9 unless stated otherwise

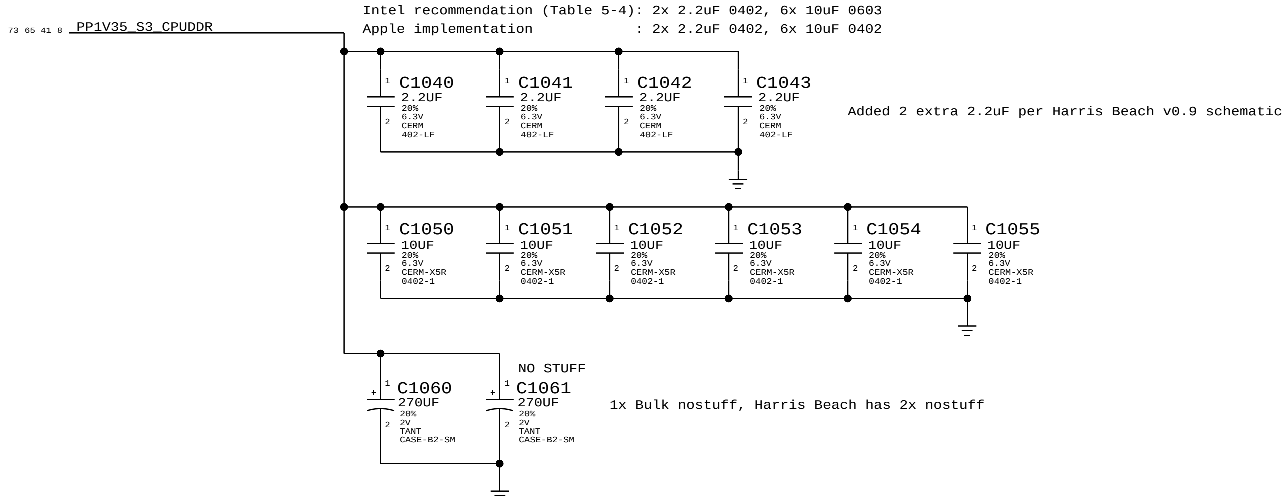
CPU VCC Decoupling

Intel recommendation (Table 5-1): 23x 22uF 0805 stuff, 7x 22uF 0805 nostuff
Apple implementation : 18x 22uF 0603 stuff, 80x 22uF 0603 nostuff



CPU VDDQ DECOUPLING

Intel recommendation (Table 5-4): 2x 2.2uF 0402, 6x 10uF 0603
Apple implementation : 2x 2.2uF 0402, 6x 10uF 0402



CPU VCC Decoupling

SYNC_MASTER=J41 SYNC_DATE=10/23/2012

PAGE TITLE: CPU Decoupling

Apple Inc.

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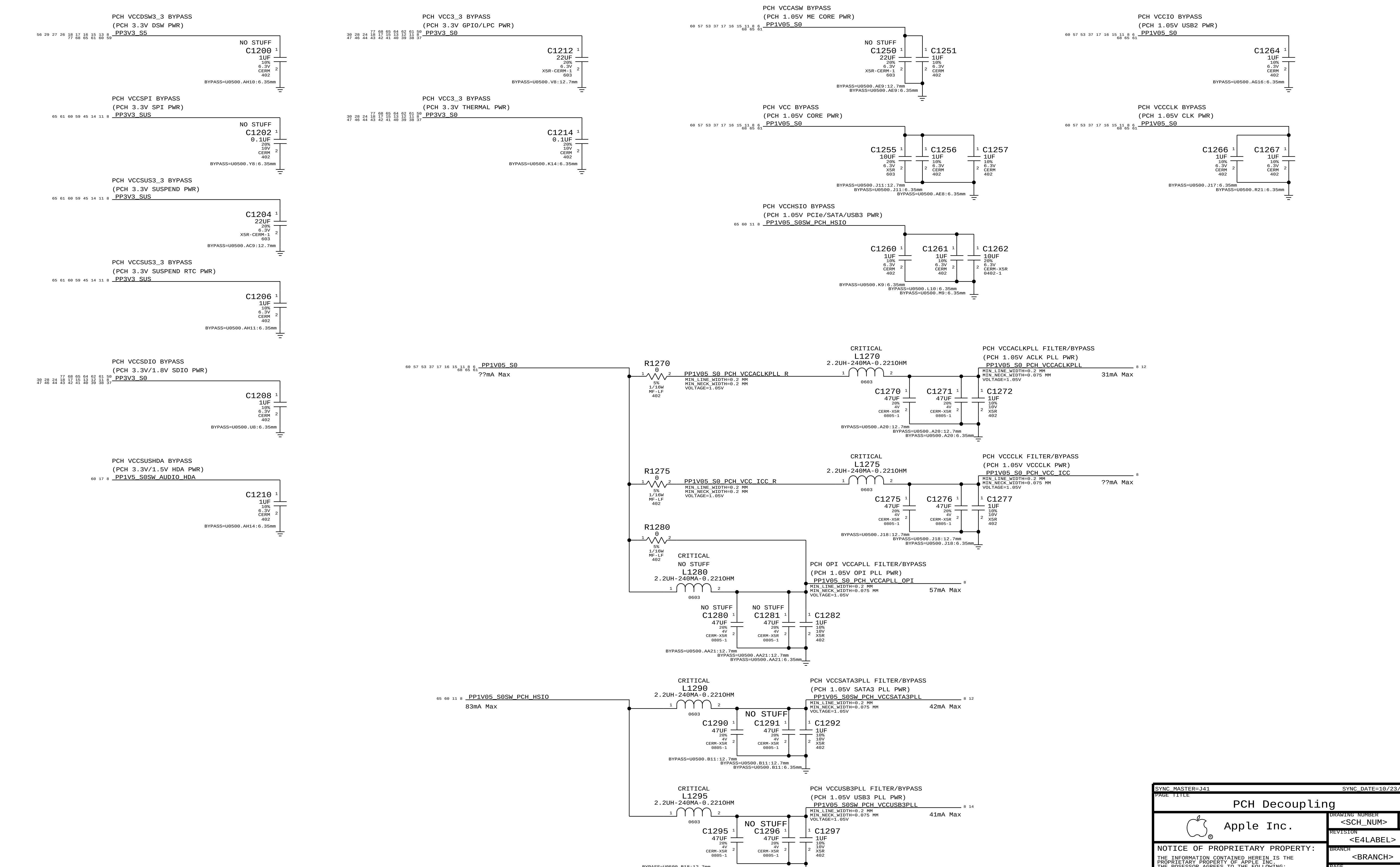

LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0
as well as from clarification email, from Srini, dated 9/10/2012 2:11pm.

SYNC_MASTER=J41 SYNC_DATE=10/23/2012

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PCH Decoupling

Apple Inc.

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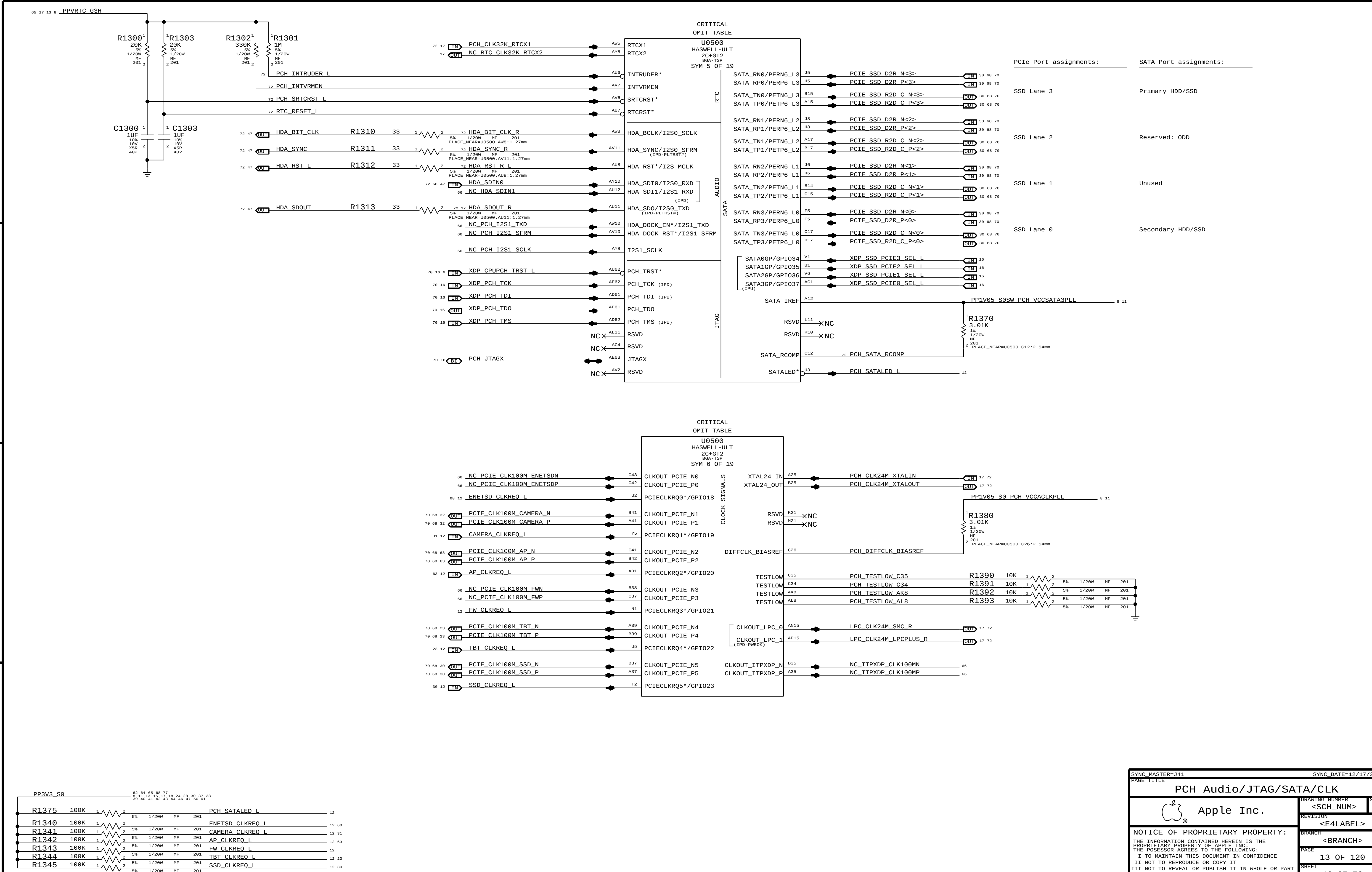

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PAGE TITLE

PCH Audio/JTAG/SATA/CLK

Apple Inc.

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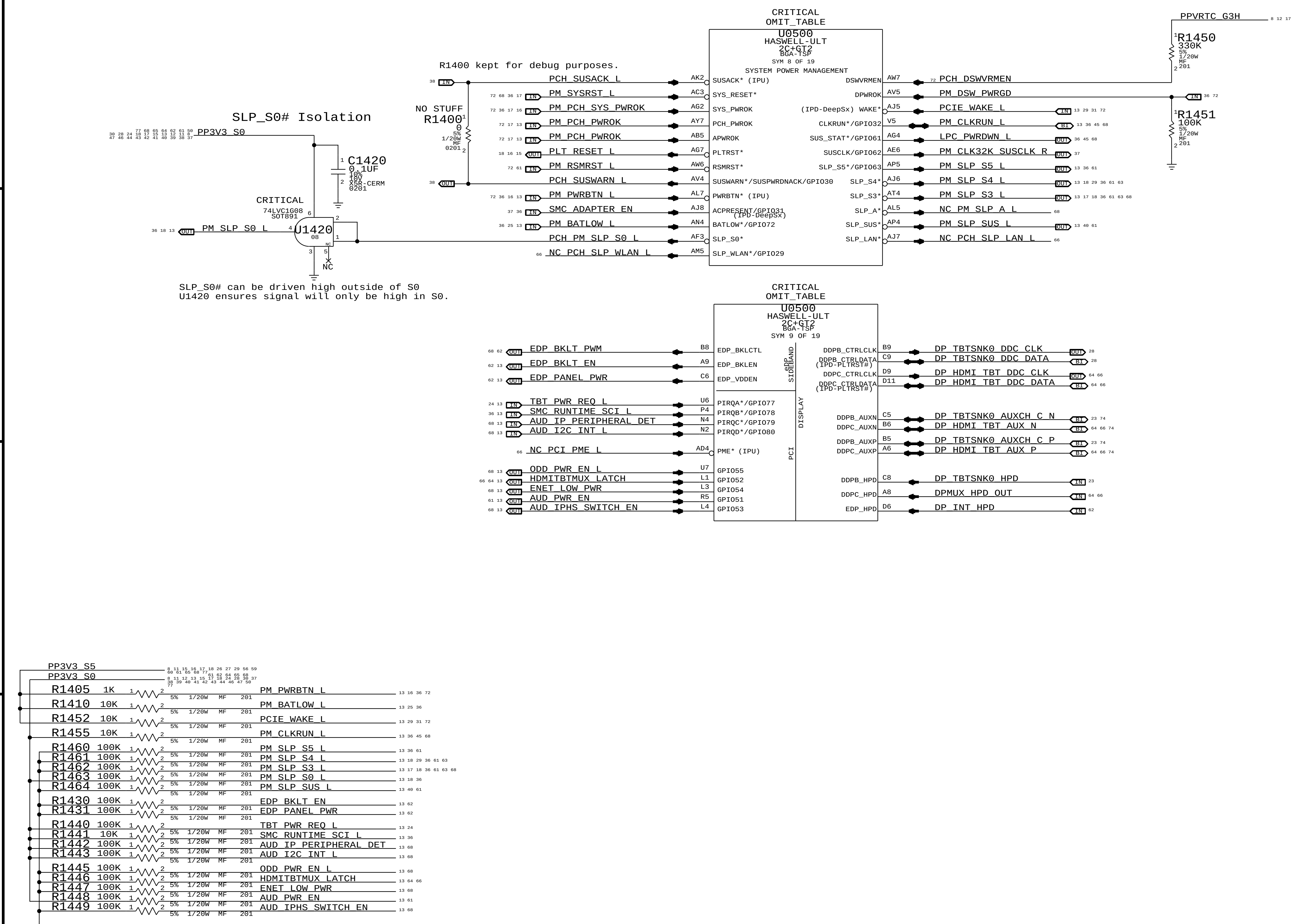

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PAGE TITLE

PCH PM/PCI/GFX

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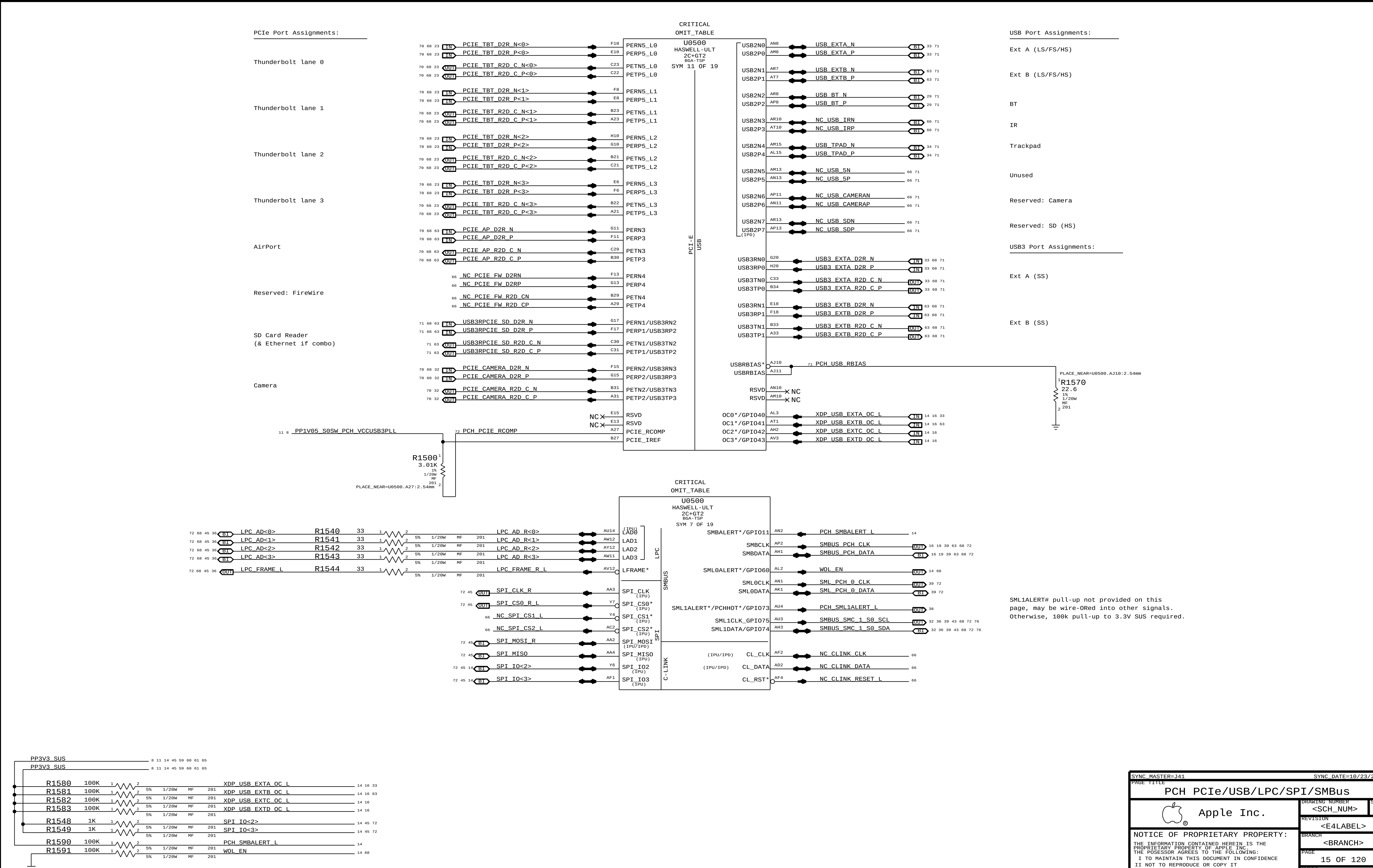

SYNC_MASTER=J41 SYNC_DATE=10/23/2012

PAGE TITLE

PCH PCIe/USB/LPC/SPI/SMBus

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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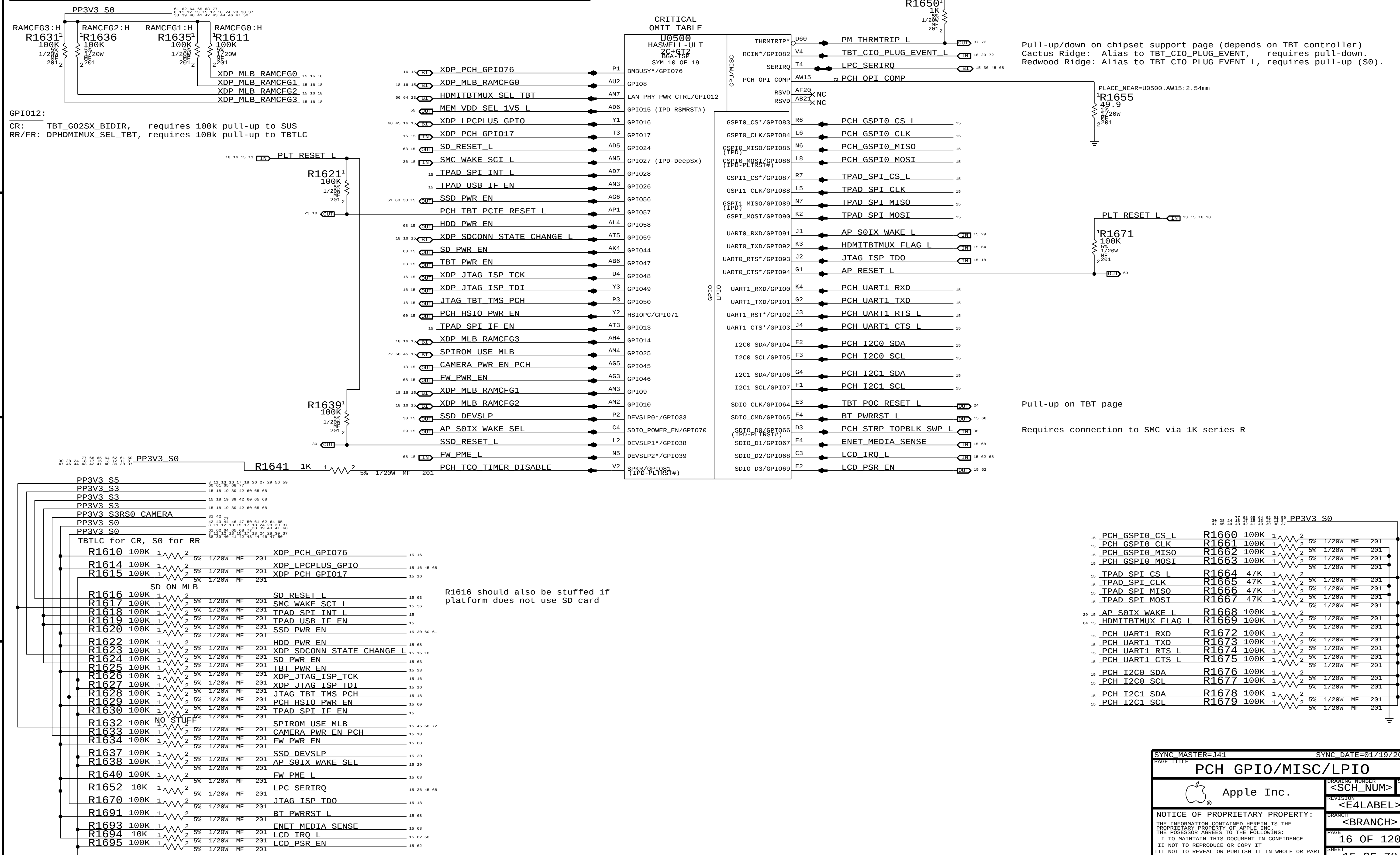
BOM GROUP	BOM OPTIONS
RAMCFG_SLOT	RAMCFG3:H,RAMCFG2:H,RAMCFG1:H,RAMCFG0:H



SYNC_MASTER=J41 SYNC_DATE=01/19/2013

PAGE TITLE

PCH GPIO/MISC/LPIO

Apple Inc.

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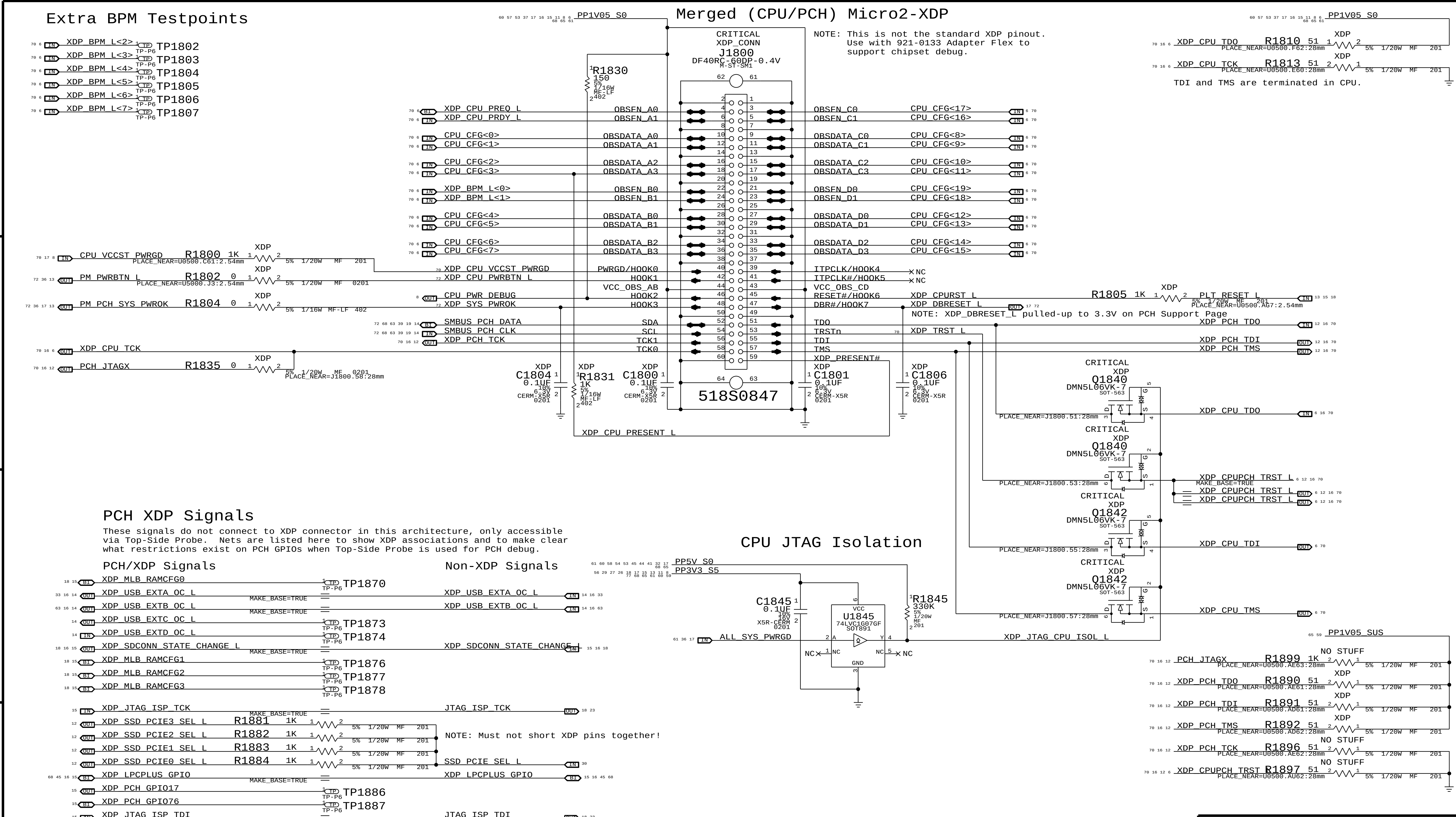

Unused & MLB_RAMCFGx GPIOs have TPs.

USB Overcurrents are aliased, do not cause USB OC# events during PCH debug.

SDCONN_STATE_CHANGE_L is aliased, do not plug/unplug SD Cards during PCH debug.

JTAG_ISP (non-TMS) nets are aliased, do not attempt bit-banged JTAG during PCH debug.

NOTE: Should force PCH GPIO47 high to ensure TBT router powered to avoid leakage/clamping of signals.

SSD_PCIEx_SEL_L straps are connected via 1K to common net.

LPCPLUS_GPIO is aliased, do not attempt use during PCH debug.

SYNC_MASTER=WFERRY_J43 SYNC_DATE=12/21/2012

PAGE TITLE
CPU/PCH Merged XDP

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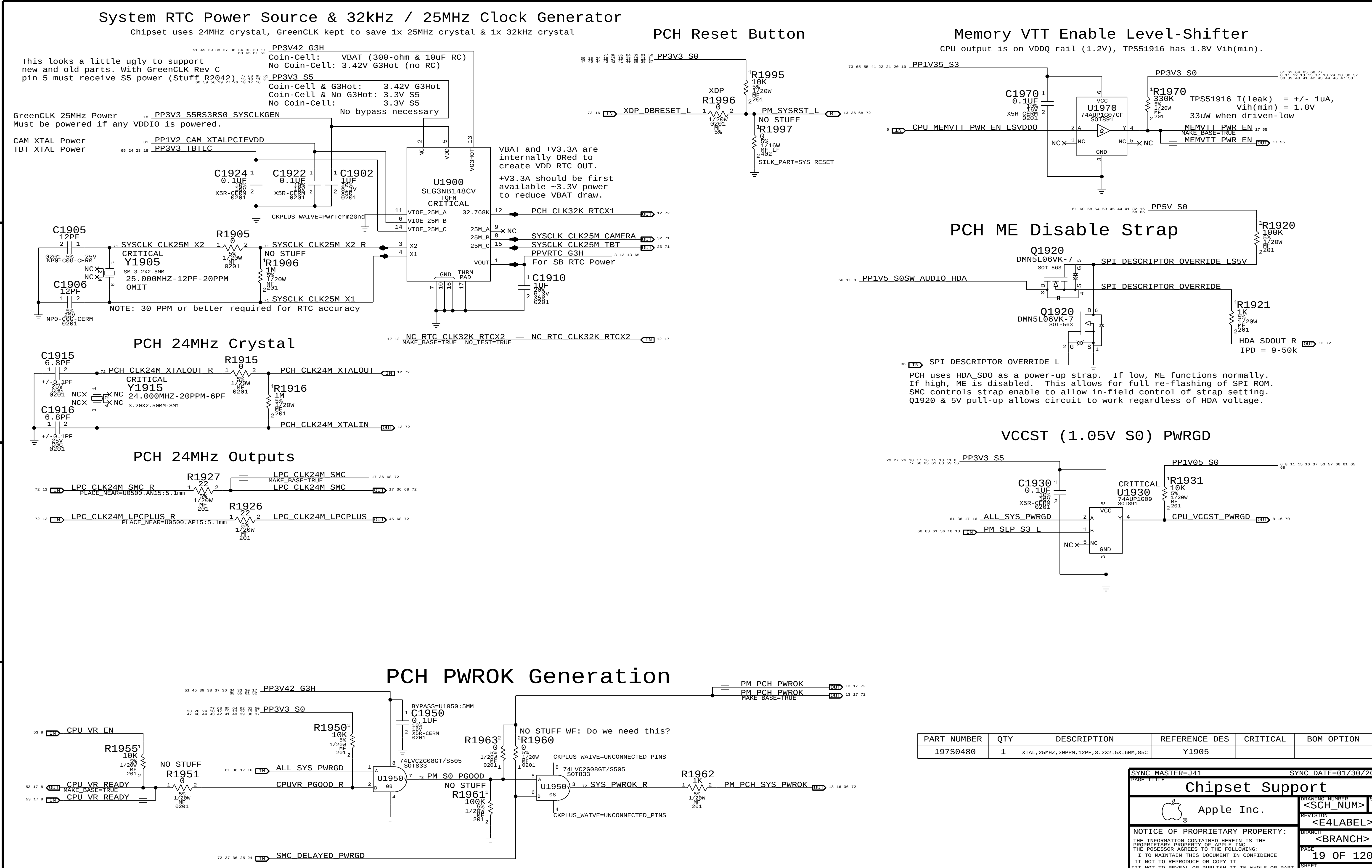

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
197S0480	1	XTAL,25MHZ,20PPM,12PF,3.2X2.5X.6MM,85C	Y1905		

SYNC_MASTER=J41 SYNC_DATE=01/30/2013

PAGE TITLE

Chipset Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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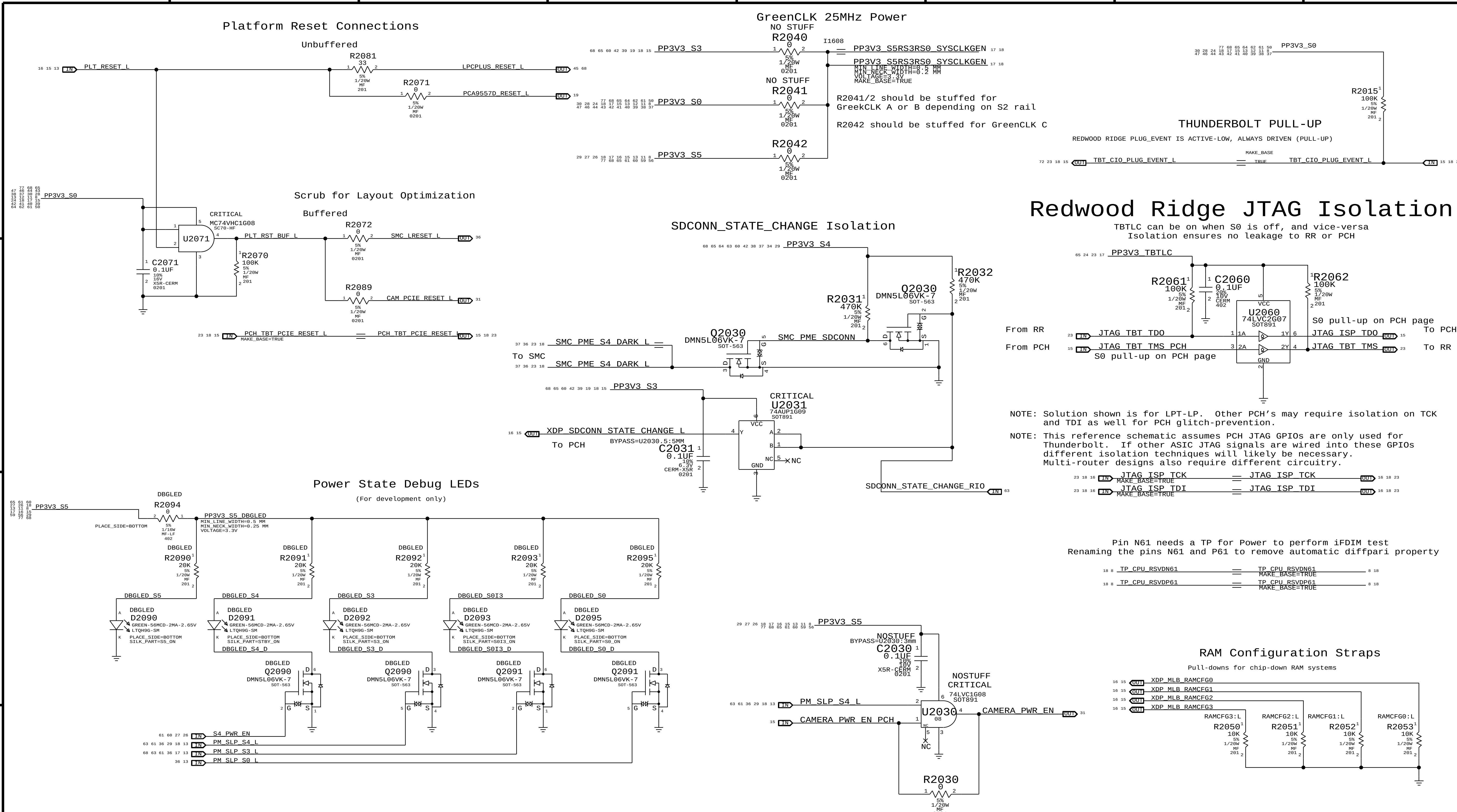


SYNC_MASTER=J41 SYNC_DATE=10/23/2012

PAGE TITLE: Project Chipset Support

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
- DDRVREF_DAC - Stuffs DAC margining circuit.

CPU-Based Margining

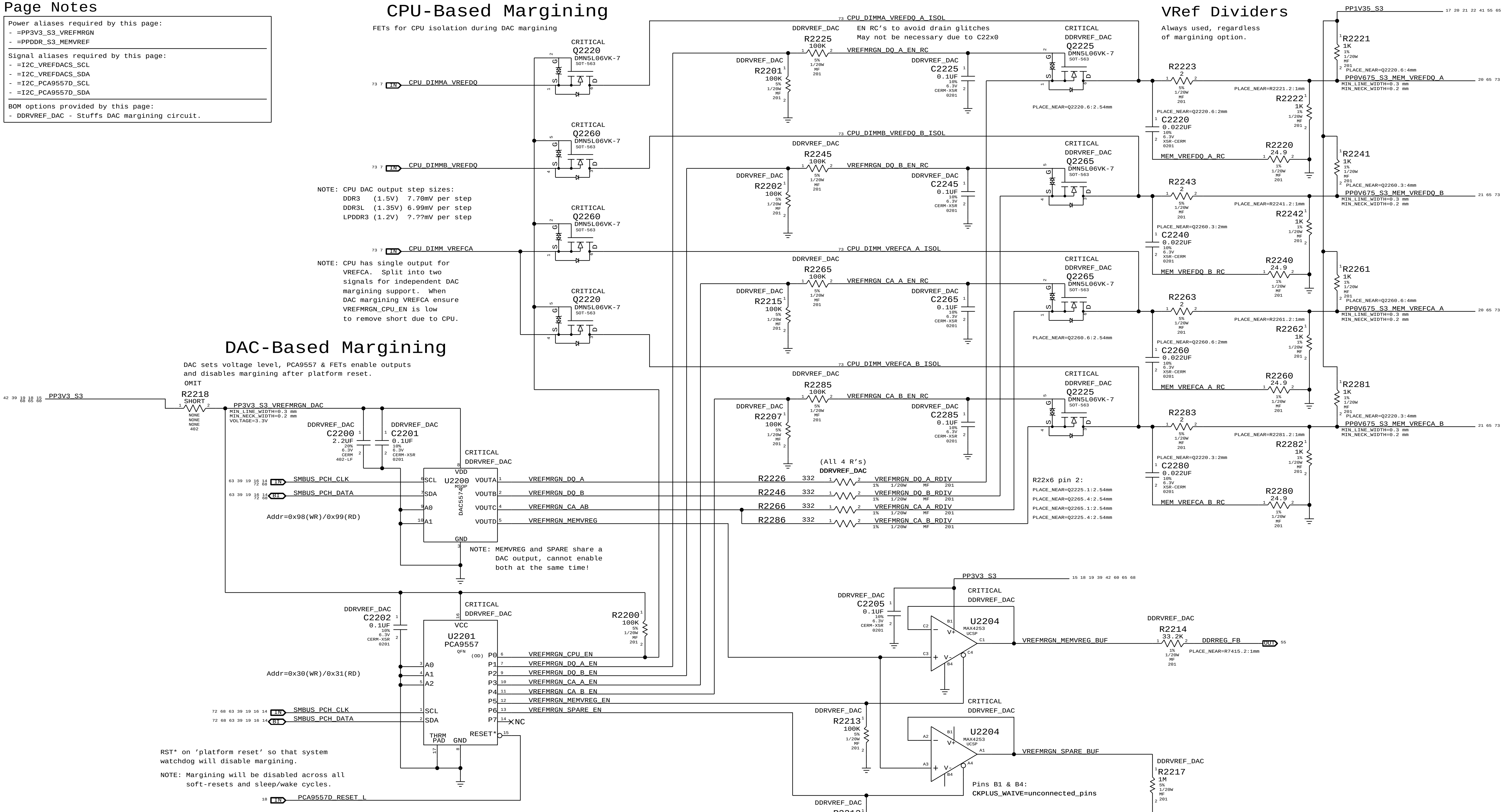
FETs for CPU isolation during DAC margining

VRef Dividers

Always used, regardless of margining option.

DAC-Based Margining

DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.



NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step
LPDDR3 (1.2V) ?.??mV per step

NOTE: CPU has single output for VREFCA. Split into two signals for independent DAC margining support. When DAC margining VREFCA ensure VREFMRGN_CPU_EN is low to remove short due to CPU.

EN RC's to avoid drain glitches
May not be necessary due to C22x0

NOTE: MEMVREG and SPARE share a DAC output, cannot enable both at the same time!

Addr=0x98(WR)/0x99(RD)

Addr=0x30(WR)/0x31(RD)

RST* on 'platform reset' so that system watchdog will disable margining.

NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

Pins B1 & B4:
CKPLUS_WAIVE=unconnected_pins

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5

	LPDDR3 (1.2V)	DDR3L (1.35V)	LPDDR3 (1.2V)	DDR3L (1.35V)
Nominal value	0.600V (DAC: 0x2E.5)	0.675V (DAC: 0x34)	1.200V (DAC: 0x5D)	1.343V (DAC: 0x68)
Margined target:	0.300V - 0.900V (+/- 300mV)	0.337V - 1.013V (+/- 337.5mV)	0.800V - 1.600V (+/- 400mV)	0.972V - 1.714V (+/- 371mV)
DAC range:	0.000V - 1.199V (0x00 - 0x5D)	0.000V - 1.354V (0x00 - 0x69)	0.000V - 2.397V (0x00 - 0xBA)	0.000V - 2.694V (0x00 - 0xD1)
VRef current:	+73uA - -73uA (- = sourced)	+82uA - -82uA (- = sourced)	+21uA - -21uA (- = sourced)	+25uA - -25uA (- = sourced)
DAC step size:	6.36mV / step @ output	6.36mV / step @ output	4.28mV / step @ output	3.53mV / step @ output

NOTE: LPDDR3 assumes TPS51916 supply with 28.7k/57.6k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=01/02/2013

PAGE TITLE
DDR3 VREF MARGINING

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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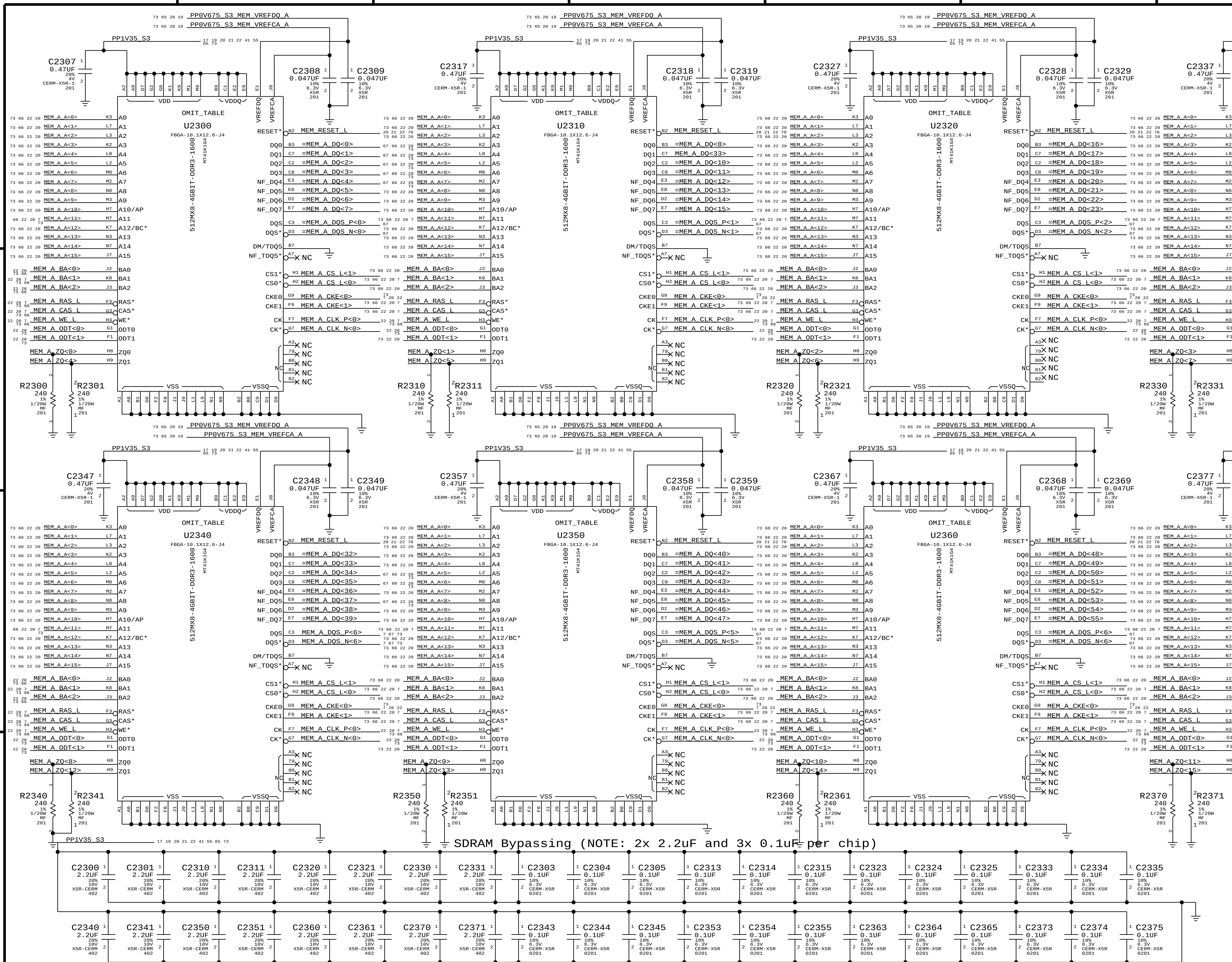

SYNC_MASTER=J14 SYNC_DATE=10/23/2012

PAGE TITLE

DDR3 SDRAM Bank A (Rank 0)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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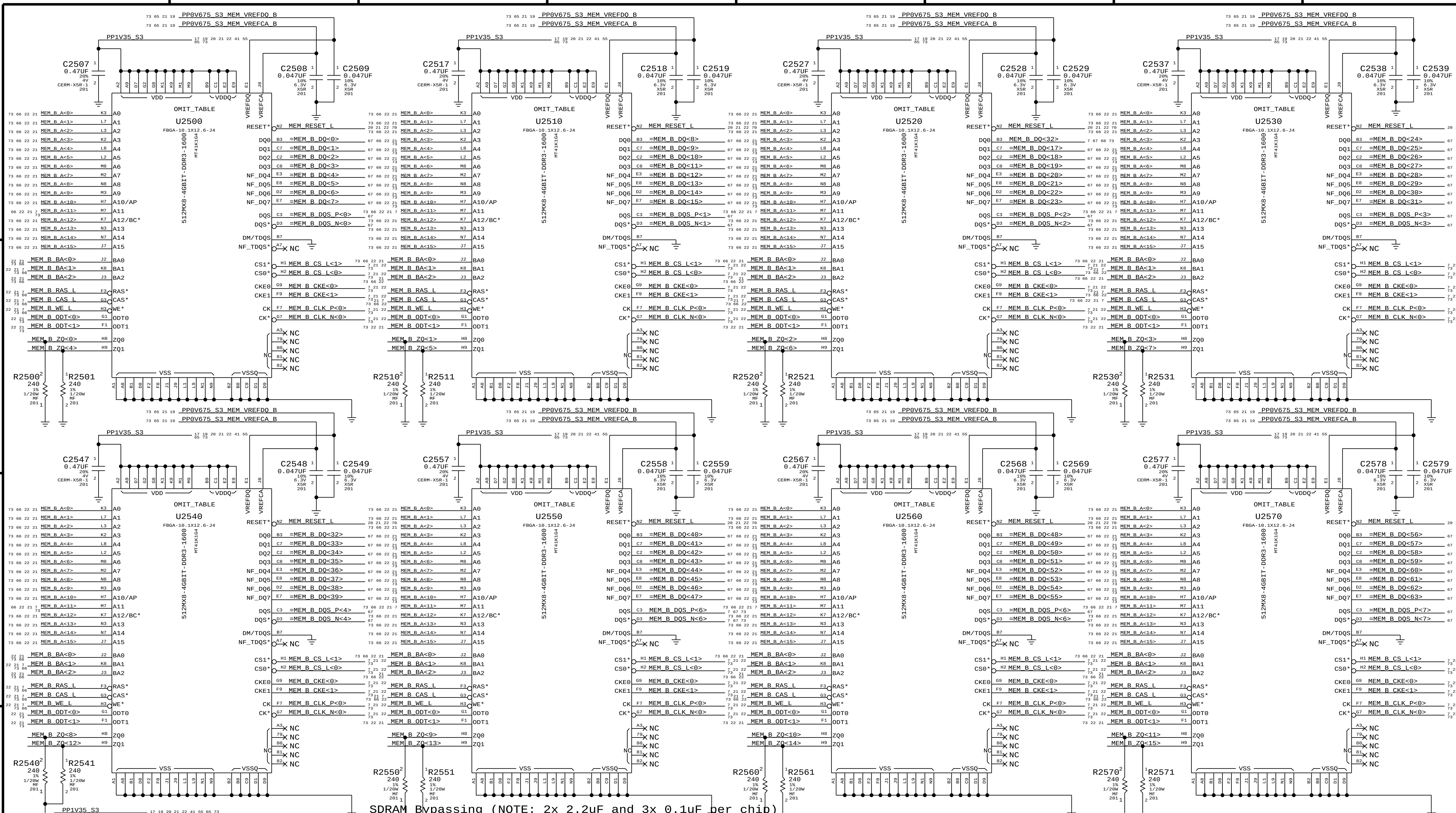
PAGE 23 OF 120

SHEET 20 OF 78



SDRAM Bypassing (NOTE: 2x 2.2uF and 3x 0.1uF per chip)

SYNC_MASTER=J14 SYNC_DATE=10/23/2012

PAGE TITLE

DDR3 SDRAM Bank B (Rank 0)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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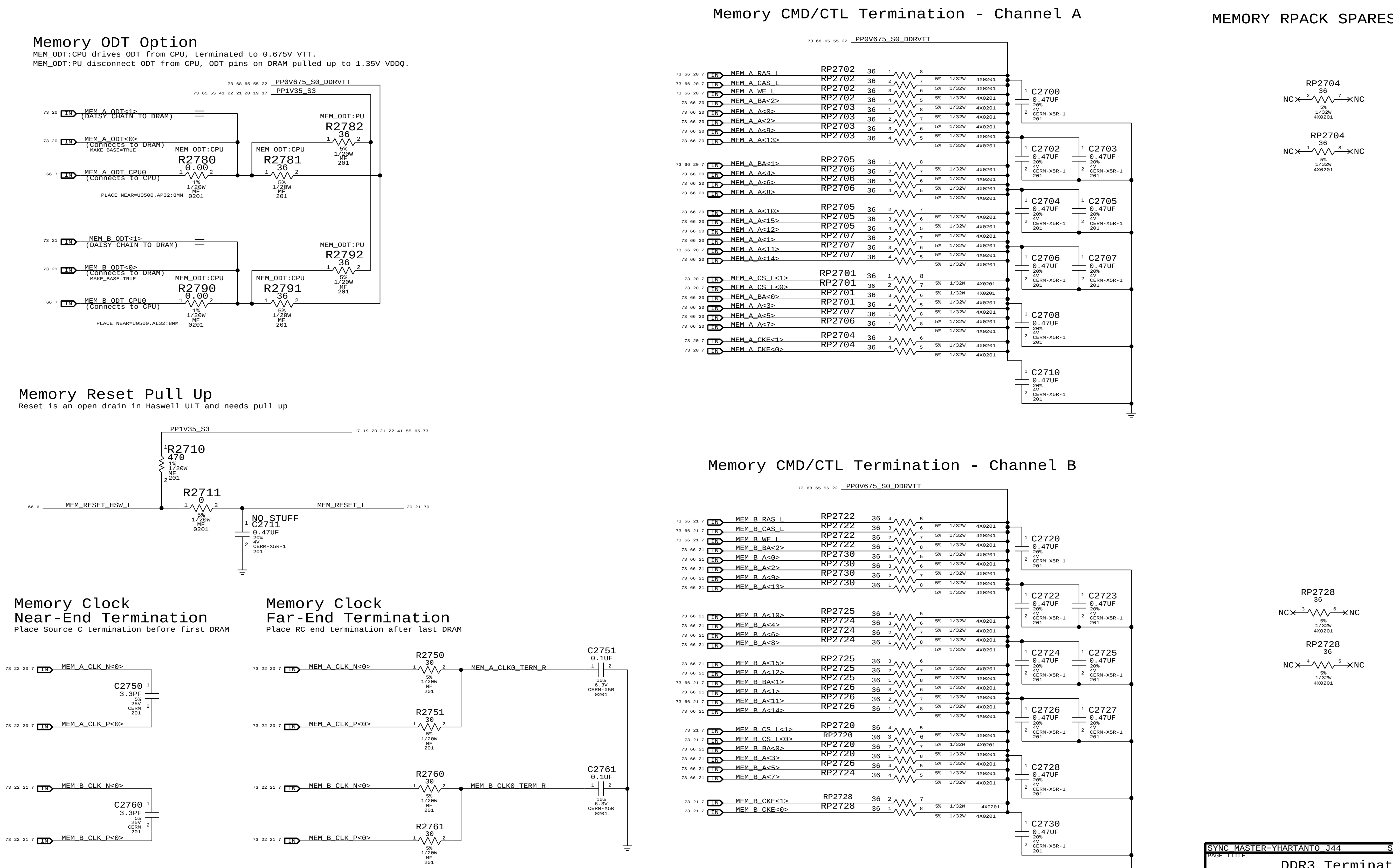
SHEET 21 OF 78



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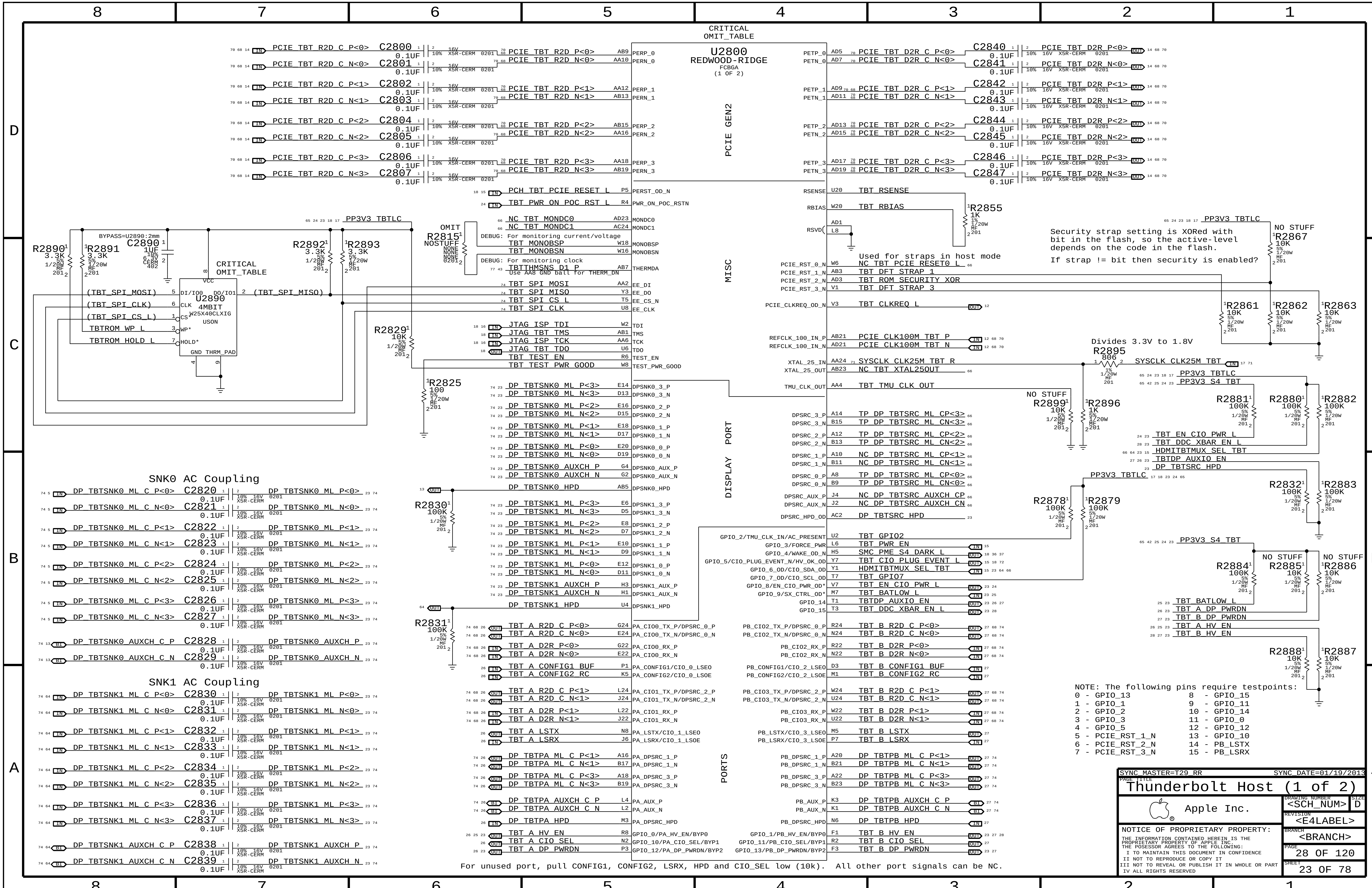
PAGE TITLE	DDR3 Termination
Apple Inc.	DRAWING NUMBER <SCH_NUM> SIZE D
	REVISION <E4LABEL>
	BRANCH <BRANCH>
	PAGE 27 OF 120
	SHEET 22 OF 78



CRITICAL
OMIT_TABLE
U2800
REDWOOD-RIDGE
FCBGA
(1 OF 2)
PCIE GEN2
MISC
DISPLAY PORT
PORTS
PCIE_TBT_R2D_C_P<0>
PCIE_TBT_R2D_C_N<0>
PCIE_TBT_D2R_P<0>
PCIE_TBT_D2R_N<0>
PCH_TBT_PCIE_RESET_L
TBT_PWR_ON_POC_RST_L
PP3V3_TBTLC
PP3V3_S4_TBT
U2890
4MBIT
W25X40CLXIG
USON
(TBT_SPI_MOSI)
(TBT_SPI_CLK)
(TBT_SPI_CS_L)
(TBT_SPI_MISO)
TBTROM_WP_L
TBTROM_HOLD_L
BYPASS=U2890:2mm
DEBUG: For monitoring current/voltage
DEBUG: For monitoring clock
Use AA8 GND ball for THERM_DN
Used for straps in host mode
Security strap setting is XORed with
bit in the flash, so the active-level
depends on the code in the flash.
If strap != bit then security is enabled?
Divides 3.3V to 1.8V
SYSCLK_CLK25M_TBT
NO STUFF
SNK0 AC Coupling
SNK1 AC Coupling
NOTE: The following pins require testpoints:
0 - GPIO_13
1 - GPIO_1
2 - GPIO_2
3 - GPIO_3
4 - GPIO_5
5 - PCIE_RST_1_N
6 - PCIE_RST_2_N
7 - PCIE_RST_3_N
8 - GPIO_15
9 - GPIO_11
10 - GPIO_14
11 - GPIO_0
12 - GPIO_12
13 - GPIO_10
14 - PB_LSTX
15 - PB_LSRX
For unused port, pull CONFIG1, CONFIG2, LSRX, HPD and CIO_SEL low (10k). All other port signals can be NC.
SYNC_MASTER=T29_RR
SYNC_DATE=01/19/2013
PAGE TITLE
Thunderbolt Host (1 of 2)
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
D
REVISION
<E4LABEL>
BRANCH
<BRANCH>
PAGE
28 OF 120
SHEET
23 OF 78
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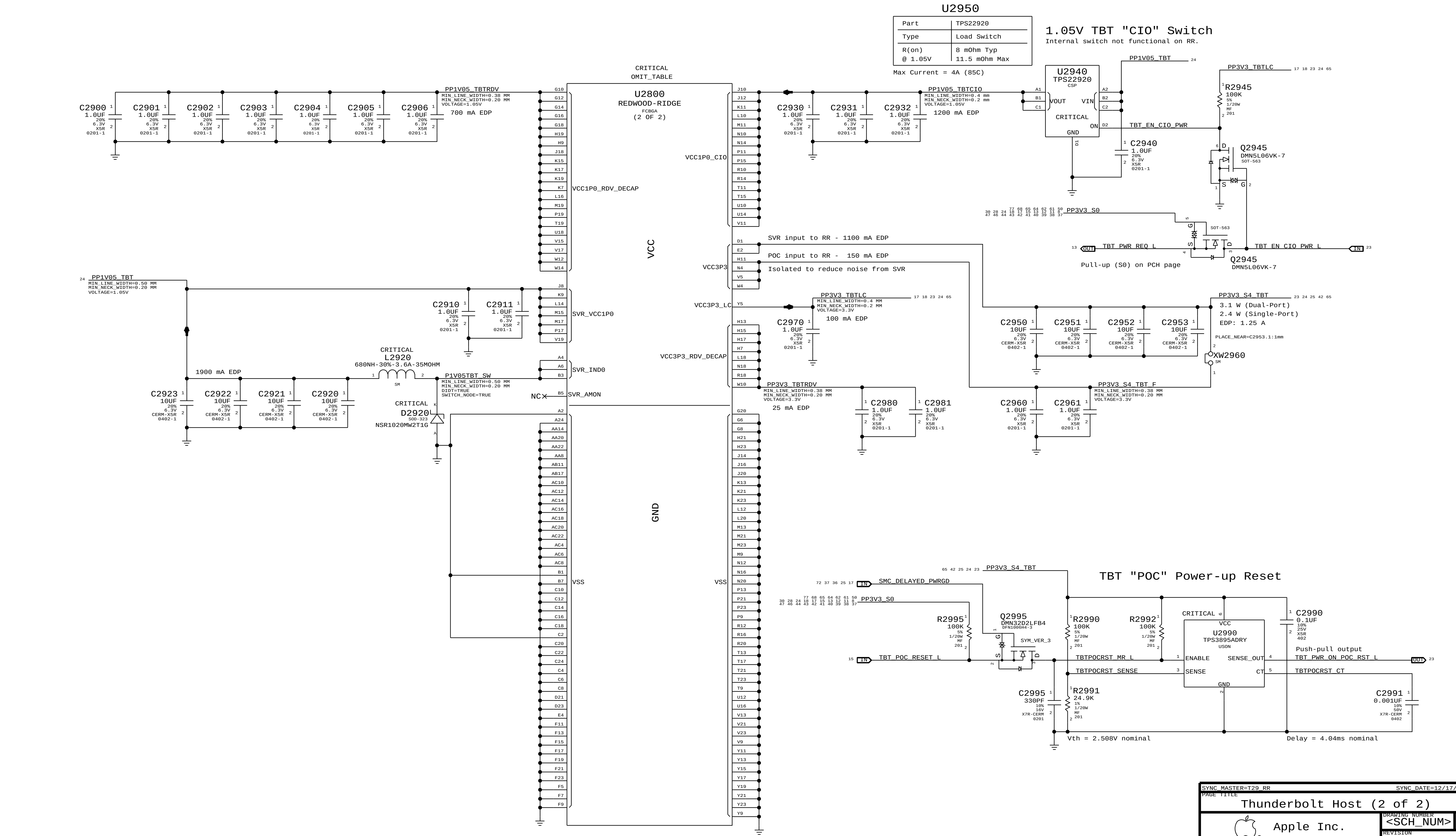
IV ALL RIGHTS RESERVED



EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

SYNC_MASTER=T29_RR SYNC_DATE=12/17/2012

PAGE TITLE: Thunderbolt Host (2 of 2)

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

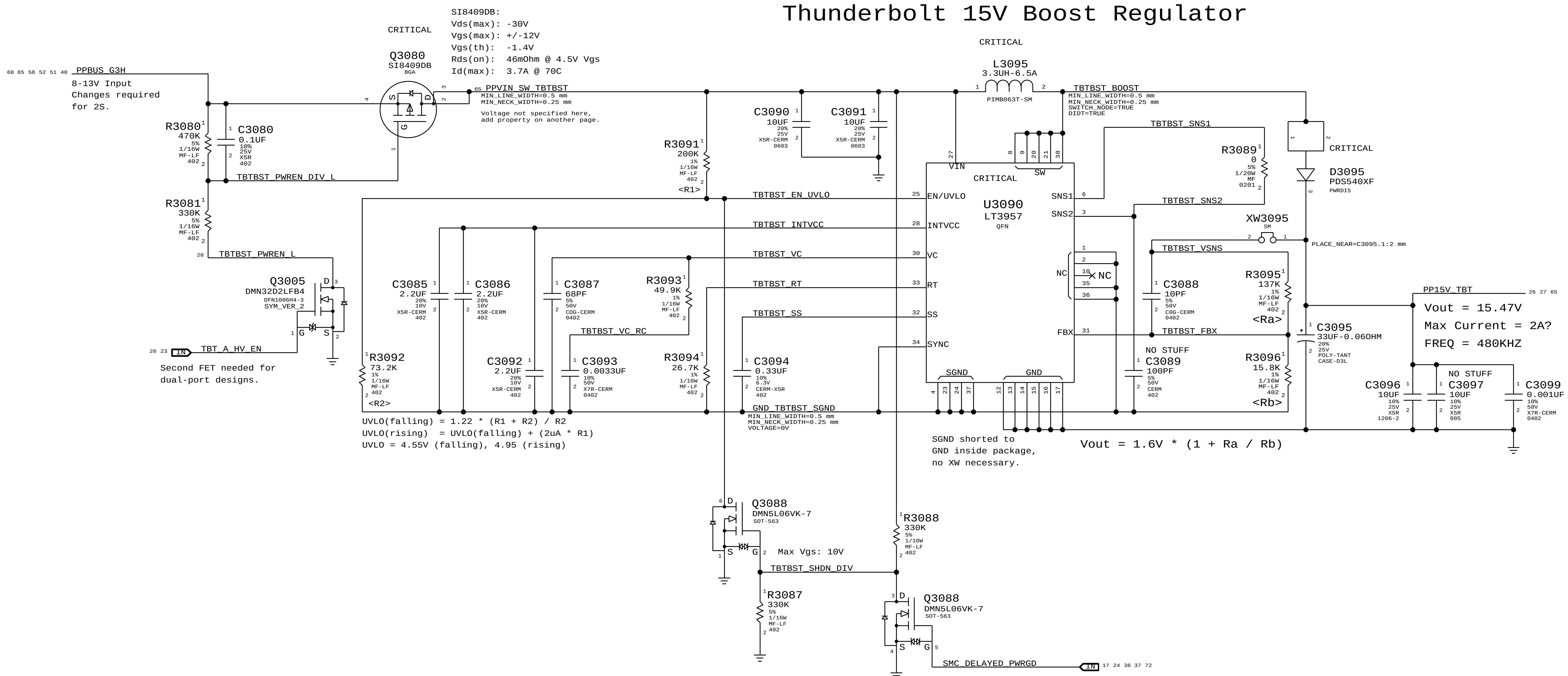
Page Notes

Power aliases required by this page:
- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)

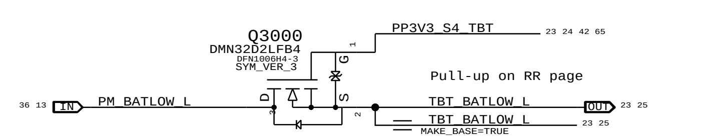
Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)

Thunderbolt 15V Boost Regulator



BATLOW# Isolation



SYNC_MASTER=T29_RR SYNC_DATE=11/19/2012

PAGE TITLE
Thunderbolt Mobile Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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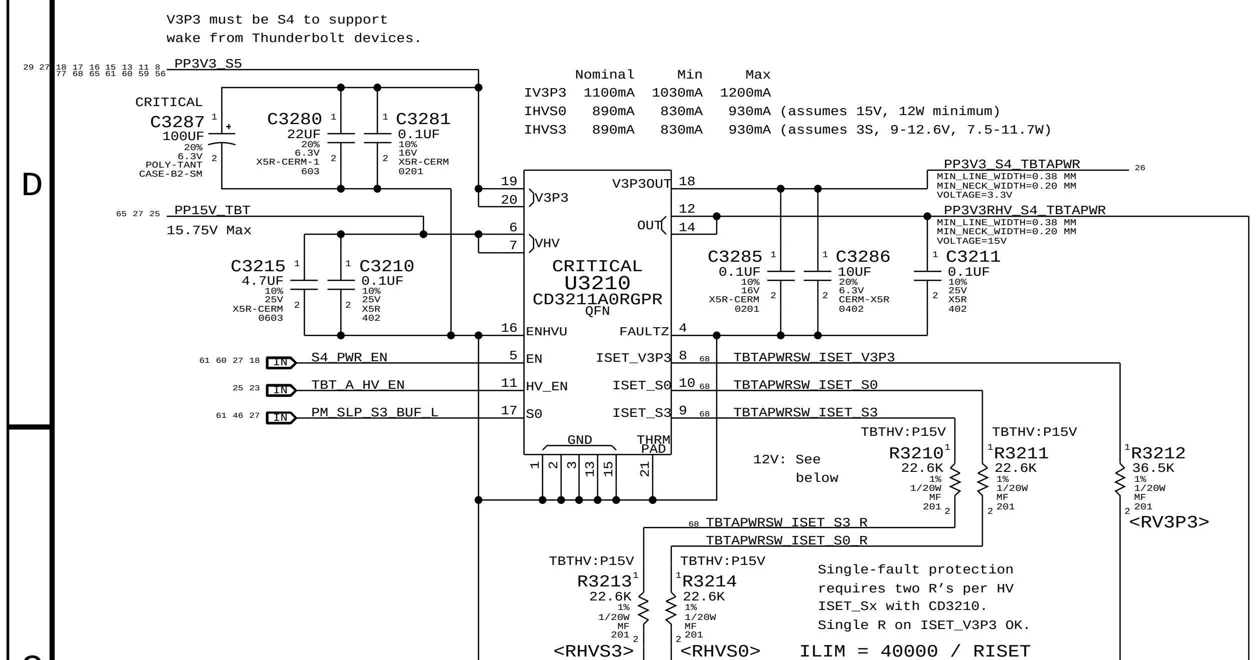
3.3V/HV Power MUX

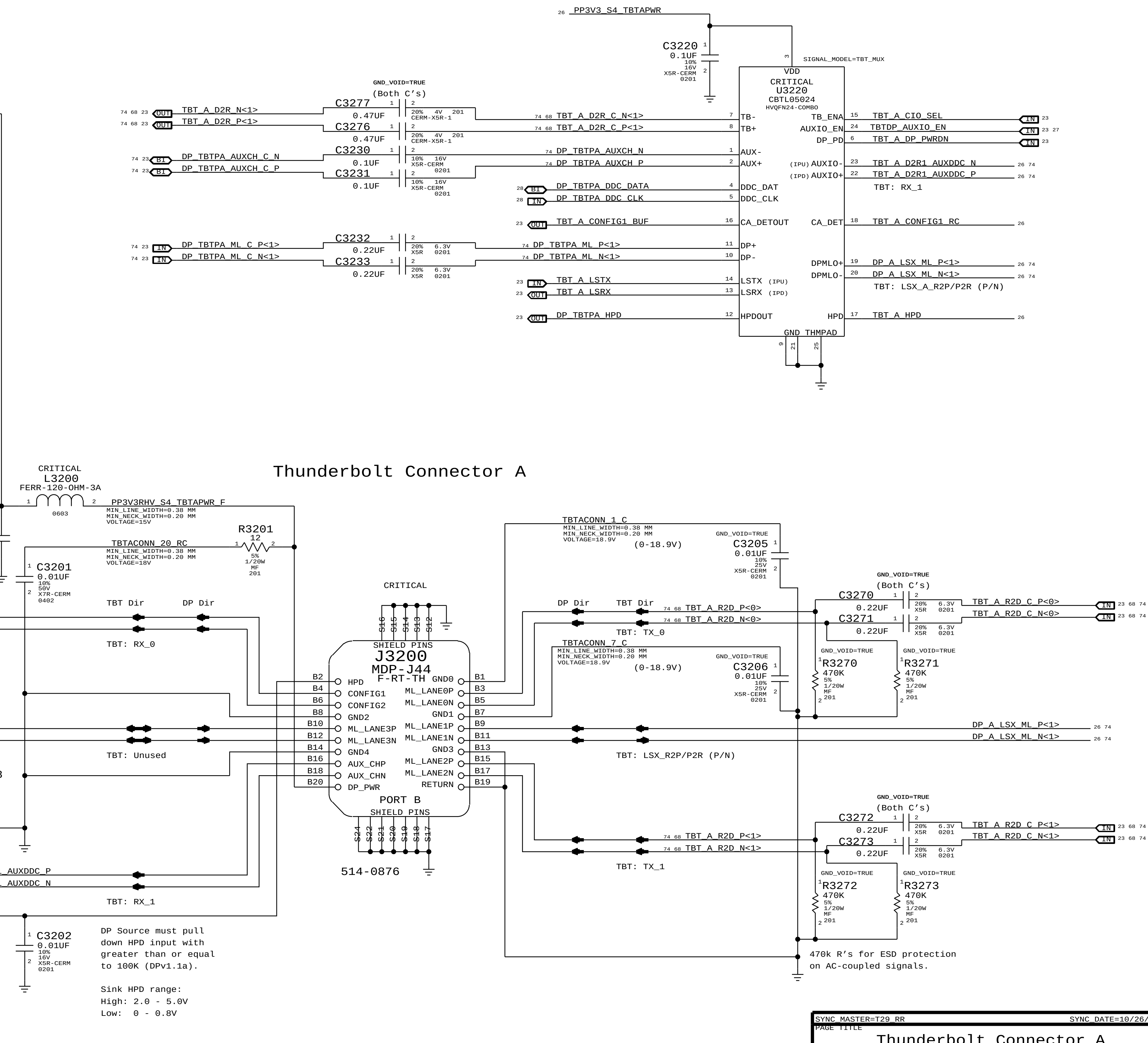


For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3210,R3213		TBTHV:P12V
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3211,R3214		TBTHV:P12V

Nominal Min Max
IHVS0/S3 1120mA 1090mA 1170mA (12W minimum)

Thunderbolt Connector A



SYNC_MASTER=T29_RR SYNC_DATE=10/26/2012

PAGE TITLE: Thunderbolt Connector A

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

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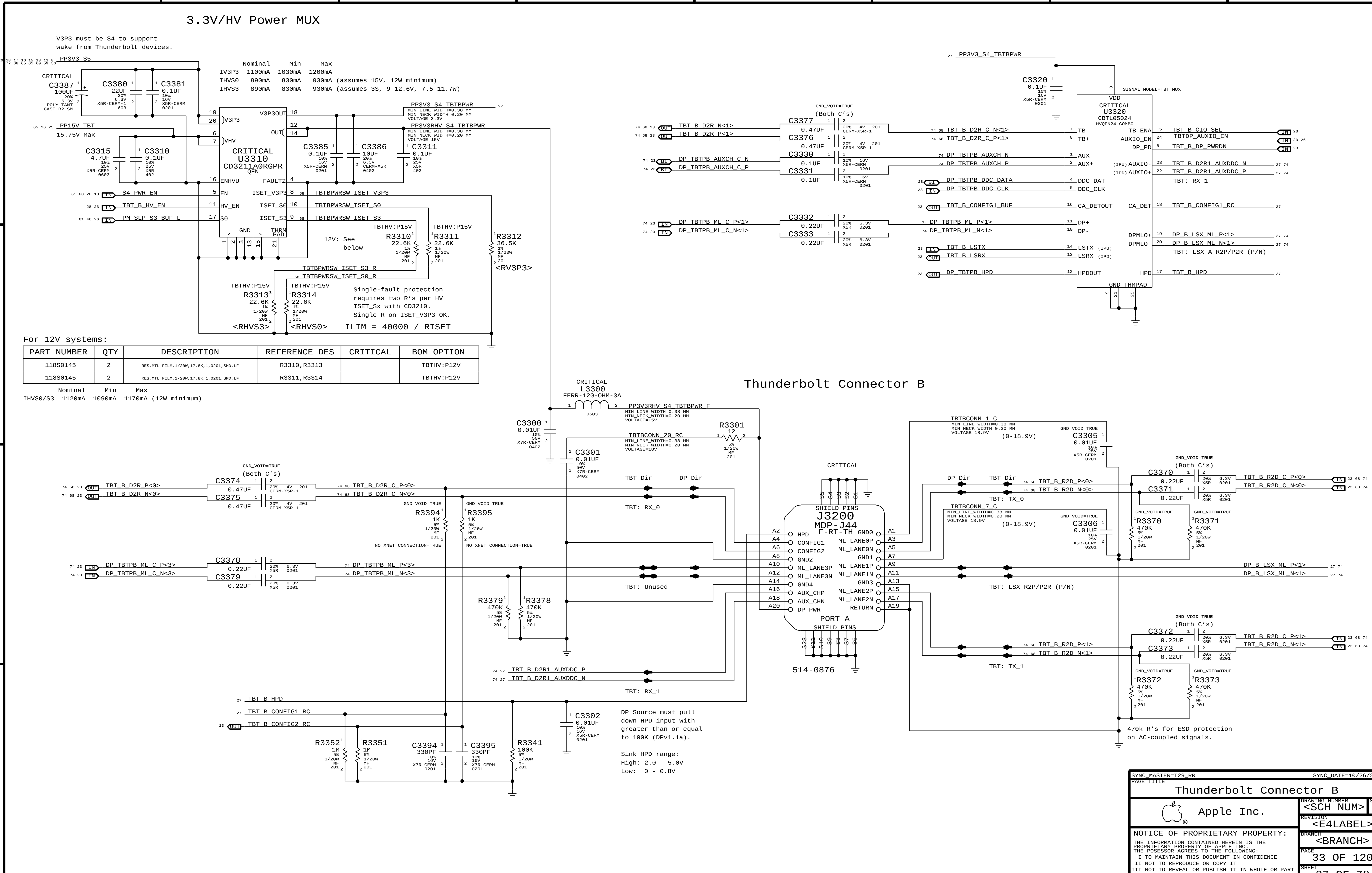
3.3V/HV Power MUX



For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3310,R3313		TBTHV:P12V
118S0145	2	RES,MTL FILM,1/20W,17.8K,1,0201,SMD,LF	R3311,R3314		TBTHV:P12V

	Nominal	Min	Max
IV3P3	1100mA	1030mA	1200mA
IHVS0	890mA	830mA	930mA (assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA (assumes 3S, 9-12.6V, 7.5-11.7W)

	Nominal	Min	Max
IHVS0/S3	1120mA	1090mA	1170mA (12W minimum)

Thunderbolt Connector B

DP Source must pull
down HPD input with
greater than or equal
to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

SYNC_MASTER=T29_RR SYNC_DATE=10/26/2012

PAGE TITLE: Thunderbolt Connector B

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

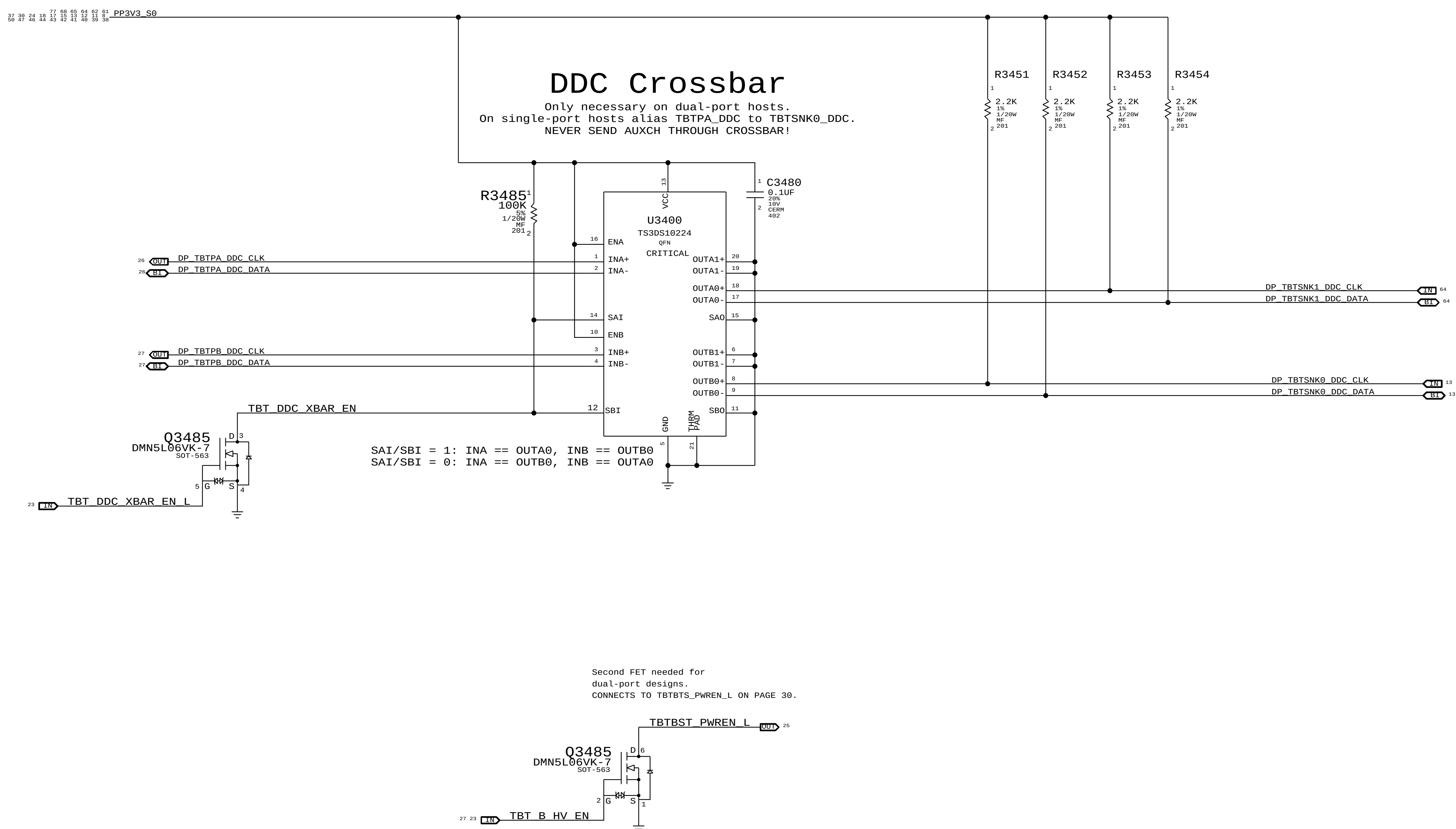
BRANCH: <BRANCH>

DDC Pull-Ups

2.2k pull-ups are required by PCH
to indicate active display interface.

DP++ spec violation, should remove!

NOTE: Only DDC_DATA is sensed, so DDC_CLK
pull-ups are unstuffed.



SYNC_MASTER=J14 SYNC_DATE=10/23/2012

PAGE TITLE

DDC Crossbar

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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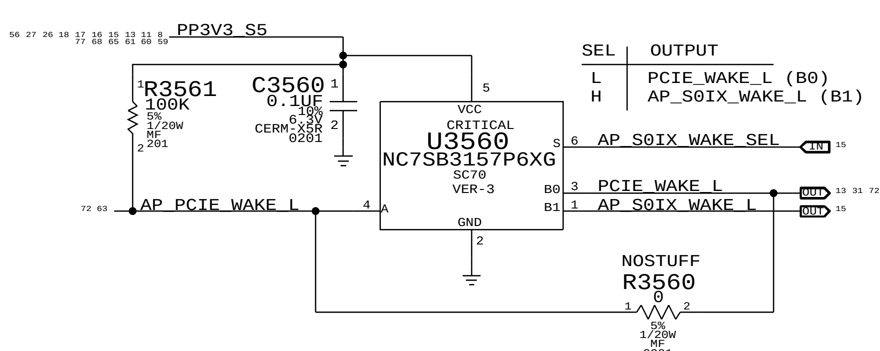
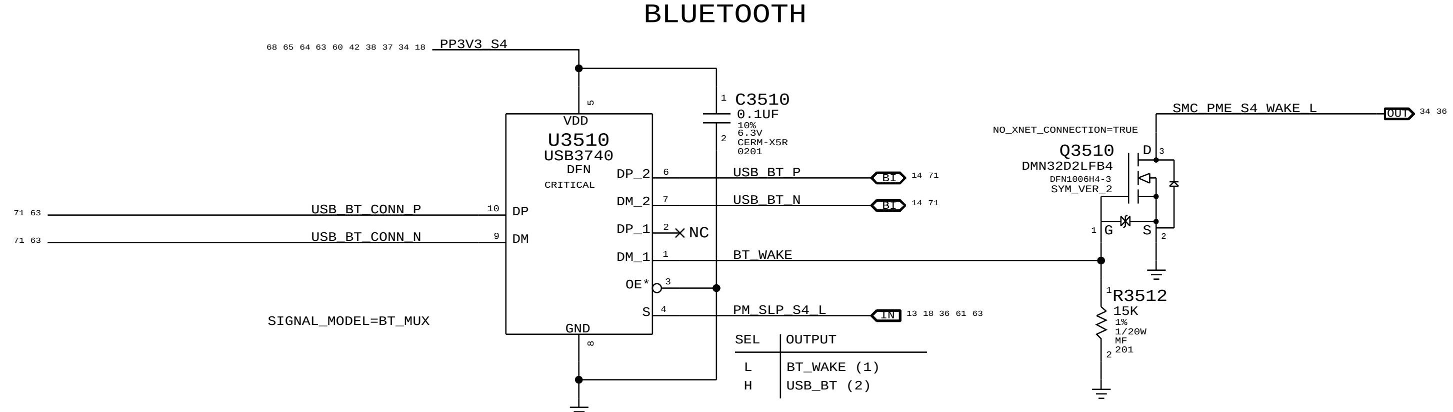
PCIe Wake Muxing



BLUETOOTH



SYNC_MASTER=J41 SYNC_DATE=11/01/2012

PAGE TITLE

WIRELESS SUPPORT

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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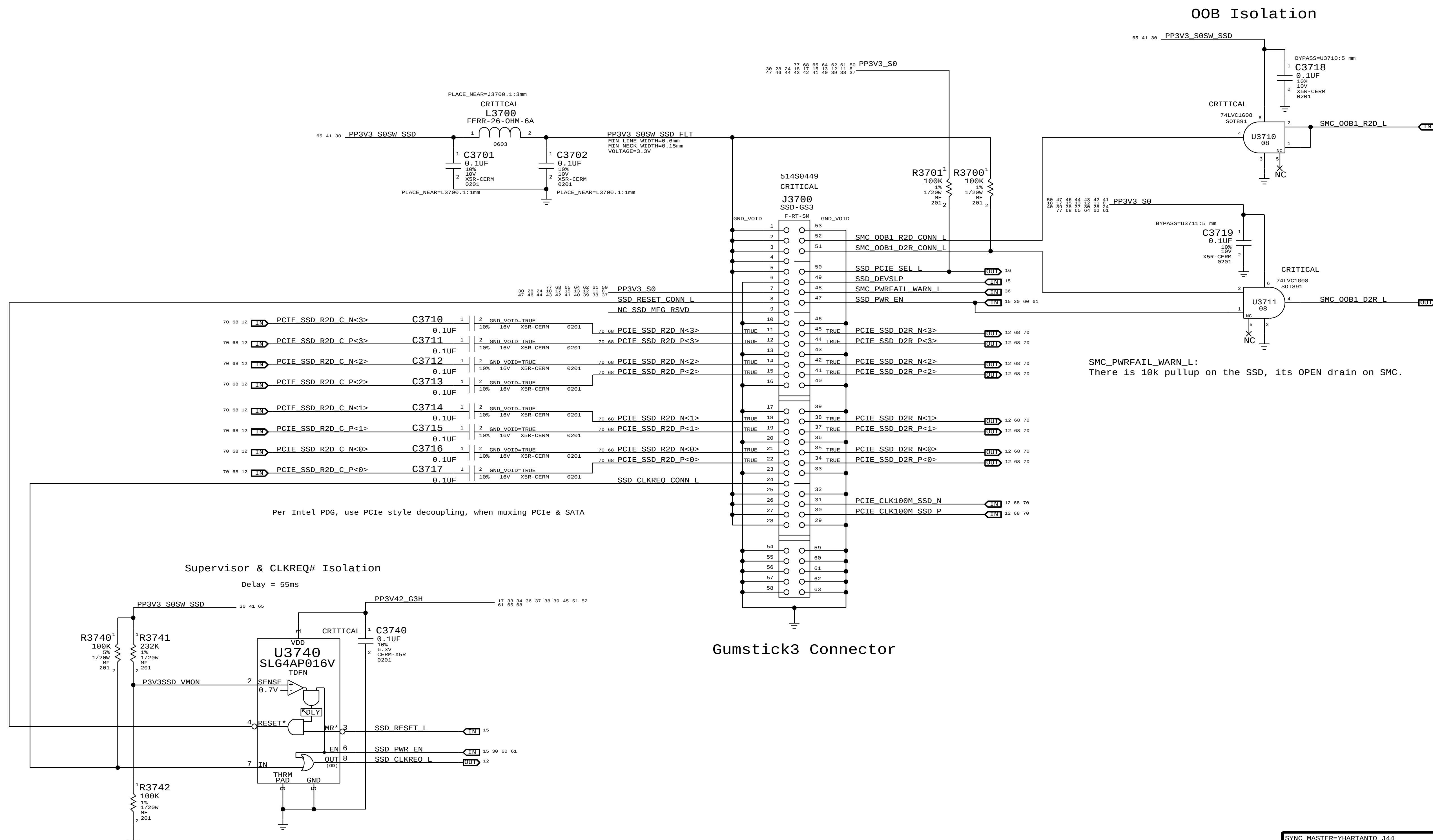

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=12/18/2012

PAGE TITLE

SSD Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

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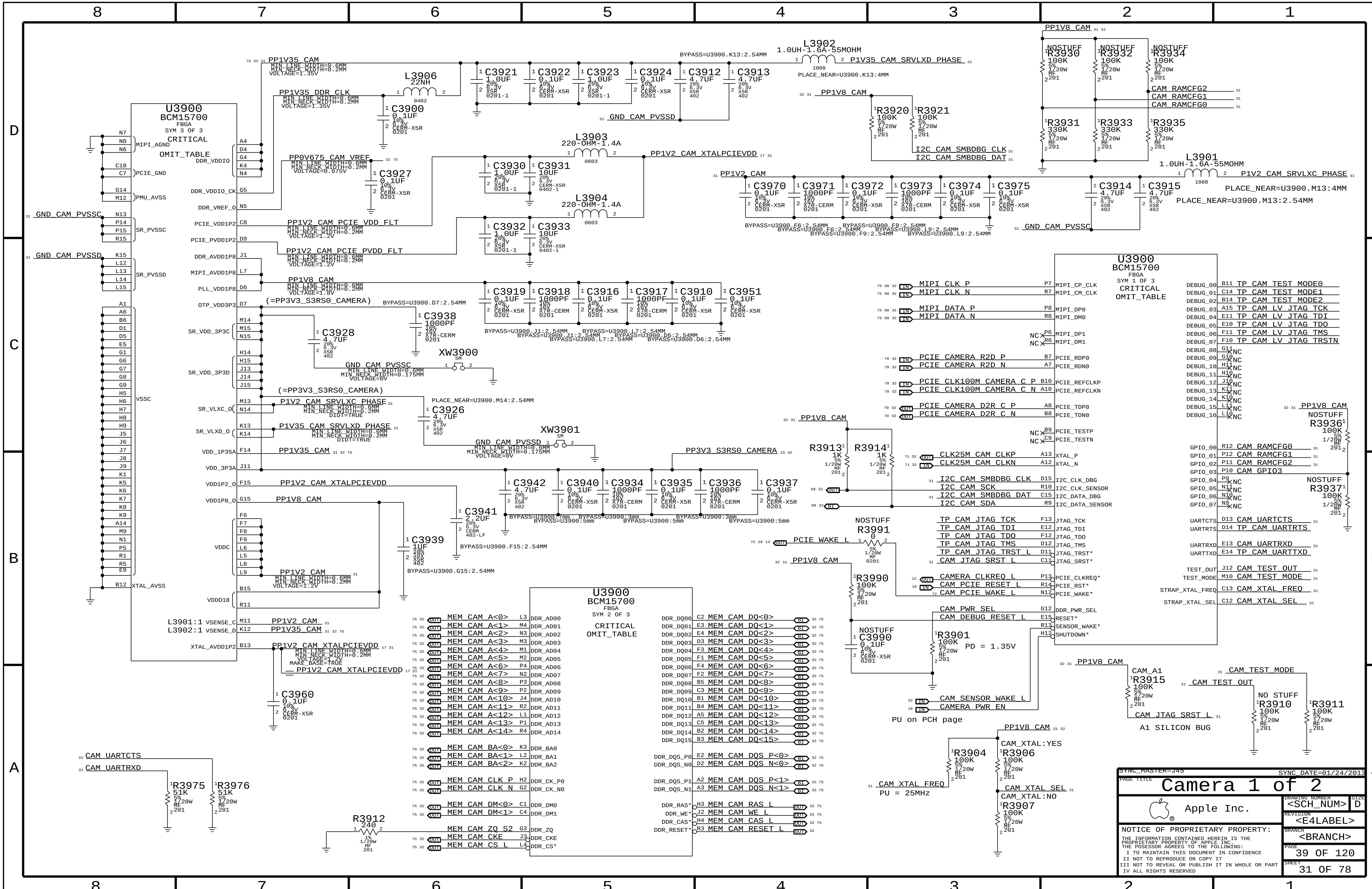

SYNC_MASTER=J45

SYNC_DATE=01/24/2013

PAGE TITLE

Camera 1 of 2

Apple Inc.

DRAWING NUMBER
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SIZE
D

REVISION
<E4LABEL>

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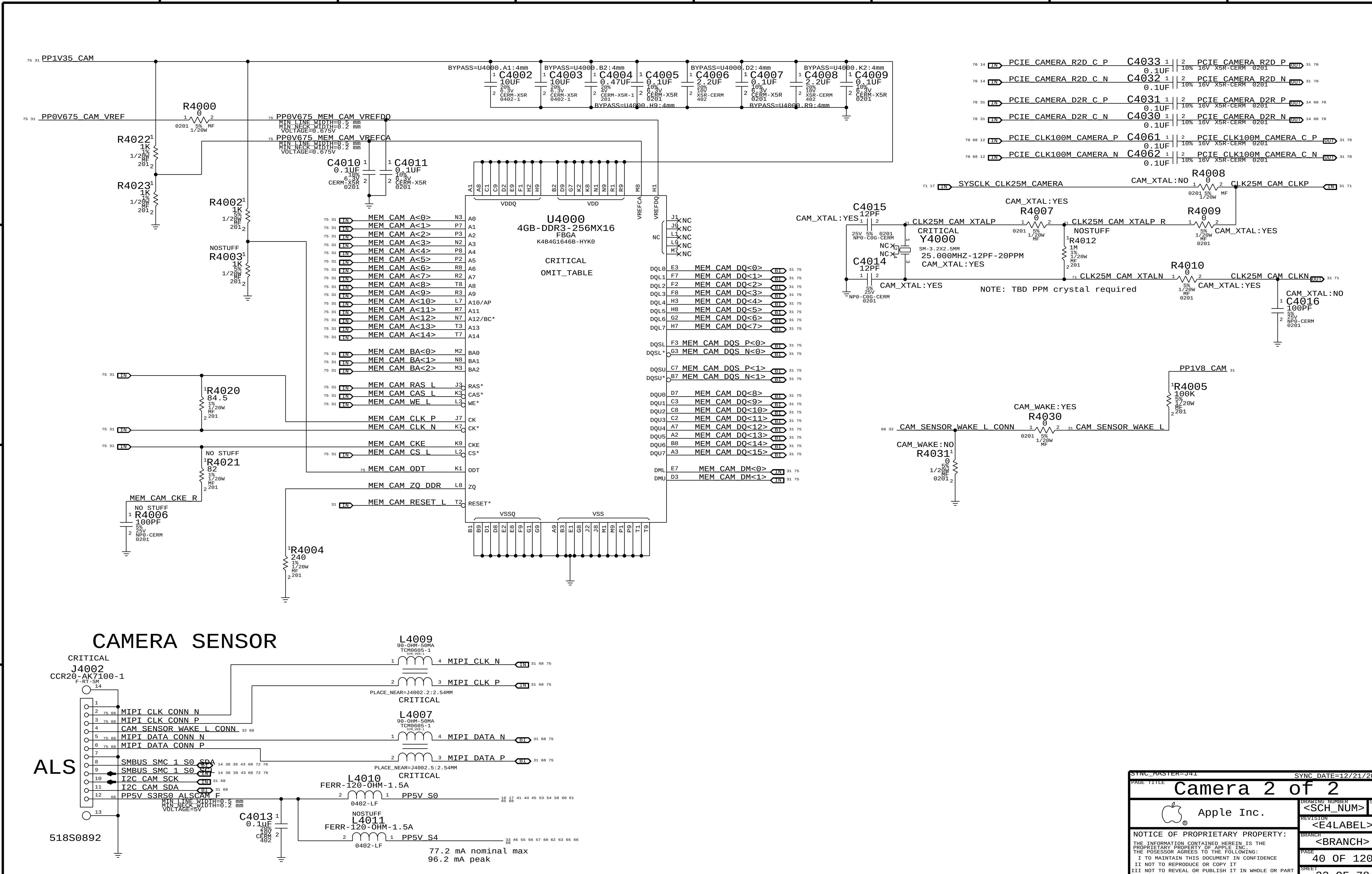

SYNC_MASTER=J41 SYNC_DATE=12/21/2012

PAGE TITLE

Camera 2 of 2

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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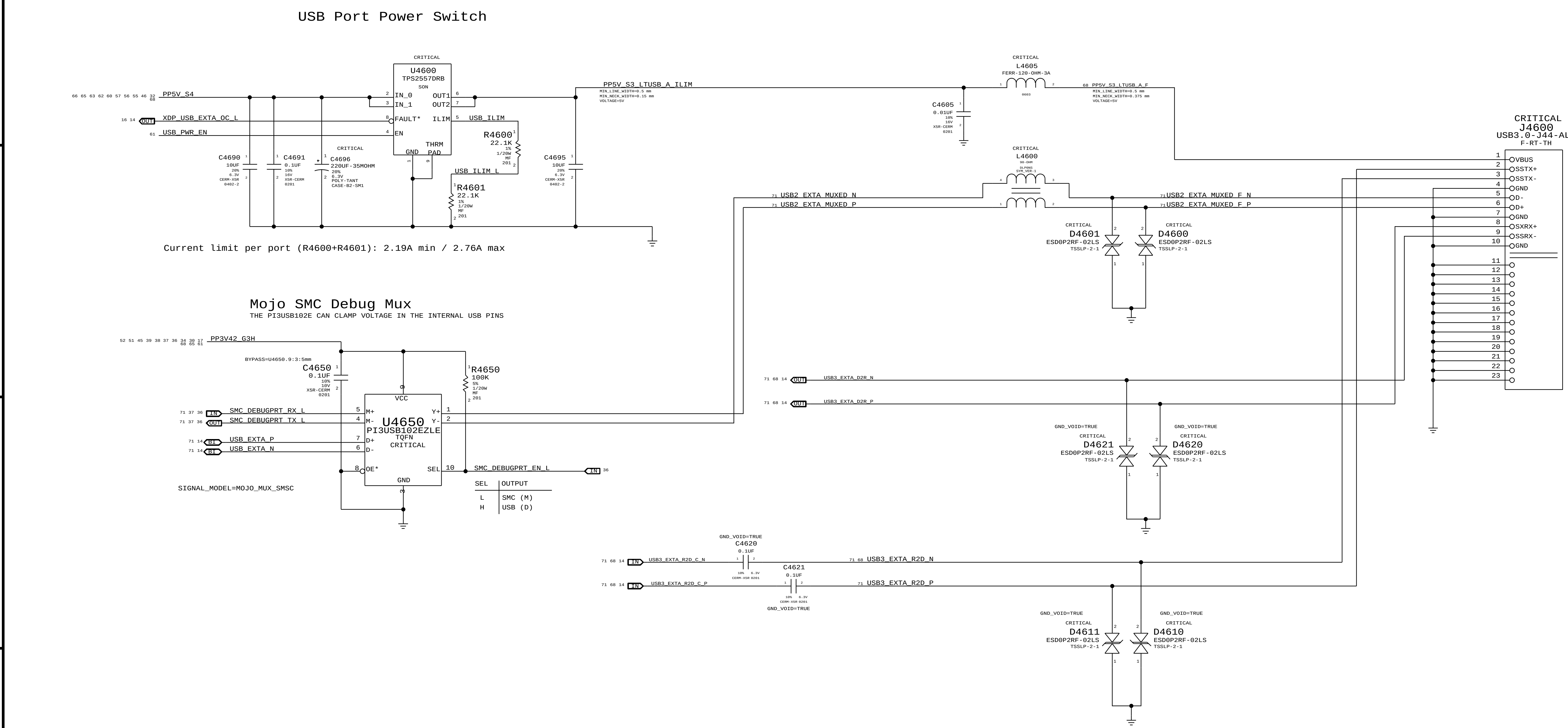
RIGHT USB PORT A



SYNC_MASTER=J41 SYNC_DATE=10/23/2012

PAGE TITLE

External A USB3 Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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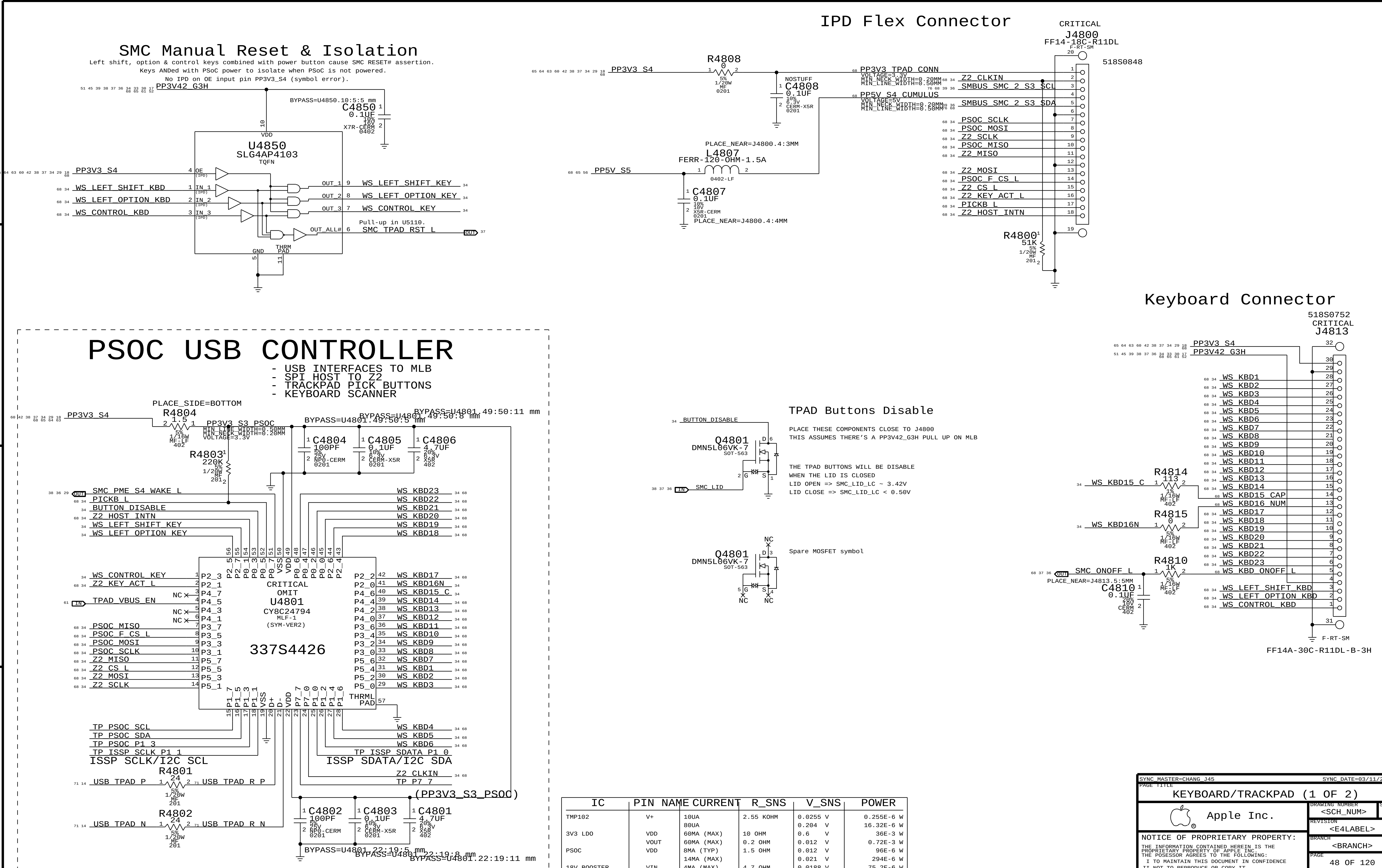


IC	PIN NAME	CURRENT	R_SNS	V_SNS	POWER
TMP102	V+	10UA	2.55 KOHM	0.0255 V	0.255E-6 W
		80UA		0.204 V	16.32E-6 W
3V3 LDO	VDD	60MA (MAX)	10 OHM	0.6 V	36E-3 W
	VOUT	60MA (MAX)	0.2 OHM	0.012 V	0.72E-3 W
PSOC	VDD	8MA (TYP)	1.5 OHM	0.012 V	96E-6 W
		14MA (MAX)		0.021 V	294E-6 W
18V BOOSTER	VIN	4MA (MAX)	4.7 OHM	0.0188 V	75.2E-6 W

SYNC_MASTER=CHANG_J45 SYNC_DATE=03/11/2013

PAGE TITLE

KEYBOARD/TRACKPAD (1 OF 2)

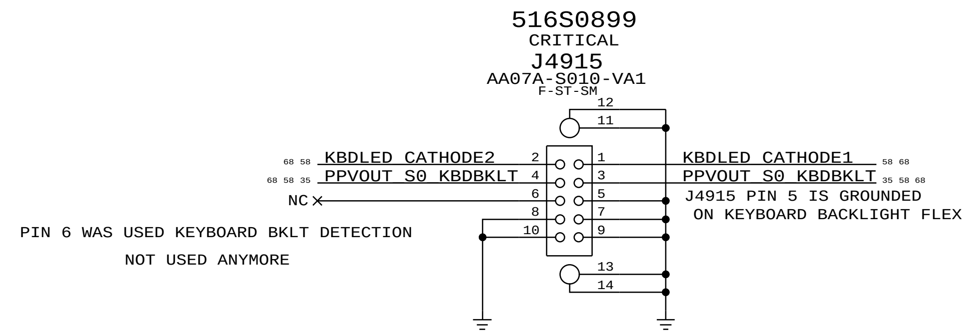
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

Keyboard Backlight Connector



SYNC_MASTER=LDUNN_J44 SYNC_DATE=03/06/2013

PAGE TITLE
KEYBOARD/TRACKPAD (2 OF 2)

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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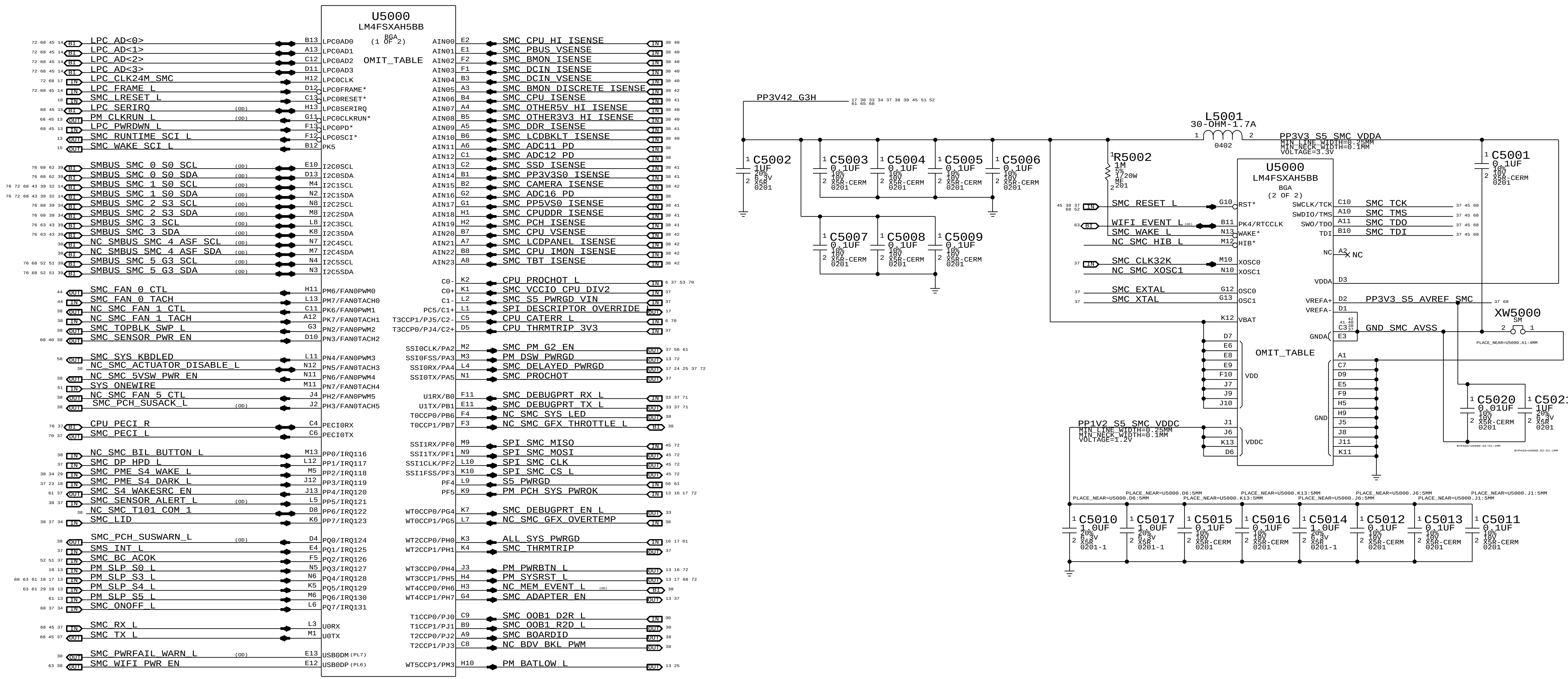

NOTE:
SMS INTERRUPT IS NOT USED, PULL UP TO SMC RAIL.

NOTE:
Unused pins have "SMC_Pxx" names. Unused
pins designed as outputs can be left floating,
those designated as inputs require pull-ups.

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=11/29/2012

PAGE TITLE

SMC

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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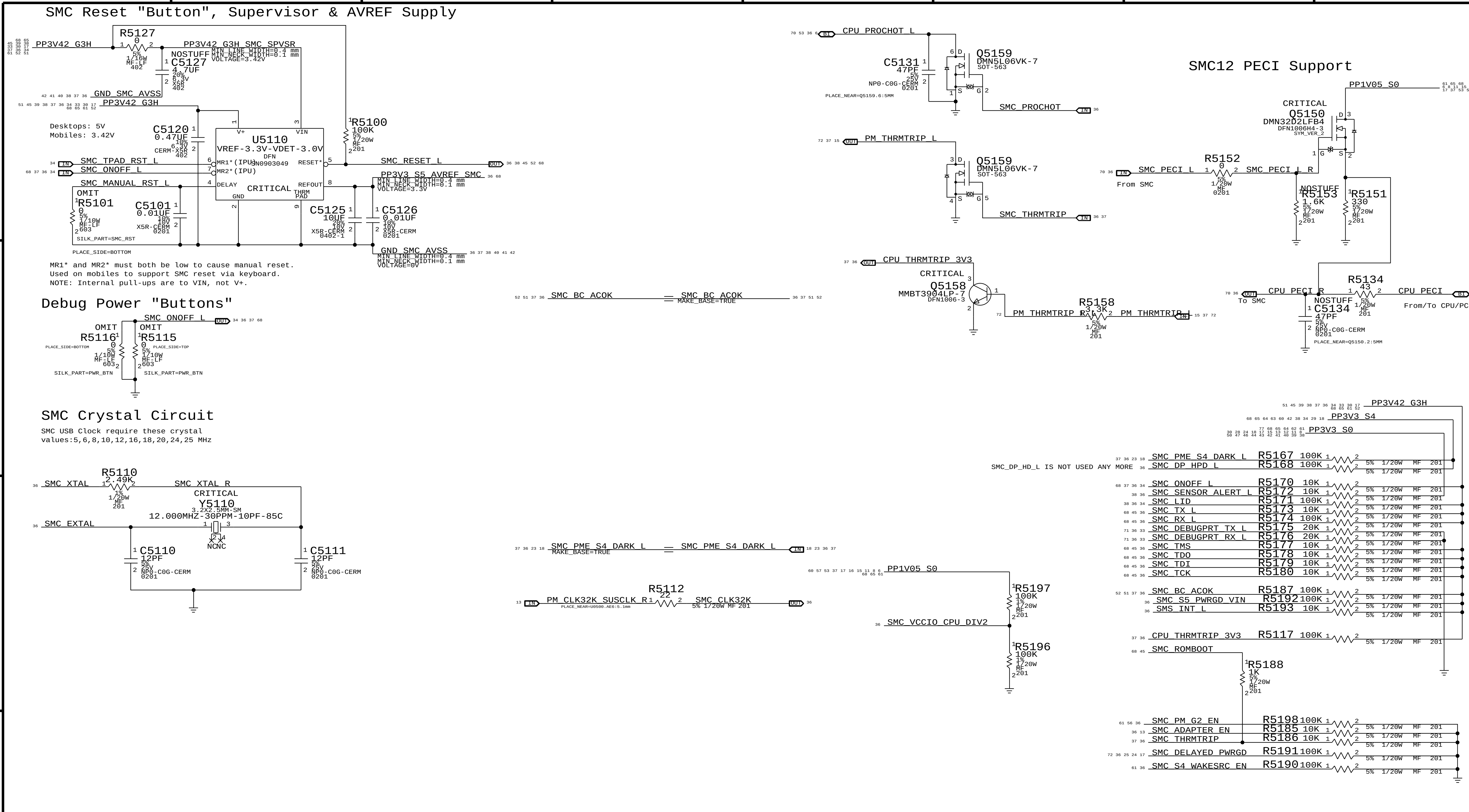

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=12/18/2012

PAGE TITLE

SMC Shared Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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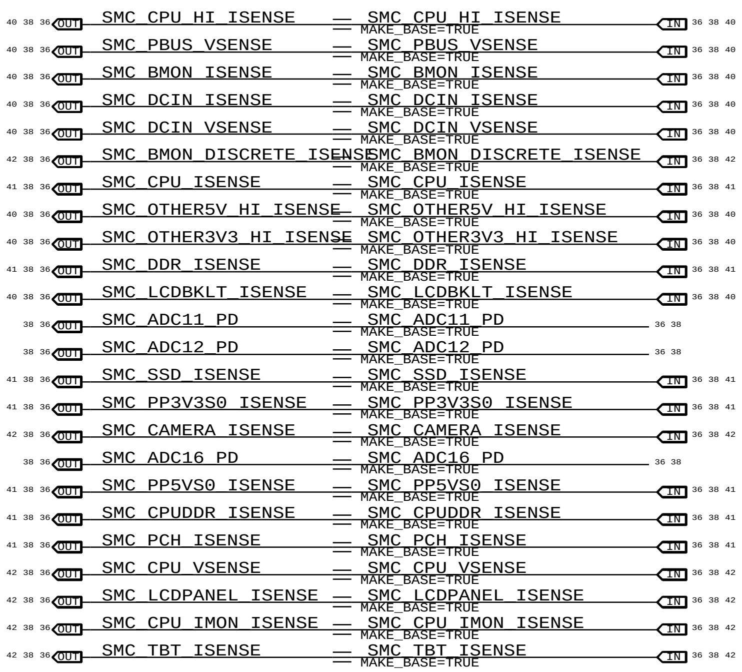
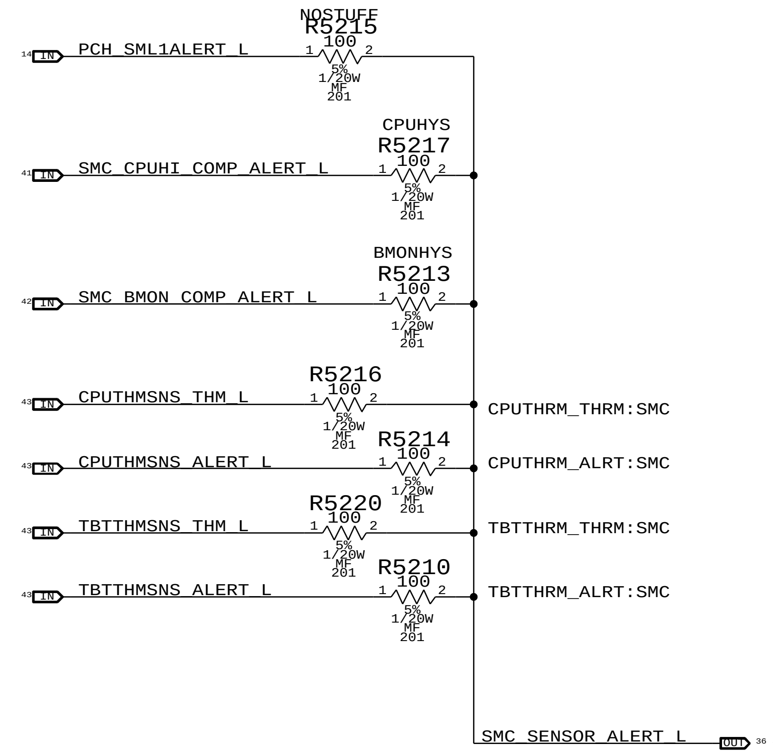
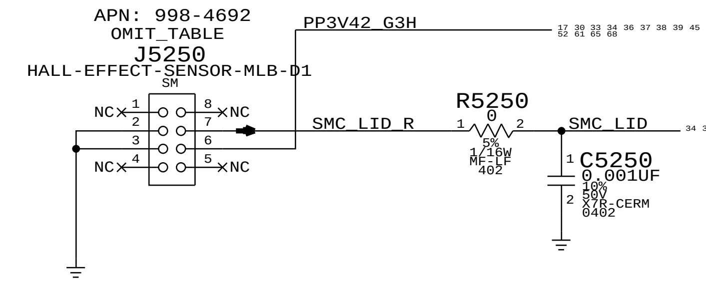
SMC12 ADC Assignments



Thermal Alerts



Hall Effect Pads



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
677-0912	1	SUBASSY,PCBA HALL EFFECT,J44	J5250	CRITICAL	

639-4502 (J44 HALL EFFECT BOARD) REPORTS TO 677-0912

Specify one of these BOM GROUPs.

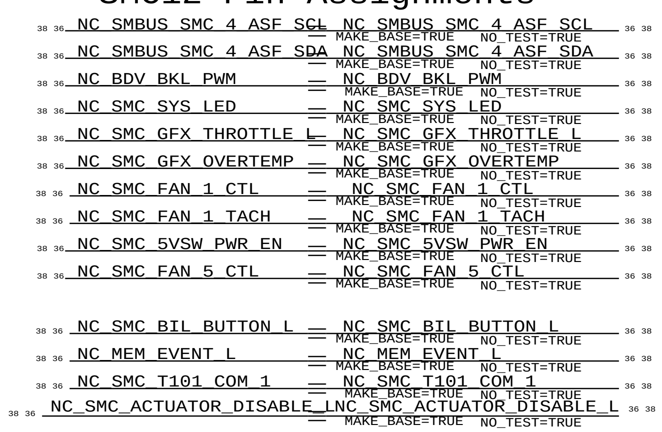
BOM GROUP	BOM OPTIONS
CPUTHRM:BOTH	CPUTHRM_THRM:SMC,CPUTHRM_ALRT:SMC
CPUTHRM:THRM	CPUTHRM_THRM:SMC,CPUTHRM_ALRT:PU
CPUTHRM:ALRT	CPUTHRM_THRM:PU,CPUTHRM_ALRT:SMC
CPUTHRM:NONE	CPUTHRM_THRM:PU,CPUTHRM_ALRT:PU

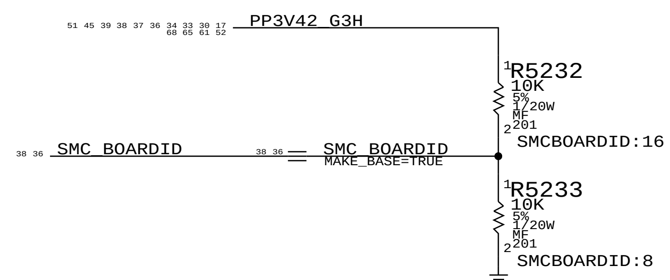
Specify one of these BOM GROUPs.

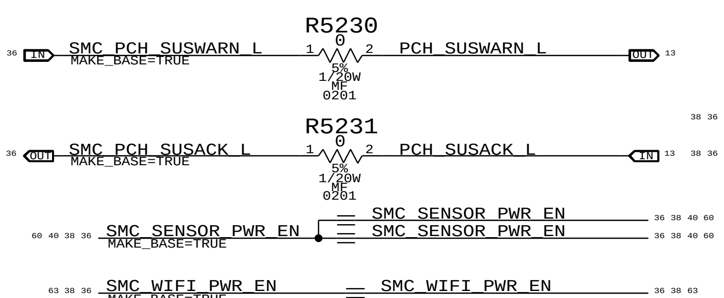
BOM GROUP	BOM OPTIONS
TBTTHRM:BOTH	TBTTHRM_THRM:SMC,TBTTHRM_ALRT:SMC
TBTTHRM:THRM	TBTTHRM_THRM:SMC,TBTTHRM_ALRT:PU
TBTTHRM:ALRT	TBTTHRM_THRM:PU,TBTTHRM_ALRT:SMC
TBTTHRM:NONE	TBTTHRM_THRM:PU,TBTTHRM_ALRT:PU
TBTTHRM:GONE	

Requires EMC1412-1 or EMC1412-2 instead of EMC1412-A, new APN needs to be created.

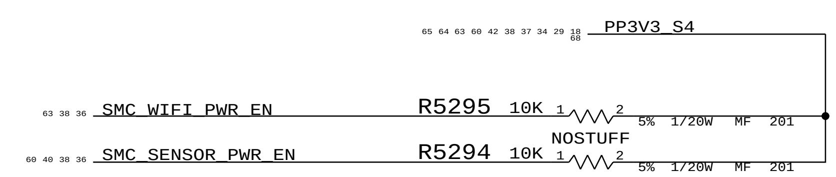
SMC12 Pin Assignments



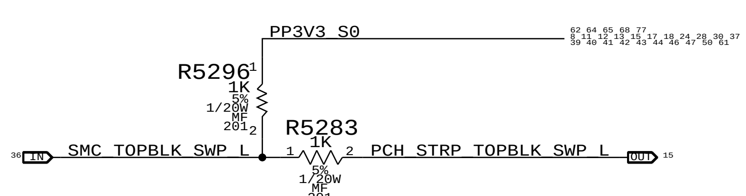


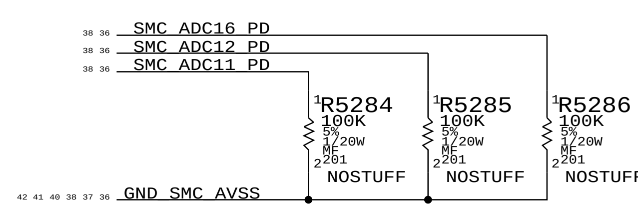


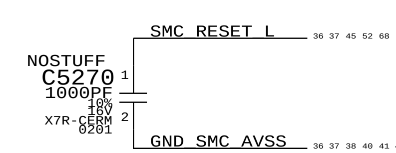
S4 SMC Wake Sources



Top Block Swap







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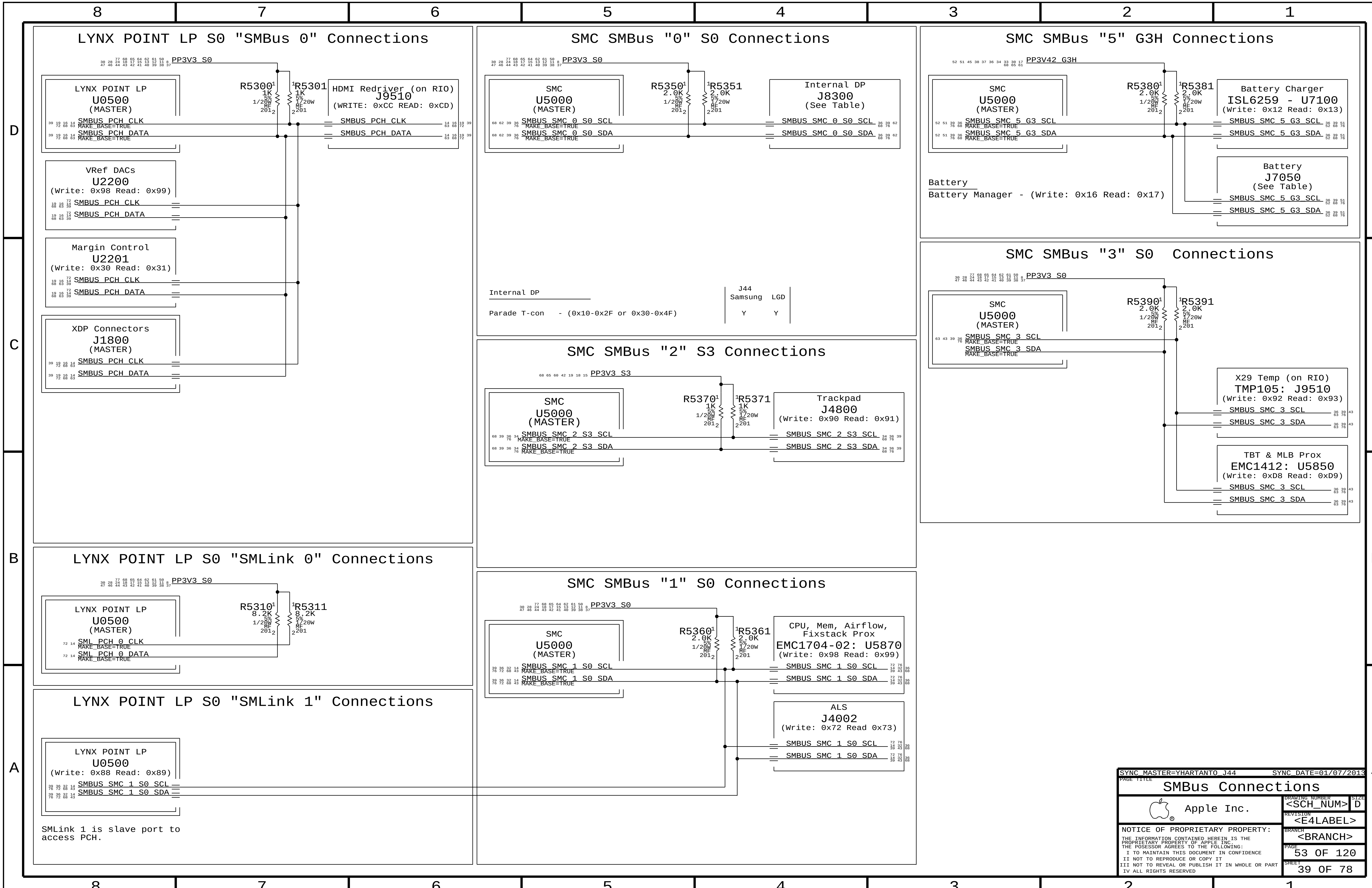
SMC Project Support

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>


LYNX POINT LP S0 "SMBus 0" Connections
PP3V3_S0
LYNX POINT LP
U0500
(MASTER)
SMBUS_PCH_CLK
MAKE_BASE=TRUE
SMBUS_PCH_DATA
MAKE_BASE=TRUE
R5300
1K
5%
1/20W
MF
201
R5301
1K
5%
1/20W
MF
201
HDMI Redriver (on RIO)
J9510
(WRITE: 0xCC READ: 0xCD)
SMBUS_PCH_CLK
SMBUS_PCH_DATA
VRef DACs
U2200
(Write: 0x98 Read: 0x99)
SMBUS_PCH_CLK
SMBUS_PCH_DATA
Margin Control
U2201
(Write: 0x30 Read: 0x31)
SMBUS_PCH_CLK
SMBUS_PCH_DATA
XDP Connectors
J1800
(MASTER)
SMBUS_PCH_CLK
SMBUS_PCH_DATA
SMC SMBus "0" S0 Connections
PP3V3_S0
SMC
U5000
(MASTER)
SMBUS_SMC_0_S0_SCL
MAKE_BASE=TRUE
SMBUS_SMC_0_S0_SDA
MAKE_BASE=TRUE
R5350
2.0K
5%
1/20W
MF
201
R5351
2.0K
5%
1/20W
MF
201
Internal DP
J8300
(See Table)
SMBUS_SMC_0_S0_SCL
SMBUS_SMC_0_S0_SDA
Internal DP
Parade T-con - (0x10-0x2F or 0x30-0x4F)
J44
Samsung
LGD
Y
Y
SMC SMBus "5" G3H Connections
PP3V42_G3H
SMC
U5000
(MASTER)
SMBUS_SMC_5_G3_SCL
MAKE_BASE=TRUE
SMBUS_SMC_5_G3_SDA
MAKE_BASE=TRUE
R5380
2.0K
5%
1/20W
MF
201
R5381
2.0K
5%
1/20W
MF
201
Battery Charger
ISL6259 - U7100
(Write: 0x12 Read: 0x13)
SMBUS_SMC_5_G3_SCL
SMBUS_SMC_5_G3_SDA
Battery
J7050
(See Table)
SMBUS_SMC_5_G3_SCL
SMBUS_SMC_5_G3_SDA
Battery
Battery Manager - (Write: 0x16 Read: 0x17)
SMC SMBus "3" S0 Connections
PP3V3_S0
SMC
U5000
(MASTER)
SMBUS_SMC_3_SCL
MAKE_BASE=TRUE
SMBUS_SMC_3_SDA
MAKE_BASE=TRUE
R5390
2.0K
5%
1/20W
MF
201
R5391
2.0K
5%
1/20W
MF
201
X29 Temp (on RIO)
TMP105: J9510
(Write: 0x92 Read: 0x93)
SMBUS_SMC_3_SCL
SMBUS_SMC_3_SDA
TBT & MLB Prox
EMC1412: U5850
(Write: 0xD8 Read: 0xD9)
SMBUS_SMC_3_SCL
SMBUS_SMC_3_SDA
SMC SMBus "2" S3 Connections
PP3V3_S3
SMC
U5000
(MASTER)
SMBUS_SMC_2_S3_SCL
MAKE_BASE=TRUE
SMBUS_SMC_2_S3_SDA
MAKE_BASE=TRUE
R5370
1K
5%
1/20W
MF
201
R5371
1K
5%
1/20W
MF
201
Trackpad
J4800
(Write: 0x90 Read: 0x91)
SMBUS_SMC_2_S3_SCL
SMBUS_SMC_2_S3_SDA
LYNX POINT LP S0 "SMLink 0" Connections
PP3V3_S0
LYNX POINT LP
U0500
(MASTER)
SML_PCH_0_CLK
MAKE_BASE=TRUE
SML_PCH_0_DATA
MAKE_BASE=TRUE
R5310
8.2K
5%
1/20W
MF
201
R5311
8.2K
5%
1/20W
MF
201
SMC SMBus "1" S0 Connections
PP3V3_S0
SMC
U5000
(MASTER)
SMBUS_SMC_1_S0_SCL
MAKE_BASE=TRUE
SMBUS_SMC_1_S0_SDA
MAKE_BASE=TRUE
R5360
2.0K
5%
1/20W
MF
201
R5361
2.0K
5%
1/20W
MF
201
CPU, Mem, Airflow,
Fixstack Prox
EMC1704-02: U5870
(Write: 0x98 Read: 0x99)
SMBUS_SMC_1_S0_SCL
SMBUS_SMC_1_S0_SDA
ALS
J4002
(Write: 0x72 Read 0x73)
SMBUS_SMC_1_S0_SCL
SMBUS_SMC_1_S0_SDA
LYNX POINT LP S0 "SMLink 1" Connections
LYNX POINT LP
U0500
(Write: 0x88 Read: 0x89)
SMBUS_SMC_1_S0_SCL
SMBUS_SMC_1_S0_SDA
SMLink 1 is slave port to
access PCH.
SYNC_MASTER=YHARTANTO_J44
SYNC_DATE=01/07/2013
PAGE TITLE
SMBus Connections
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
D
REVISION
<E4LABEL>
BRANCH
<BRANCH>
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53 OF 120
SHEET
39 OF 78
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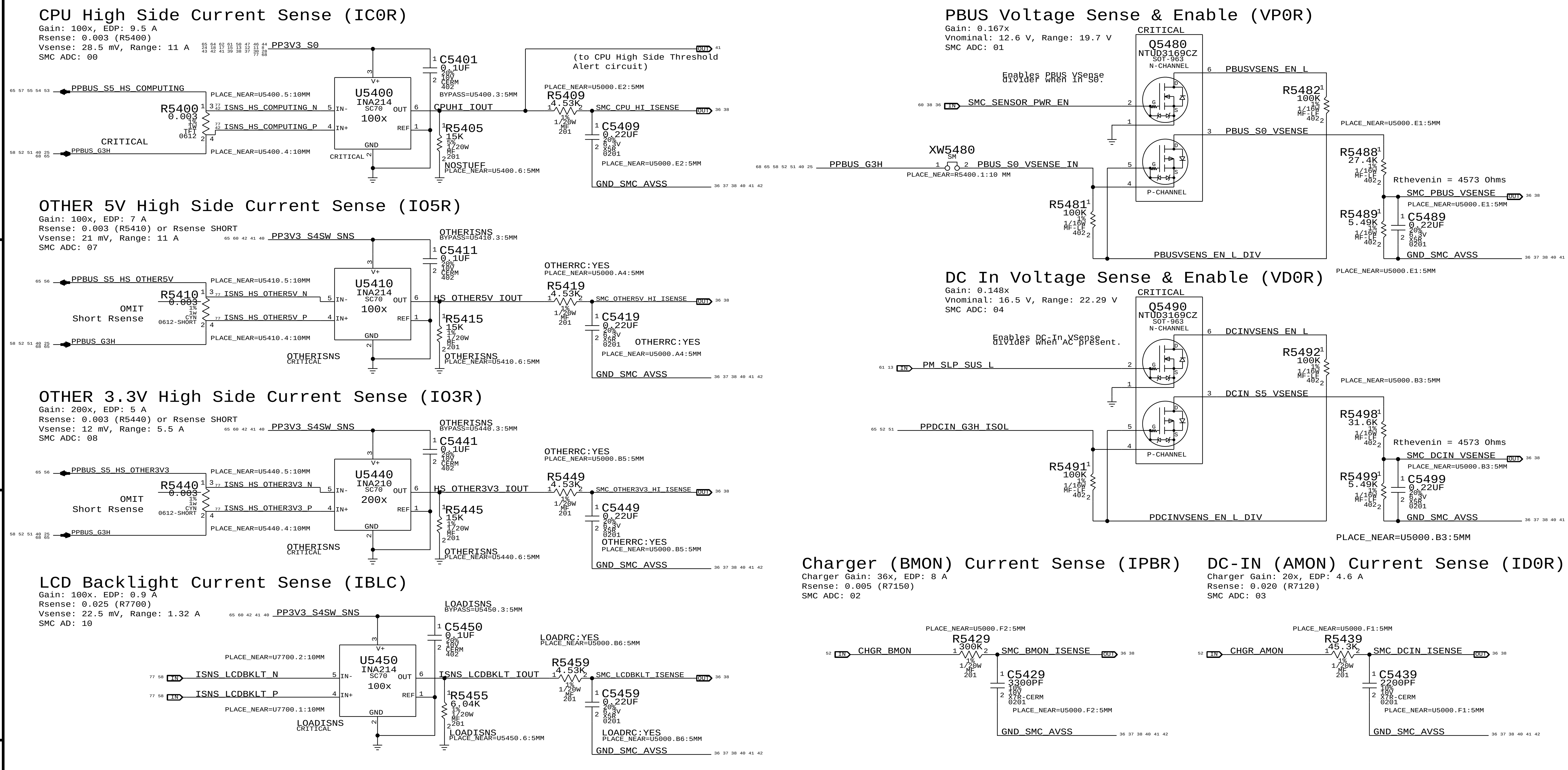


PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5419,C5449		OTHERRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5459		LOADRC:NO

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=03/25/2013

PAGE TITLE
Power Sensors: High Side

Apple Inc.

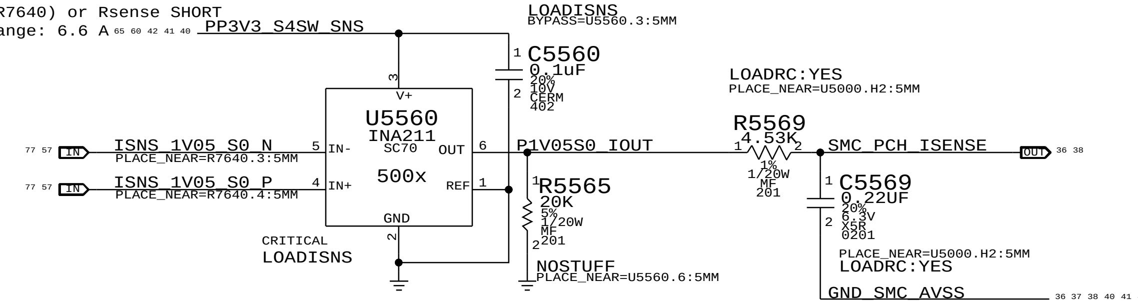
DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

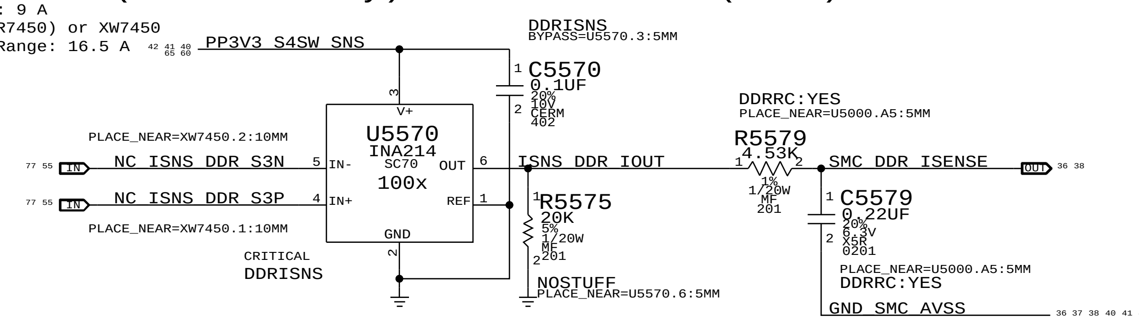
PCH 1.05V Current Sense (IC1C)

Gain: 500x, EDP: 5 A
Rsense: 0.001 (R7640) or Rsense SHORT
Vsense: 5 mV, Range: 6.6 A
SMC ADC: 19



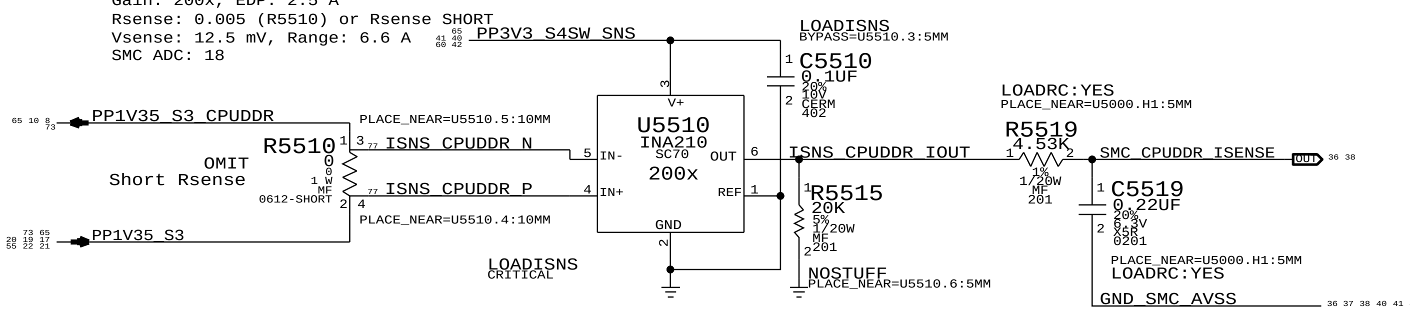
DDR 1.35V S3 (CPU & Memory) Current Sense (IM0C)

Gain: 100x, EDP: 9 A
Rsense: 0.002 (R7450) or XW7450
Vsense: 21 mV, Range: 16.5 A
SMC ADC: 09



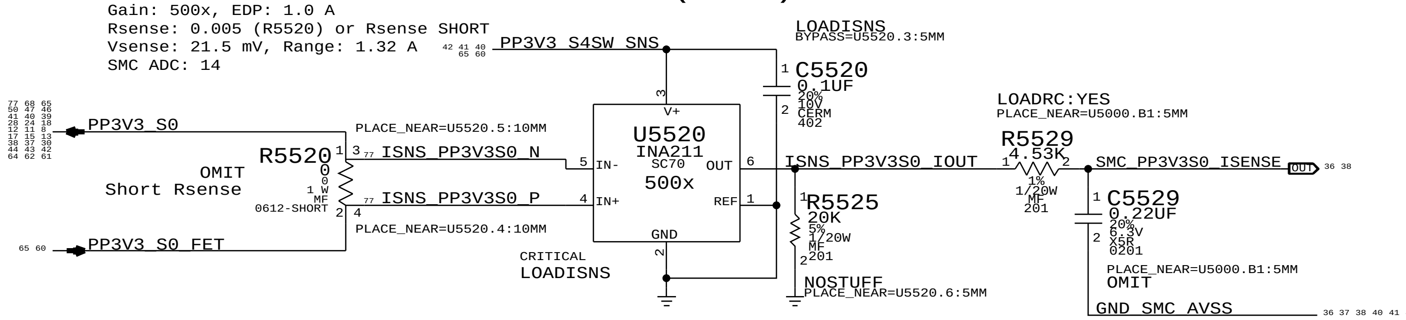
CPU DDR 1.35V S3 (CPU Only) Current Sense (IM1C)

Gain: 200x, EDP: 2.5 A
Rsense: 0.005 (R5510) or Rsense SHORT
Vsense: 12.5 mV, Range: 6.6 A
SMC ADC: 18



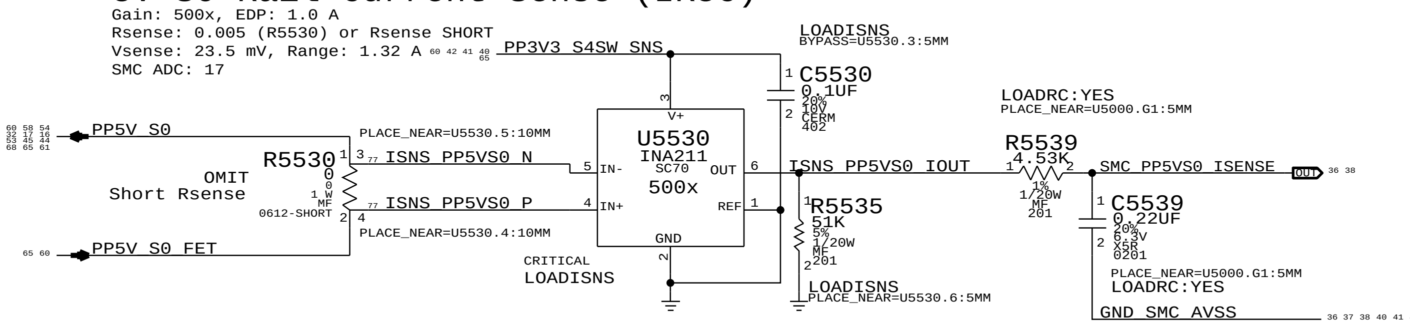
3.3V S0 Rail Current Sense (IR3C)

Gain: 500x, EDP: 1.0 A
Rsense: 0.005 (R5520) or Rsense SHORT
Vsense: 21.5 mV, Range: 1.32 A
SMC ADC: 14



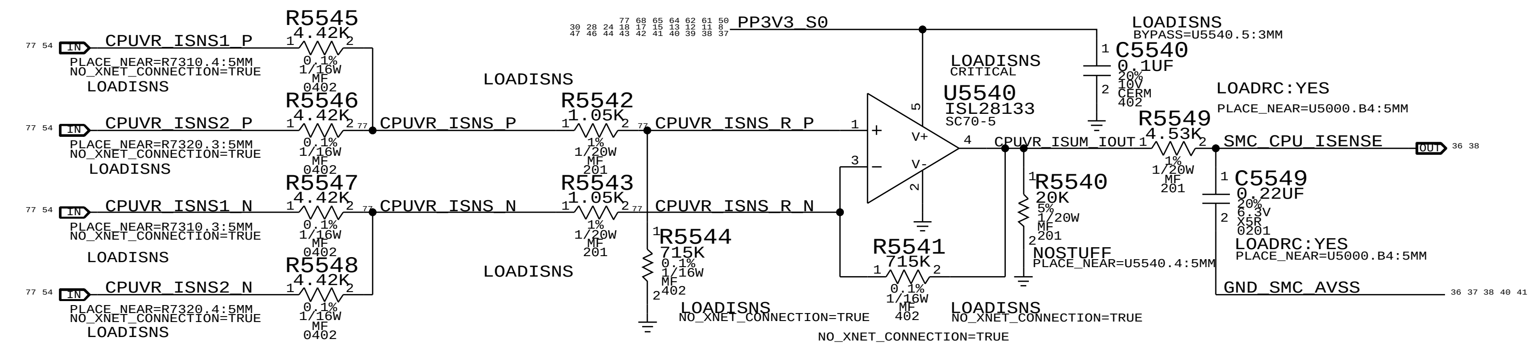
5V S0 Rail Current Sense (IR5C)

Gain: 500x, EDP: 1.0 A
Rsense: 0.005 (R5530) or Rsense SHORT
Vsense: 23.5 mV, Range: 1.32 A
SMC ADC: 17



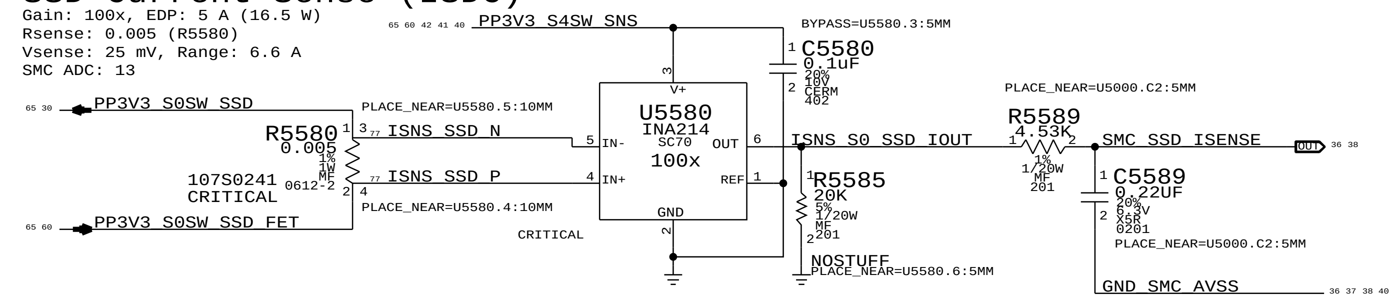
CPU Fixed Current Sense (IC0C)

Gain: 219.33x, EDP: 40 A
Rsense: 2x of 0.00075 (R7310, R7320), Rsum: 0.000375
Vsense: 15 mV, Range: 40.12 A
SMC ADC: 06



SSD Current Sense (ISDC)

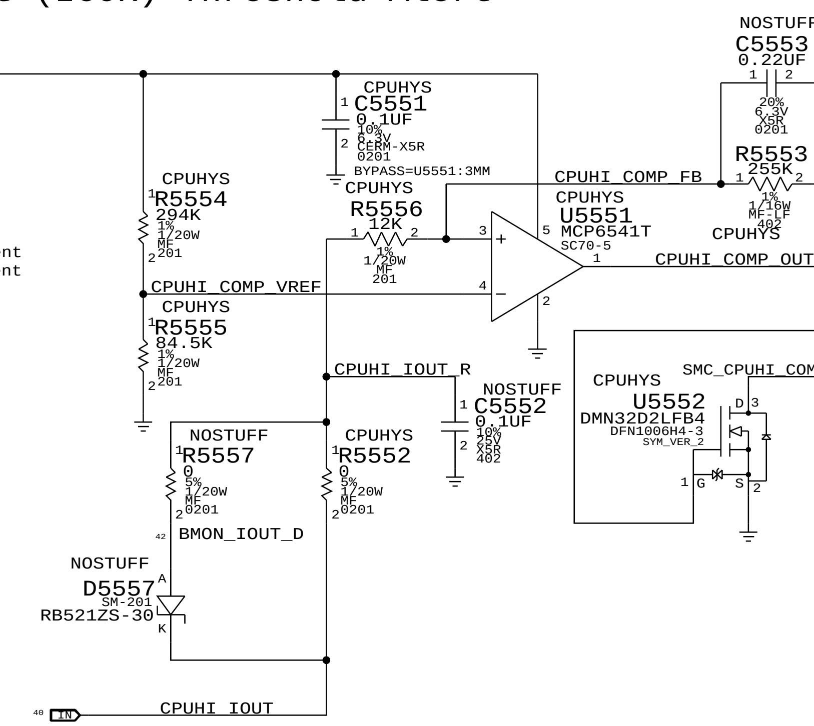
Gain: 100x, EDP: 5 A (16.5 W)
Rsense: 0.005 (R5580)
Vsense: 25 mV, Range: 6.6 A
SMC ADC: 13



CPU High Side Current (IC0R) Threshold Alert

Gain: 100x
Rsense: 0.003 (R5400)

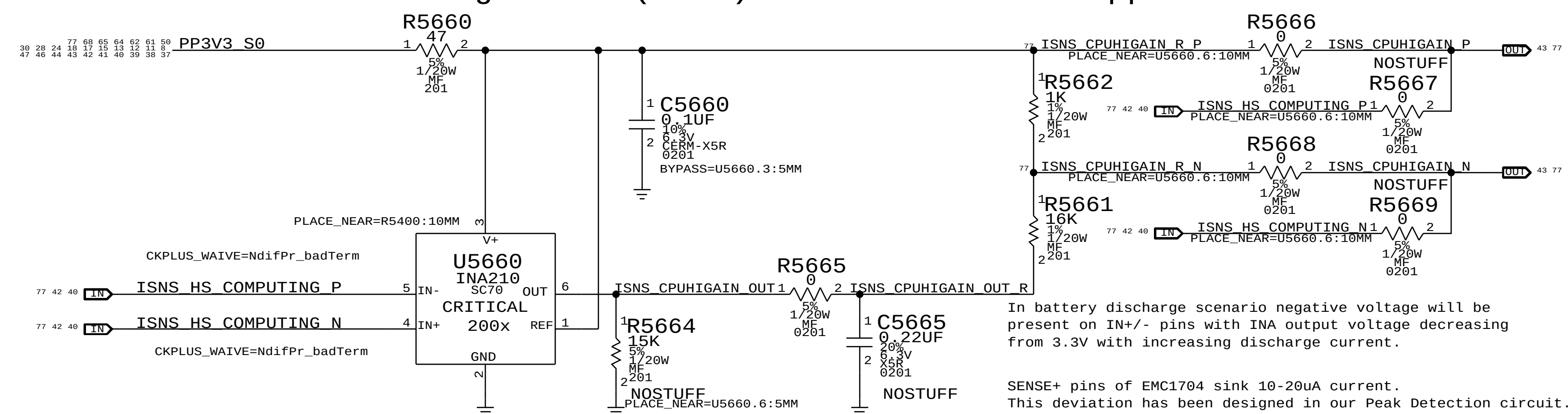
Trip Target on CPU High current: 2.5 A
Hysteresis Circuit:
Vref = 0.737 V
Vth = 0.616 V -> 2.054 A on CPU High current
Vtl = 0.771 V -> 2.571 A on CPU High current
Hysteresis Margin = 0.518 A



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5569,C5519		LOADRC:NO
117S0008	3	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5529,C5539,C5549		LOADRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5579		DDRRC:NO

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=03/27/2013
PAGE TITLE
Power Sensors: Load Side
DRAWING NUMBER <SCH_NUM> SIZE D
Apple Inc.
REVISION <E4LABEL>
BRANCH <BRANCH>

CPU High Side (IC0R) Peak Detection Support

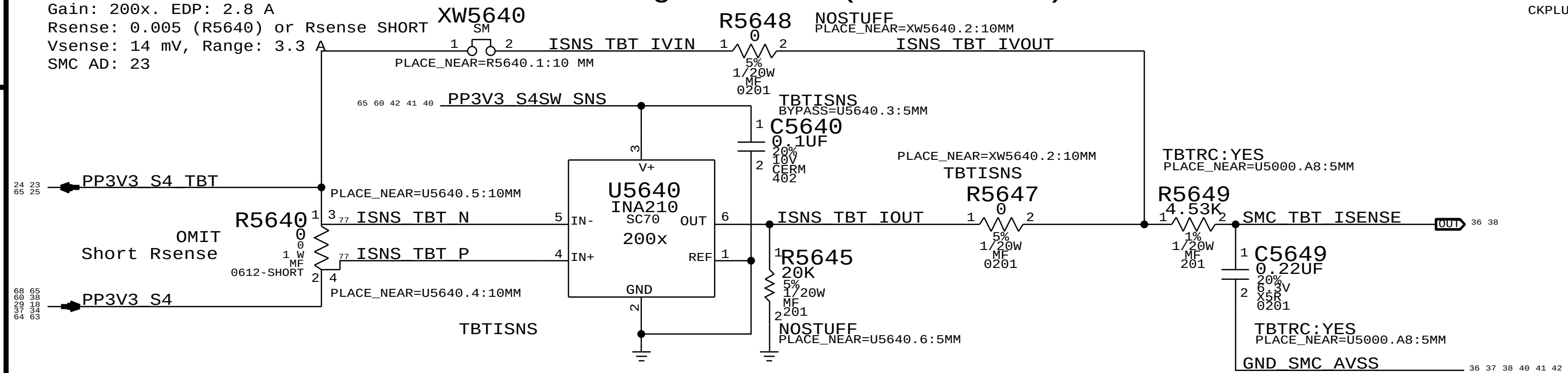


In battery discharge scenario negative voltage will be present on IN+/- pins with INA output voltage decreasing from 3.3V with increasing discharge current.

SENSE+ pins of EMC1704 sink 10-20uA current.
This deviation has been designed in our Peak Detection circuit.
With 10uA sink: 0.125A - 2.1A -> 13mV - 83 mV
With 20uA sink: 0.125A - 2.1A -> 23mV - 92 mV

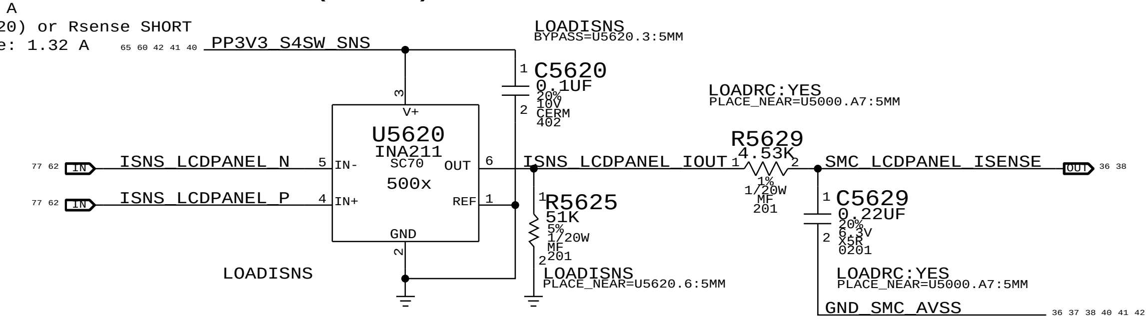
Thunderbolt TBT Current/Voltage Sense (IHSC/VHSC)

Gain: 200x. EDP: 2.8 A
Rsense: 0.005 (R5640) or Rsense SHORT
Vsense: 14 mV, Range: 3.3 A
SMC AD: 23



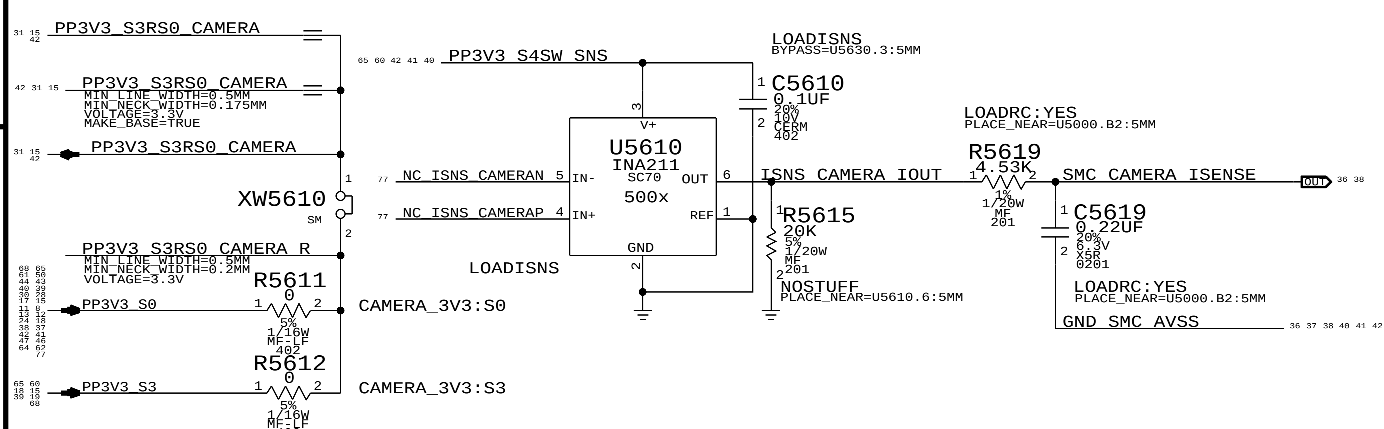
LCD Panel Current Sense (ILDC)

Gain: 500x. EDP: 1 A
RSENSE: 0.005 (R8320) or Rsense SHORT
Vsense: 5 mV, Range: 1.32 A
SMC AD: 21



Camera (S2 Controller) Current Sense (ICMC)

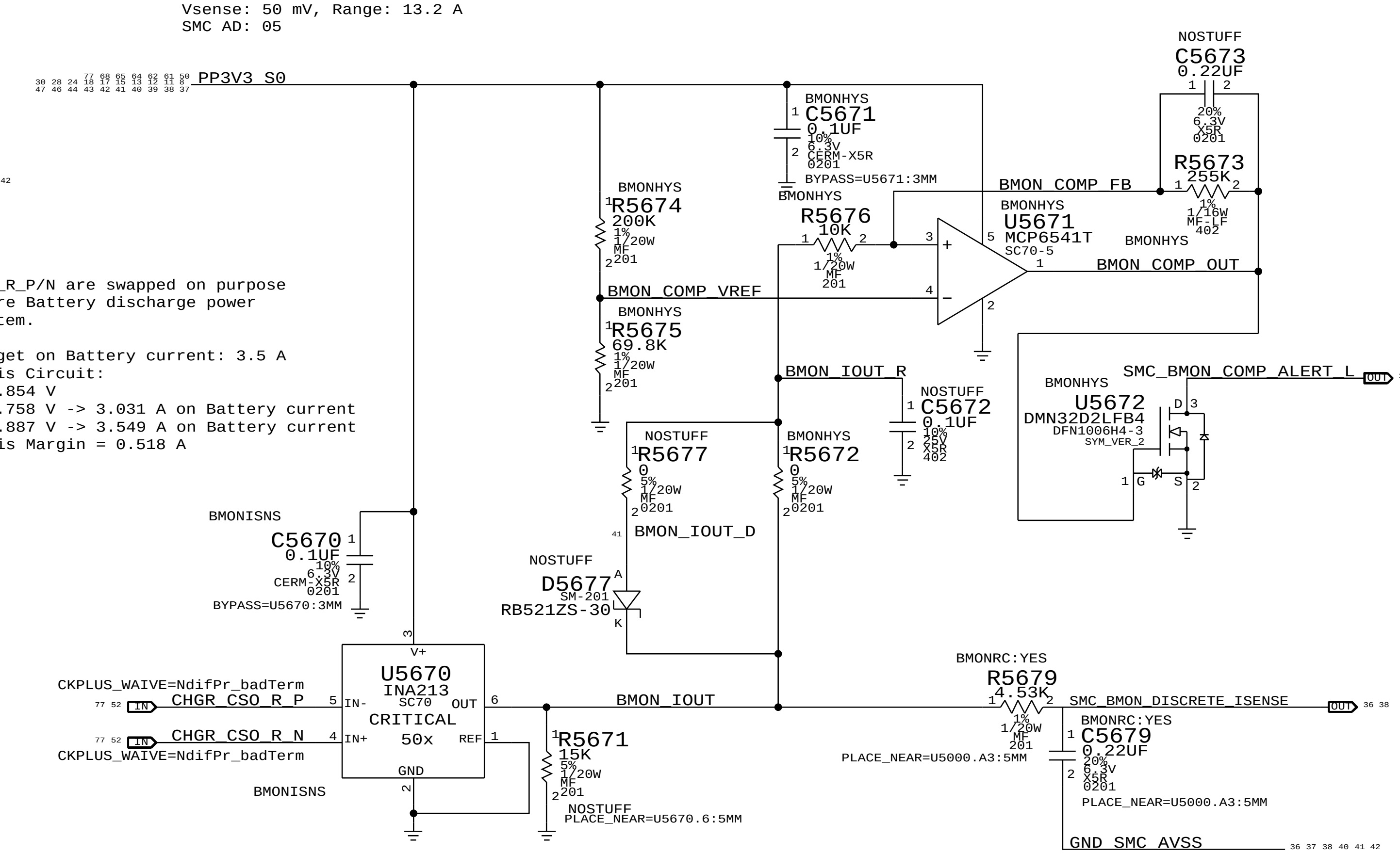
Gain: 500x. EDP: 0.82 A
Rsense: 0.005 (R5610) or XW5610
Vsense: 4.1 mV, Range: 1.32 A
SMC AD: 15



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	4	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5619,C5629,C5649		
117S0008	1	RES,MTL FILM,100K,1/16W,0201,SMD,LF	C5679		

Battery BMON Discrete Current Sense (IP0R) & Threshold Alert

Gain: 50x. EDP: 8 A
Rsense: 0.005 (R7150)
Vsense: 50 mV, Range: 13.2 A
SMC AD: 05

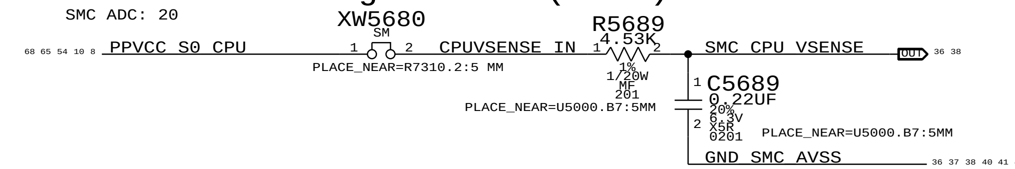


CHGR_CSO_R_P/N are swapped on purpose
to measure Battery discharge power
into system.

Trip Target on Battery current: 3.5 A
Hysteresis Circuit:
Vref = 0.854 V
Vth = 0.758 V -> 3.031 A on Battery current
Vtl = 0.887 V -> 3.549 A on Battery current
Hysteresis Margin = 0.518 A

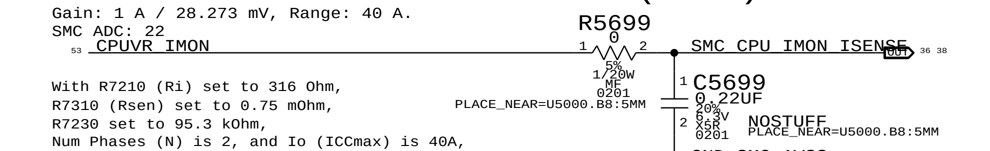
CPU Core Voltage Sense (VC0C)

SMC ADC: 20



CPU Core IMON Current Sense (IC2C)

Gain: 1 A / 28.273 mV, Range: 40 A.
SMC ADC: 22



With R7210 (Ri) set to 316 Ohm,
R7310 (Rsen) set to 0.75 mOhm,
R7230 set to 95.3 kOhm,
Num Phases (N) is 2, and Io (ICCmax) is 40A,
then 1A of Io gives 28.273mV at the Vimon.

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=03/27/2013

PAGE TITLE
Power Sensors: Extended

Apple Inc.

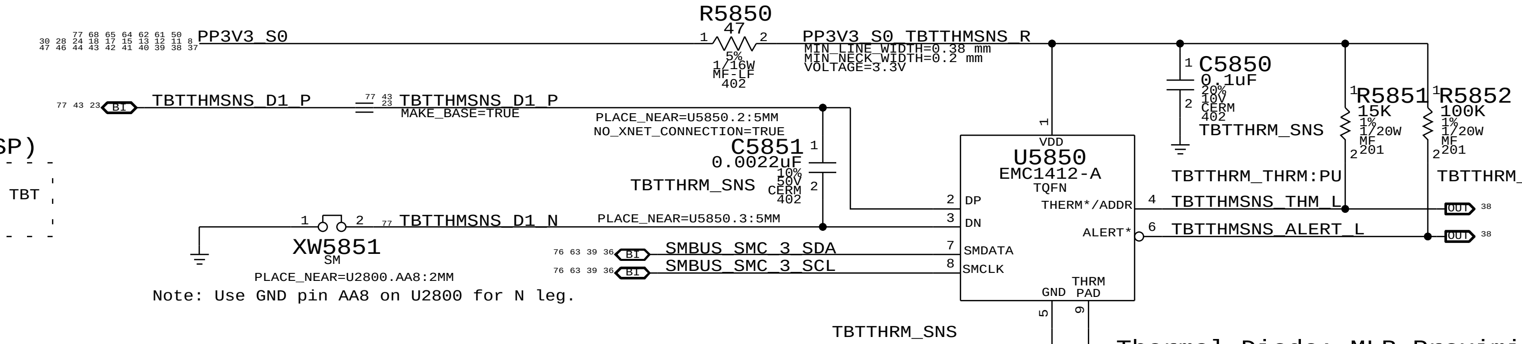
DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>

Thermal Sensor A:
Thunderbolt Die, MLB Proximity

I2C Write: 0xD8, I2C Read: 0xD9

Thermal Diode: TBT Die (THSP)

Placement Note:
The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AA8.



U5850 I2C Address:
By setting R5851 to 15k, I2C address for U5850 is 0xD8/0xD9.

Thermal Diode: MLB Proximity (TMLB)

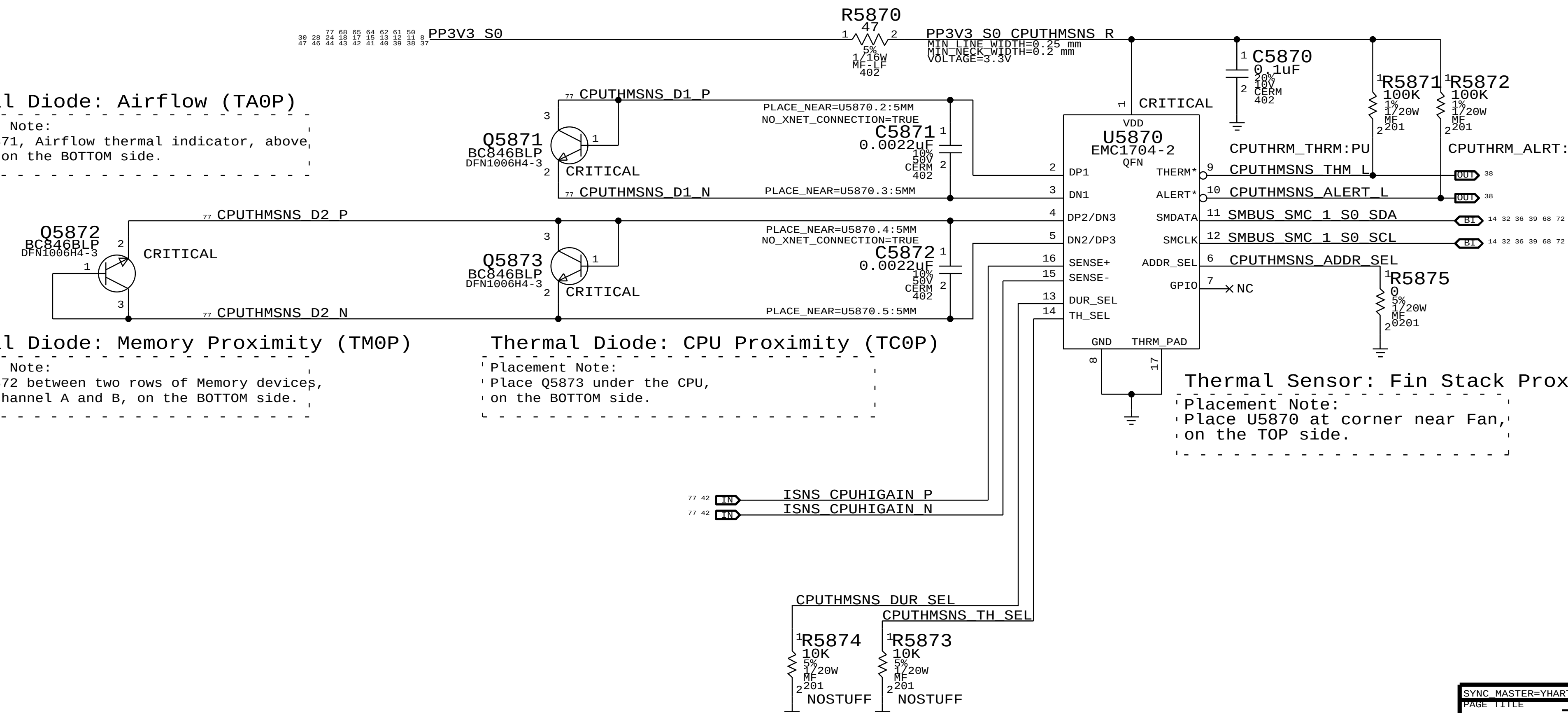
Placement Note:
Place U5850 on the TOP side, on the left portion of the board, 1" to the right of USB connector.

Thermal Sensor B & CPU High Peak Detection:
CPU Proximity, Memory Proximity, Airflow, Fin Stack Proximity

I2C Write: 0x98, I2C Read: 0x99

Thermal Diode: Airflow (TA0P)

Placement Note:
Place Q5871, Airflow thermal indicator, above the SSD, on the BOTTOM side.



Thermal Diode: Memory Proximity (TM0P)

Placement Note:
Place Q5872 between two rows of Memory devices, between channel A and B, on the BOTTOM side.

Thermal Diode: CPU Proximity (TC0P)

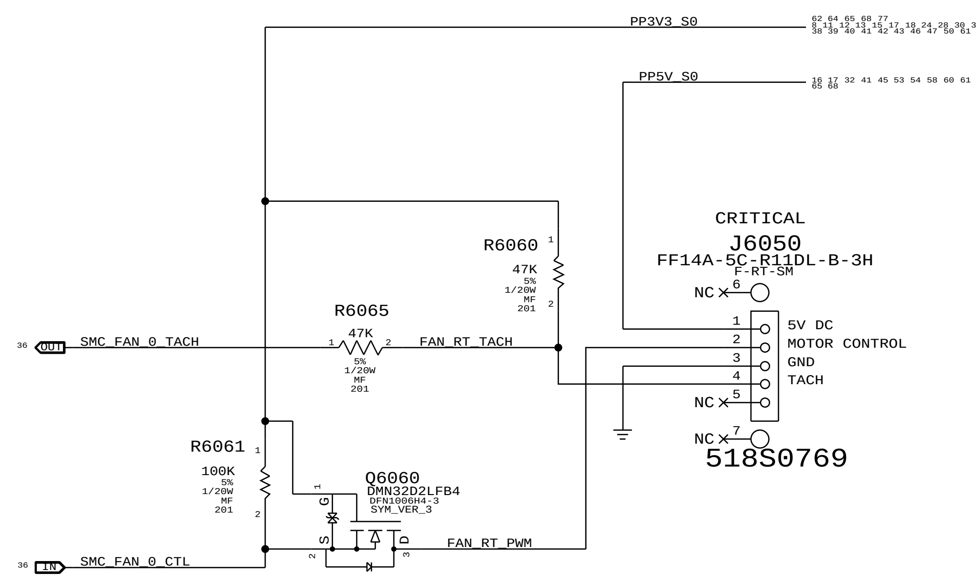
Placement Note:
Place Q5873 under the CPU, on the BOTTOM side.

Thermal Sensor: Fin Stack Proximity (Th1H)

Placement Note:
Place U5870 at corner near Fan, on the TOP side.

FAN CONNECTOR

KEEP THE 5 PIN CONNECTOR FROM D1



SYNC_MASTER=J41		SYNC_DATE=10/23/2012
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	REVISION: <E4LABEL>	
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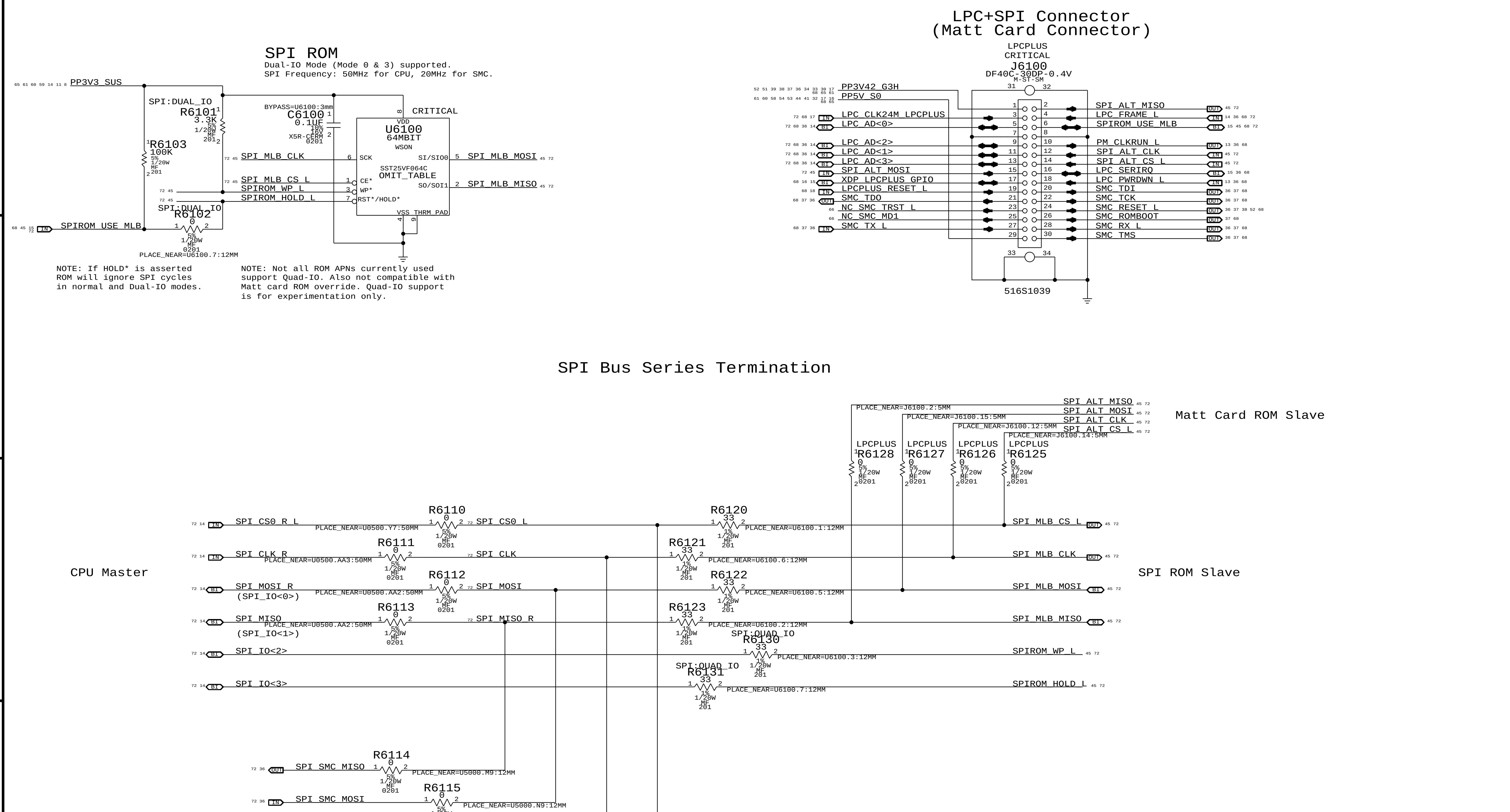
	SHEET: 44 OF 78	



SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=01/09/2013

PAGE TITLE

LPC+SPI Debug Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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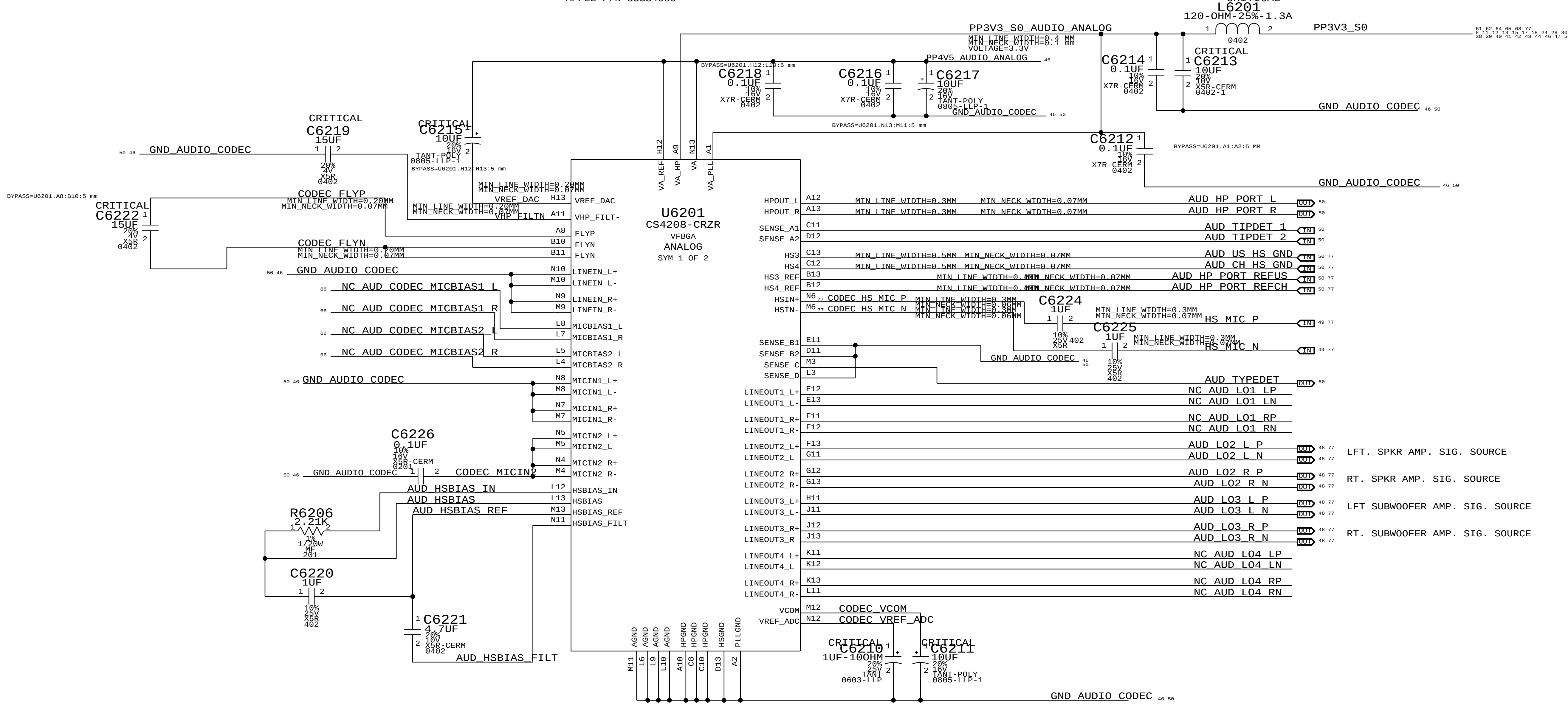
AUDIO CODEC, ANALOG BLOCKS

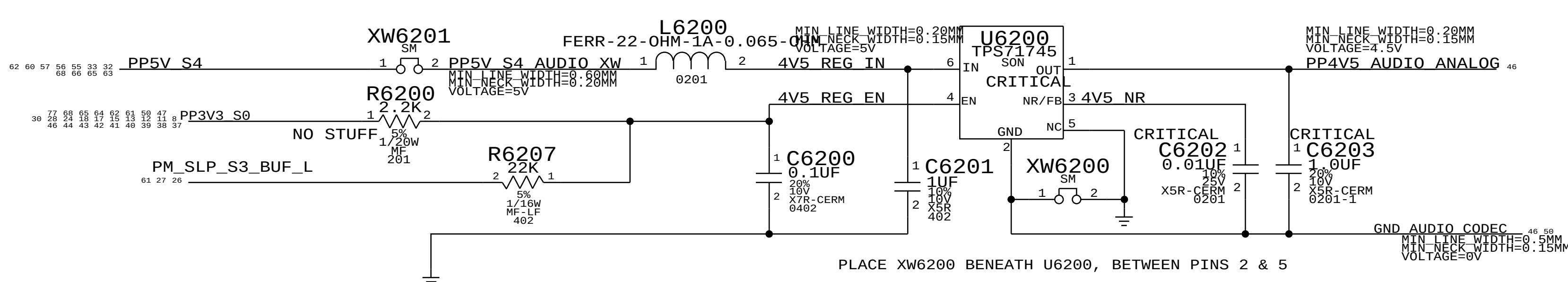
APPLE P/N 353S4080



4.5V POWER SUPPLY FOR CODEC

APPLE P/N 353S2456

PLACE XW6201 NEAR 5V SOURCE



PLACE XW6200 BENEATH U6200, BETWEEN PINS 2 & 5

SYNC_MASTER=JCURCIO_J44 SYNC_DATE=05/13/2013

PAGE TITLE

AUDIO:CODEC, ANALOG

Apple Inc.

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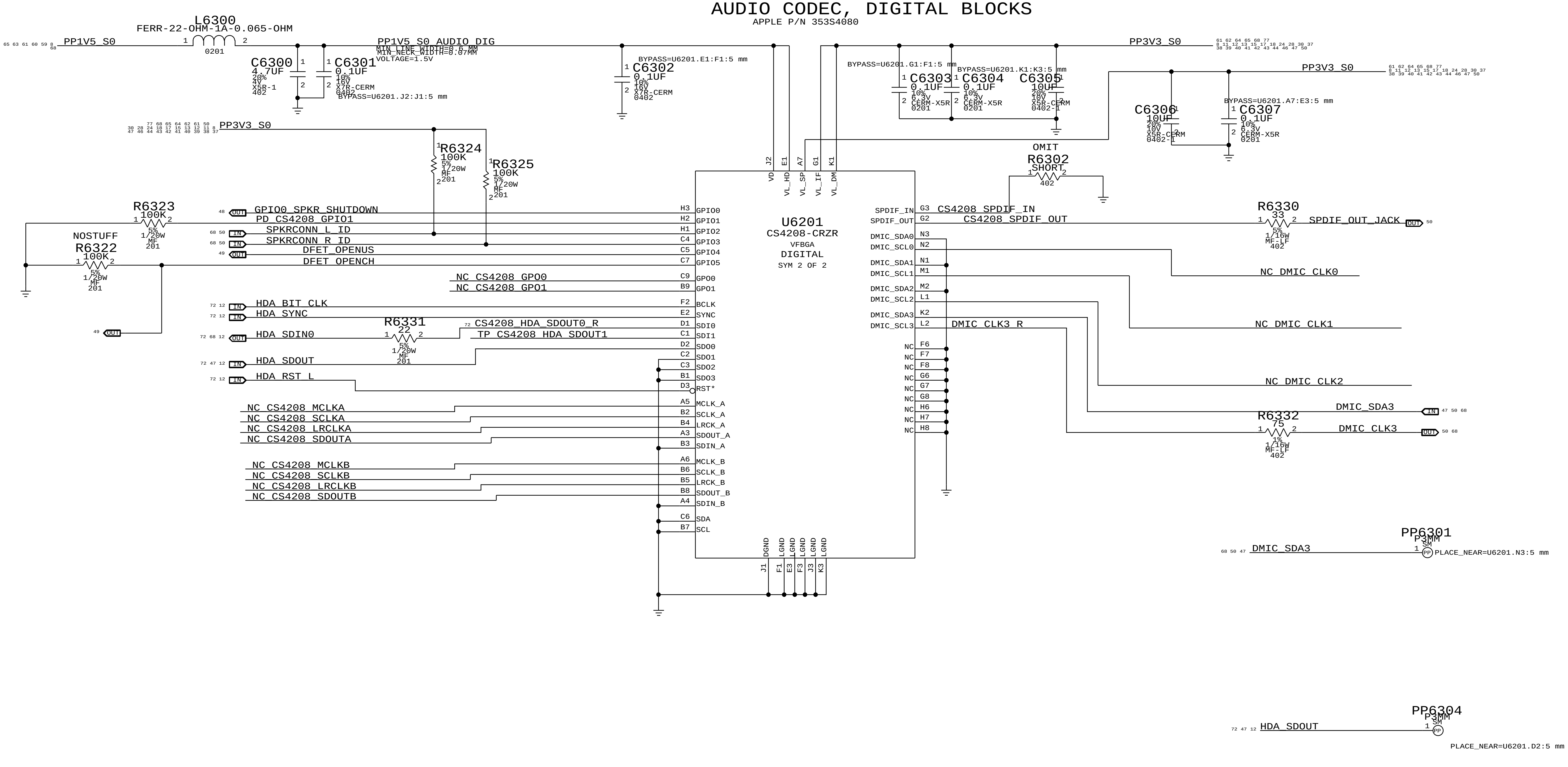
AUDIO CODEC, DIGITAL BLOCKS

APPLE P/N 353S4080



SYNC_MASTER=JCURCIO_J44 SYNC_DATE=07/25/2013

PAGE TITLE

AUDIO:CODEC, DIGITAL

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

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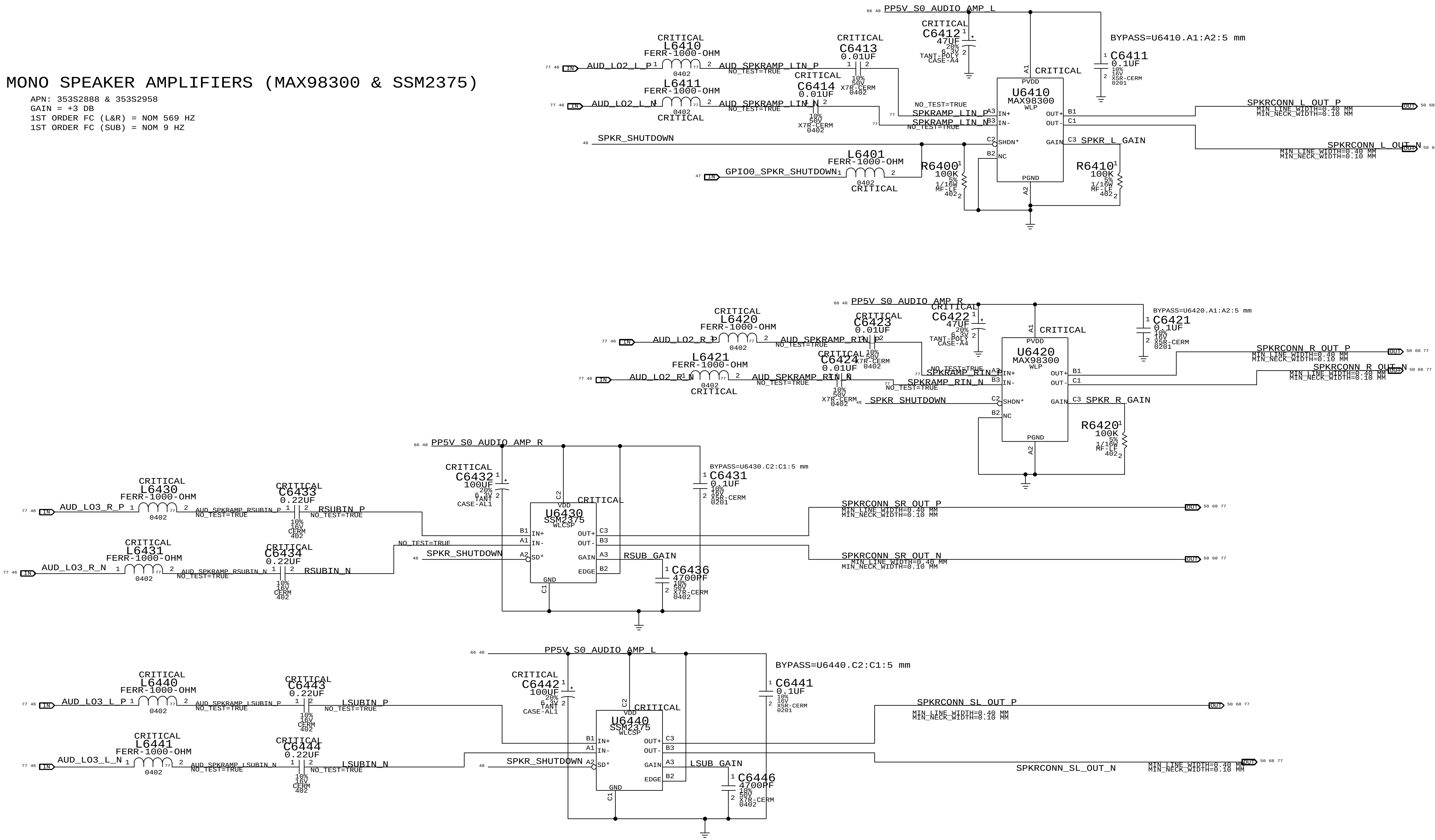
4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)

APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ



SYNC_MASTER=DIRK_J44	SYNC_DATE=01/09/2013
PAGE TITLE	AUDIO: SPEAKER AMP
Apple Inc.	DRAWING NUMBER <SCH_NUM> SIZE D
	REVISION <E4LABEL>
	BRANCH <BRANCH>
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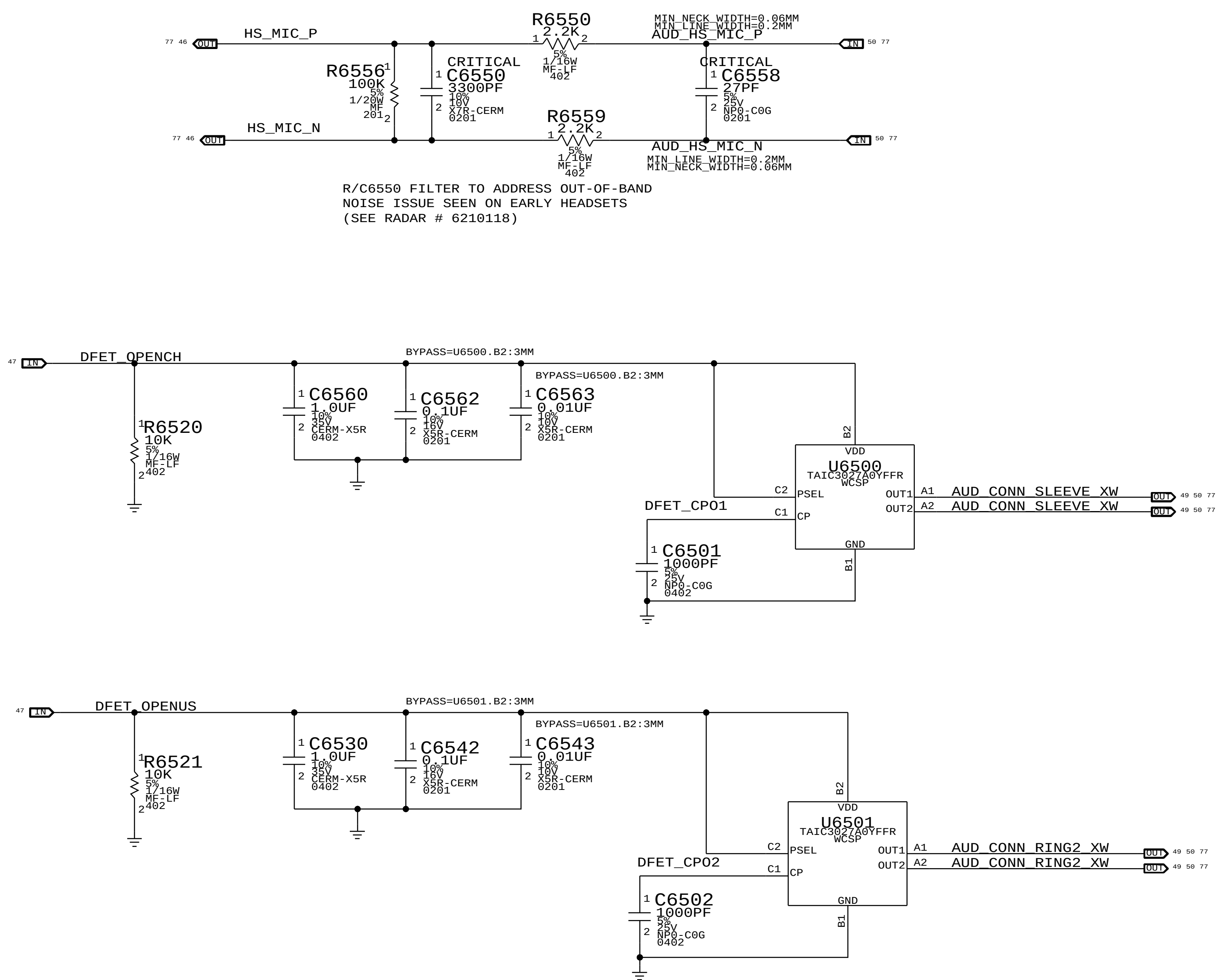
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART



SYNC_MASTER=JCURCIO_J44 SYNC_DATE=07/25/2013

PAGE TITLE

AUDIO: JACK

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

CODEC OUTPUT SIGNAL PATHS

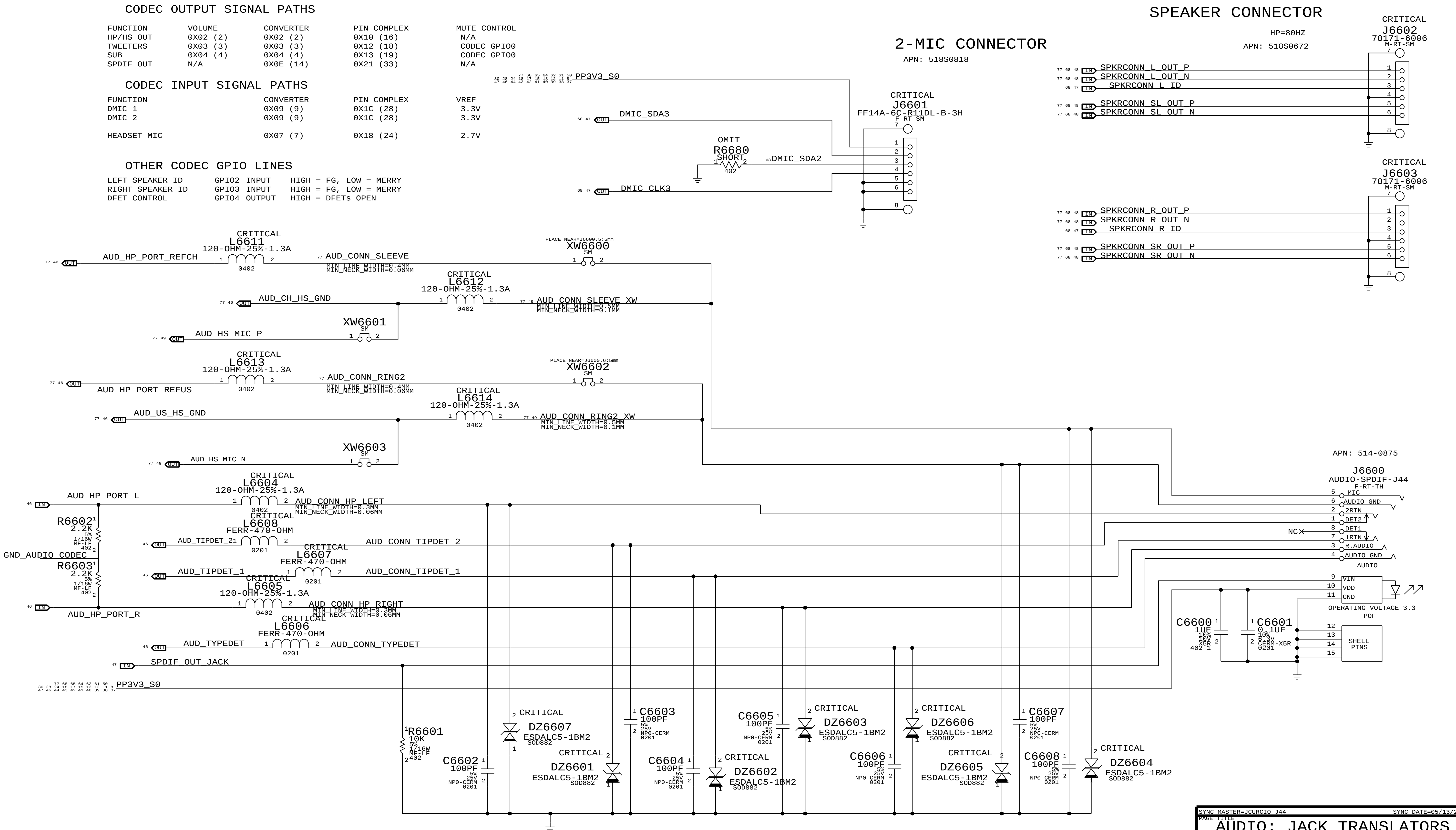
FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL
HP/HS OUT	0X02 (2)	0X02 (2)	0X10 (16)	N/A
TWEETERS	0X03 (3)	0X03 (3)	0X12 (18)	CODEC GPIO0
SUB	0X04 (4)	0X04 (4)	0X13 (19)	CODEC GPIO0
SPDIF OUT	N/A	0X0E (14)	0X21 (33)	N/A

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF
DMIC 1	0X09 (9)	0X1C (28)	3.3V
DMIC 2	0X09 (9)	0X1C (28)	3.3V
HEADSET MIC	0X07 (7)	0X18 (24)	2.7V

OTHER CODEC GPIO LINES

LEFT SPEAKER ID	GPIO2 INPUT	HIGH = FG, LOW = MERRY
RIGHT SPEAKER ID	GPIO3 INPUT	HIGH = FG, LOW = MERRY
DFET CONTROL	GPIO4 OUTPUT	HIGH = DFETs OPEN



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Apple Inc.

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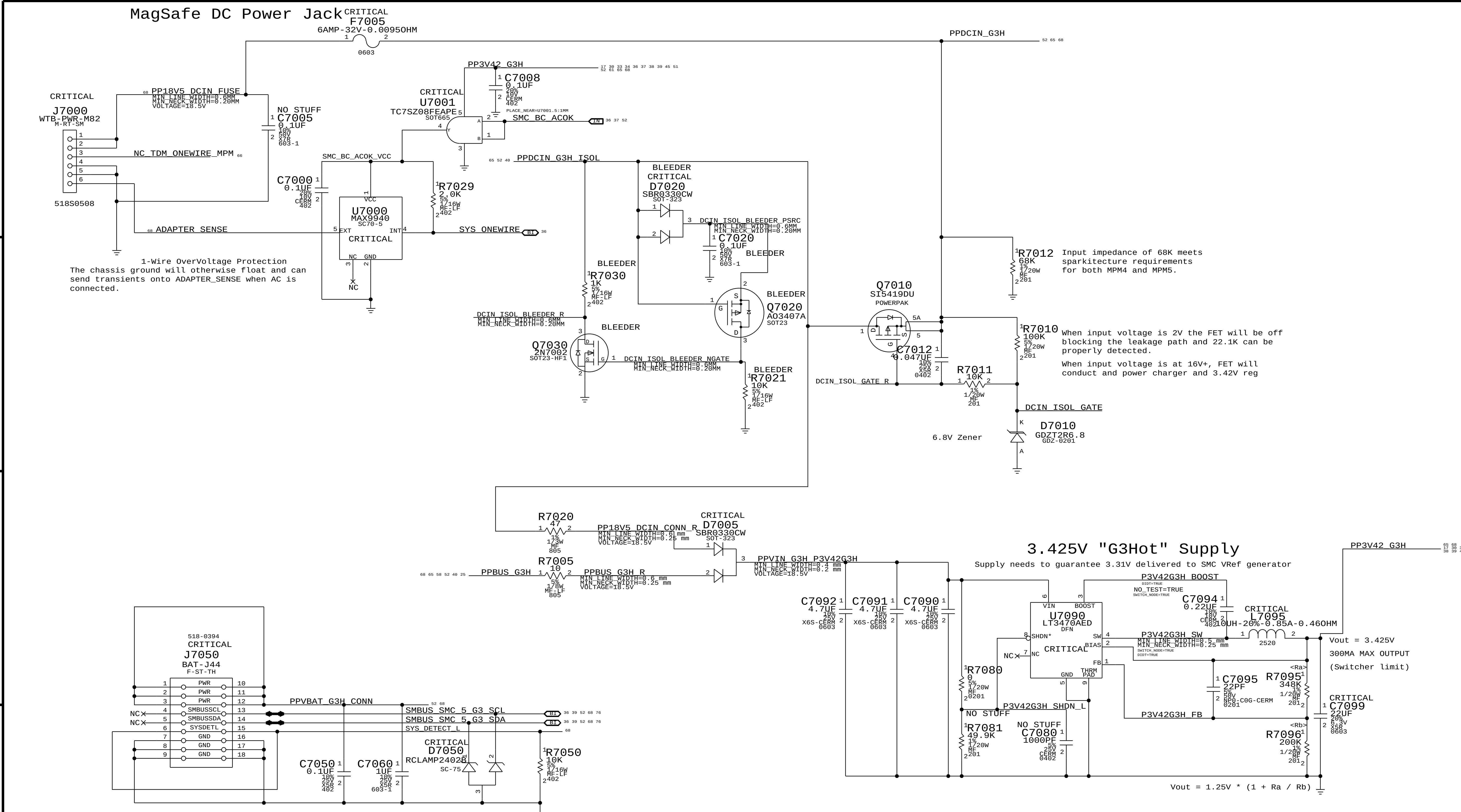

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DC-In & Battery Connectors

Apple Inc.

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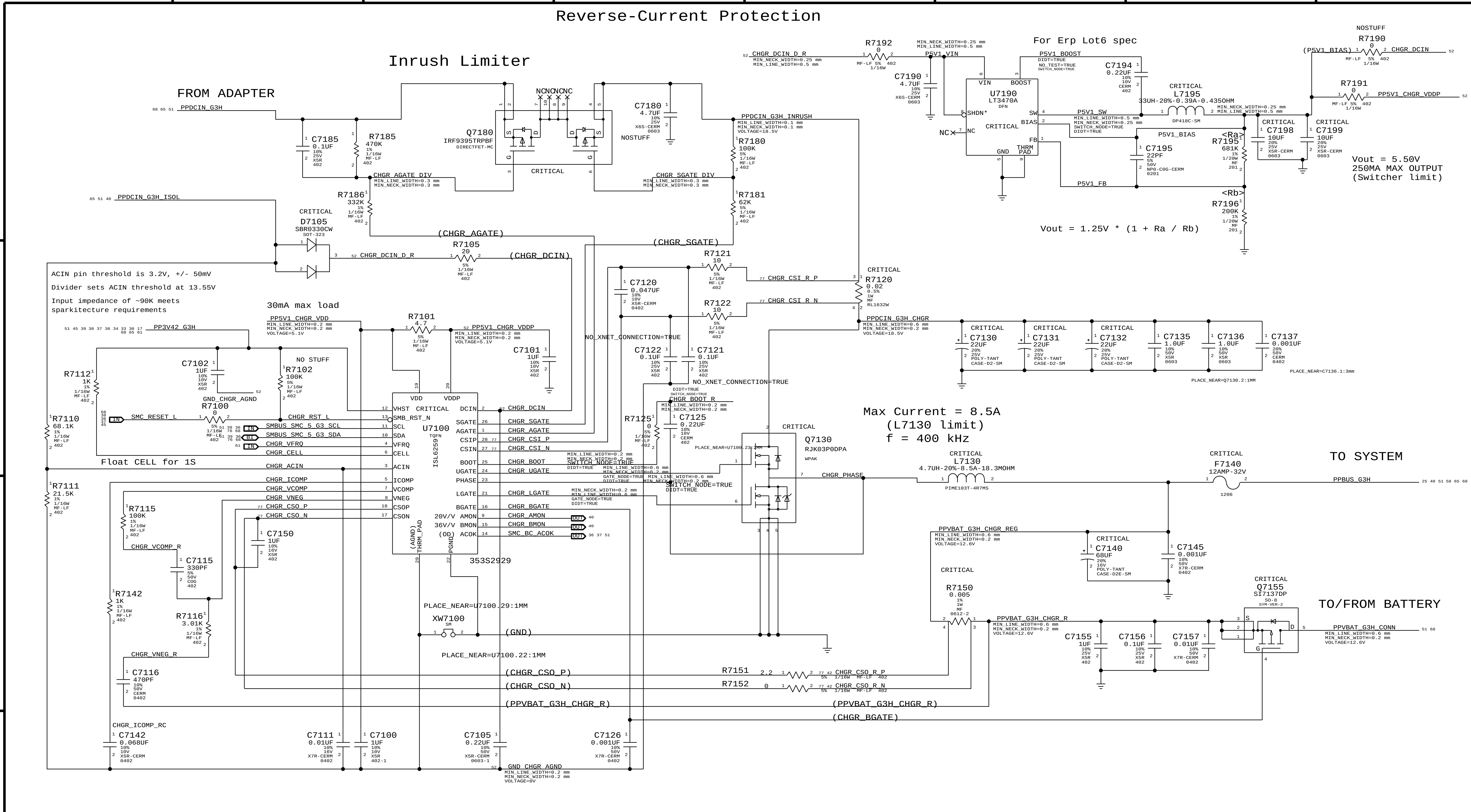


SYNC_MASTER=J14 SYNC_DATE=10/23/2012

PAGE TITLE

PBus Supply & Battery Charger

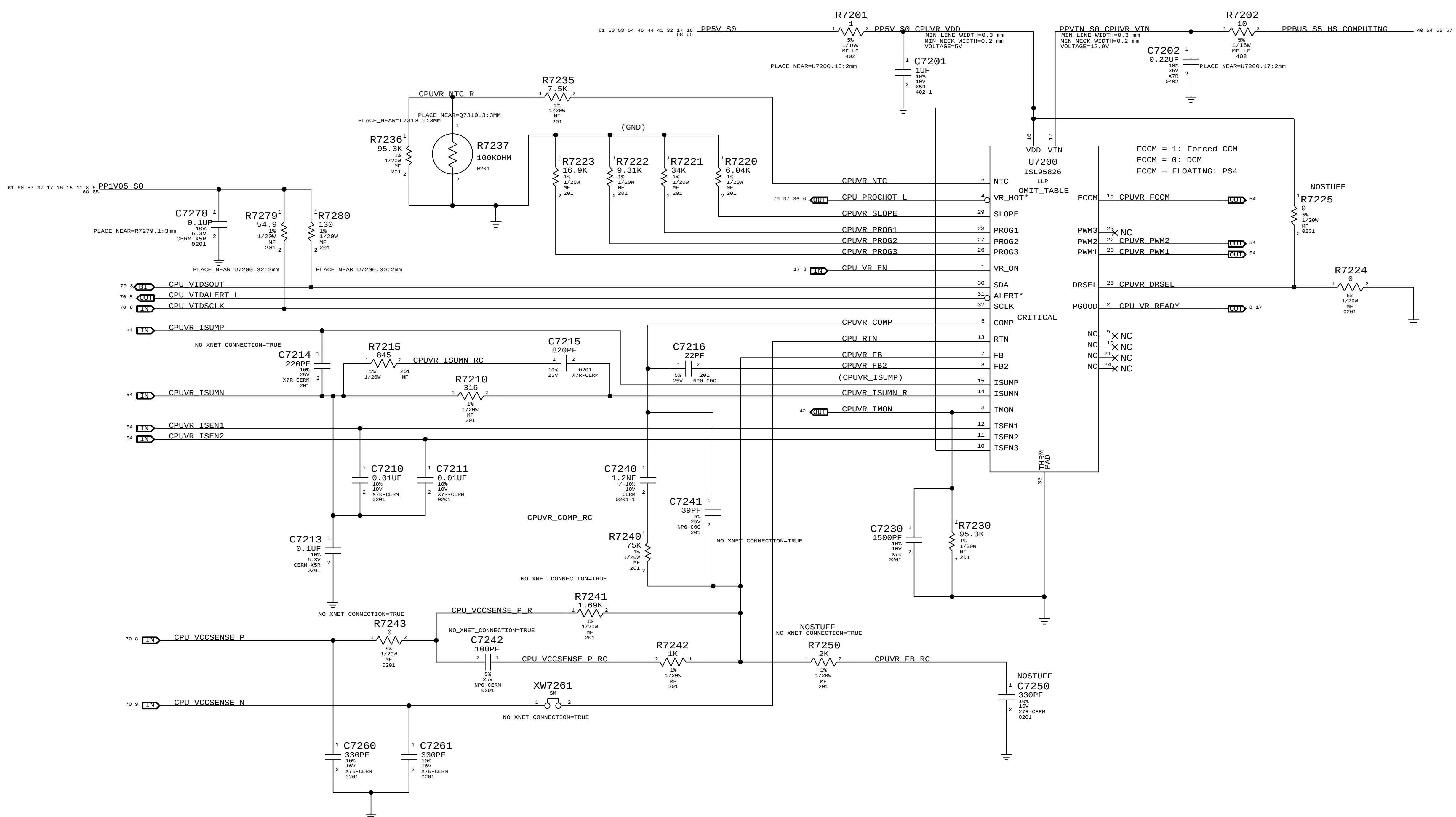
Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S4170	1	IC,ISL95826R6200,PWM,PGOOD,SCREN,32P,QFN	U7200	CRITICAL	



SYNC_MASTER=J41
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PAGE TITLE

CPU VR12.6 VCC Regulator IC

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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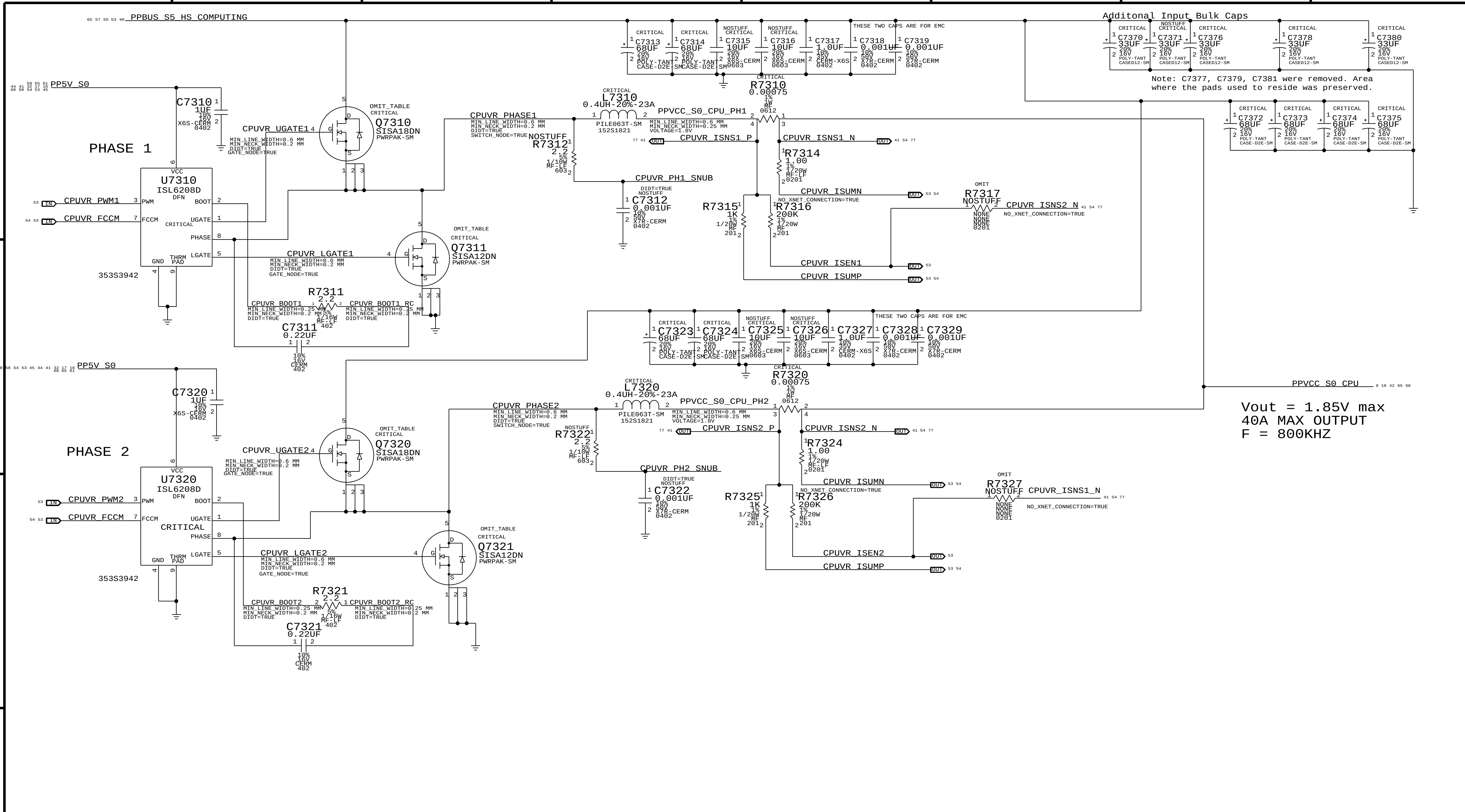

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PAGE TITLE

CPU VR12.5 VCC Power Stage

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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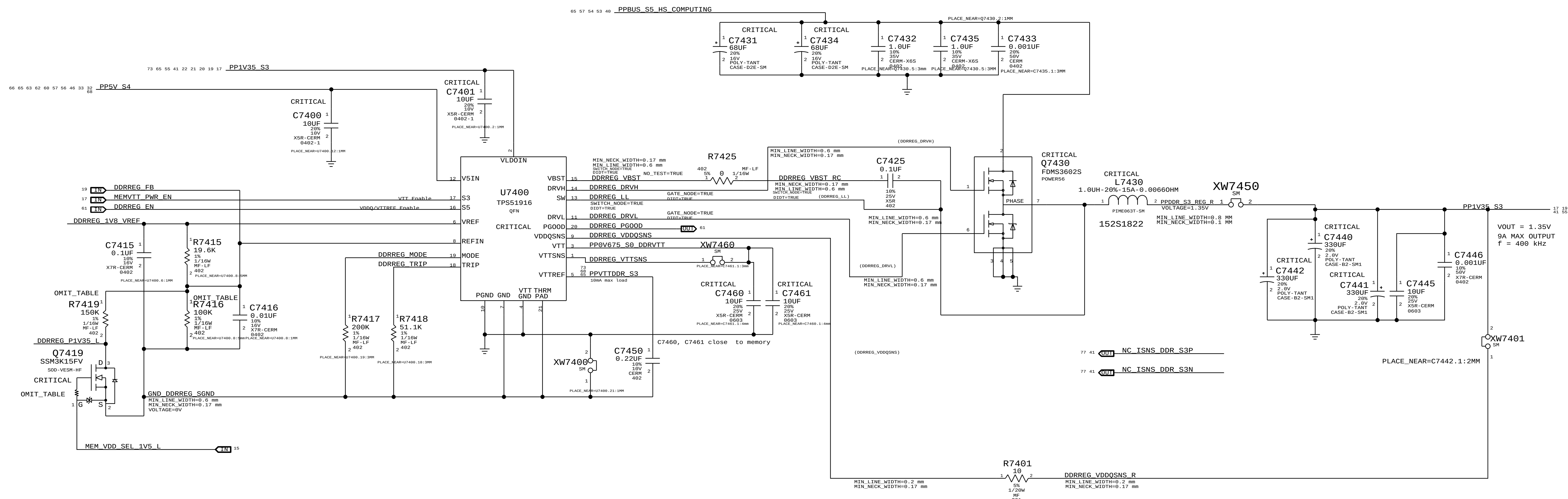
DDR3L (1V35 S3) REGULATOR



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0411	1	RES,MTL FILM,1/16W,100K,1,0402,SMD,LF	R7416	CRITICAL	PPDDR:1V5
114S0391	1	RES,MTL FILM,1/16W,60.4K,1,0402,SMD,LF	R7416	CRITICAL	PPDDR:1V35
376S0612	1	MOSFET,N-CH,30V,100MA,7.0OHM,SOT-723,HF	Q7419	CRITICAL	PPDDR:1V5
114S0428	1	RES, MTL FILM,1/16W,150k,0402,SMD,LF	R7419	CRITICAL	PPDDR:1V5

SYNC_MASTER=J14 SYNC_DATE=10/23/2012

PAGE TITLE

1.35V DDR3 SUPPLY

Apple Inc.

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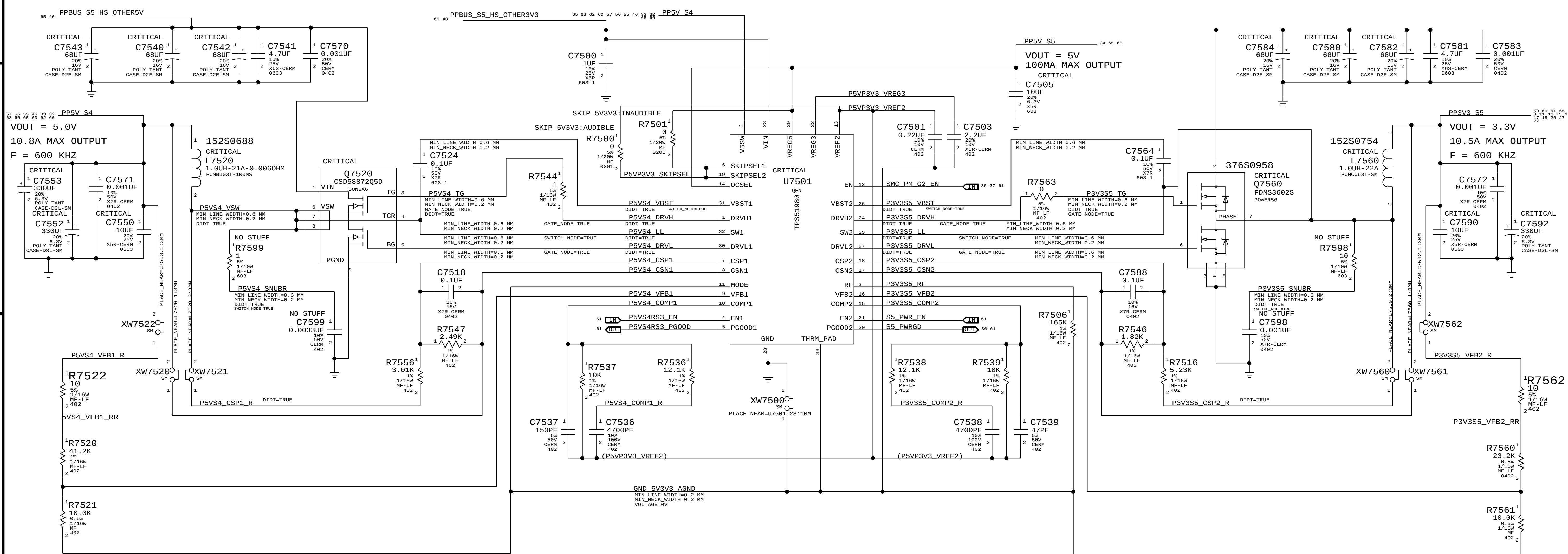

SYNC_MASTER=J14 SYNC_DATE=10/23/2012

PAGE TITLE

5V / 3.3V Power Supply

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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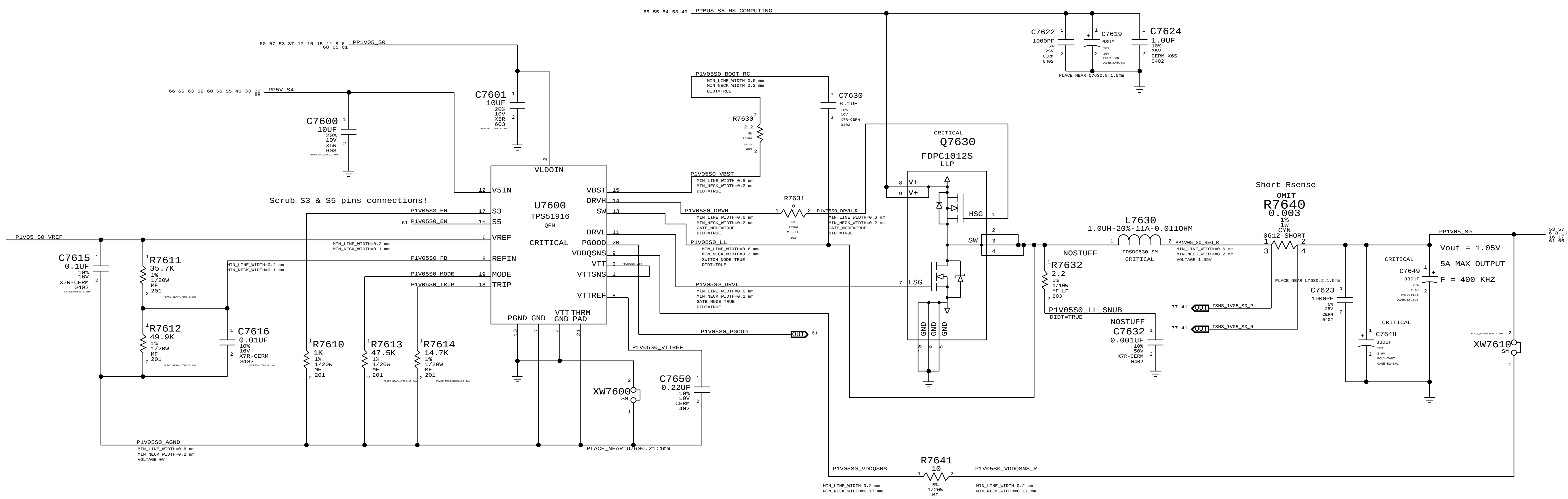
PAGE 75 OF 120

SHEET 56 OF 78

1.05V S0 Regulator



SYNC_MASTER=J41 SYNC_DATE=10/23/2012

PAGE TITLE

1.05V S0 Power Supply

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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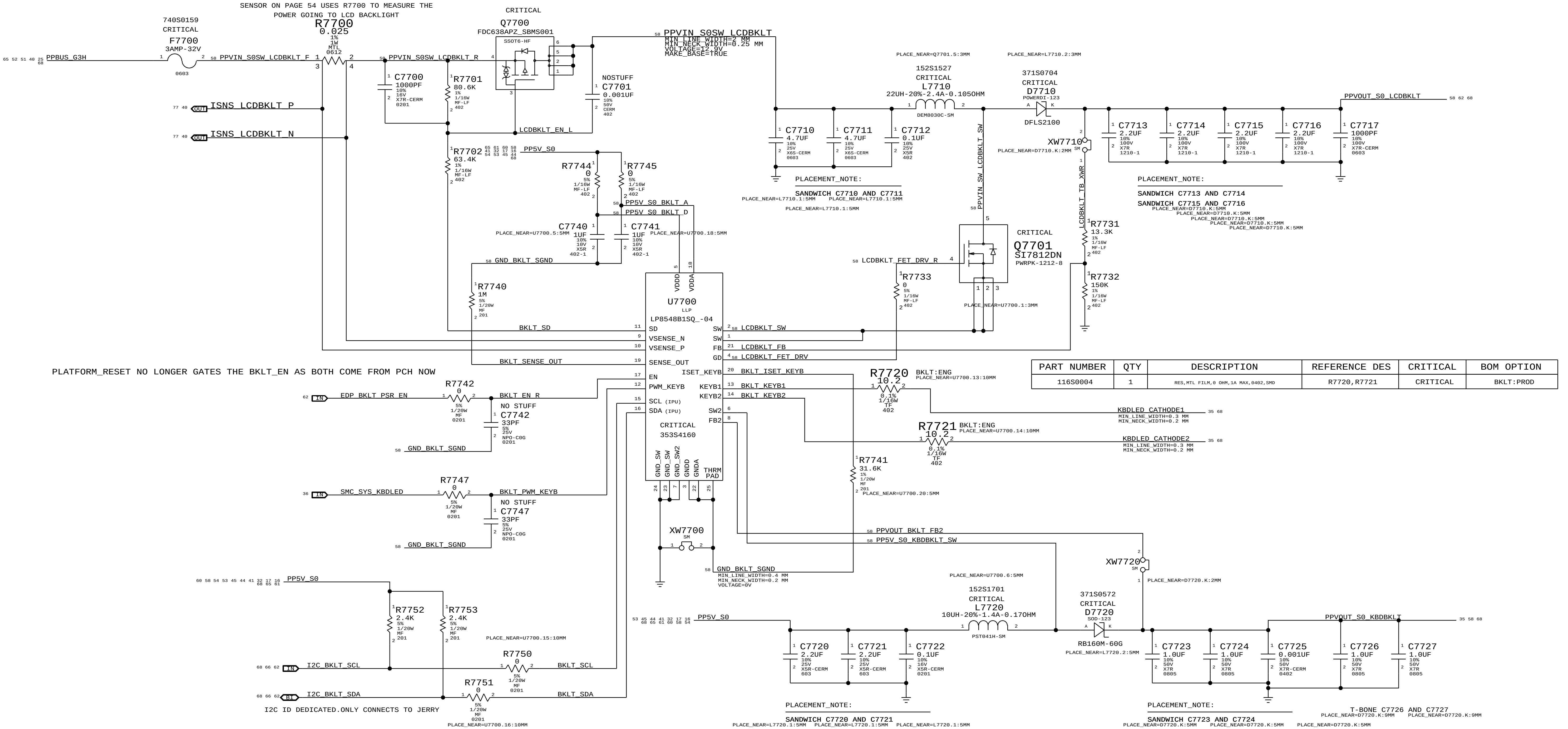
Page Notes

Power aliases required by this page:
- =PPVIN_S0SW_LCDBKLTFET (9-12.6V LCD BACKLIGHT INPUT)
- =PP5V_S0_BKLT (5V BACKLIGHT DRIVER INPUT)
- =PP5V_S0SW_KBDLED (5V KEYBOARD BACKLIGHT INPUT)

BOM options provided by this page:
BKLT:ENG - Stuffs 10.2 ohm series R for engineering builds
BKLT:PROD - Stuffs 0 ohm series R for production



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	1	RES,MTL FILM,0 OHM,1A MAX,0402,SMD	R7720,R7721	CRITICAL	BKLT:PROD

PBUS LINE WIDTHS

PP5V_S0_BKLT_A — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=5V
PP5V_S0_BKLT_D — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=5V
PPVIN_S0SW_LCDBKLT_F — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=12.9V
PPVIN_S0SW_LCDBKLT_R — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=12.9V
PPVIN_S0SW_LCDBKLT_FET — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=12.9V
PPVIN_S0SW_LCDBKLT — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=12.9V

LCD BKLT LINE WIDTHS

LCDBKLT_FET_DRV_R — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=5V, GATE_NODE=TRUE, DIDT=TRUE
LCDBKLT_FET_DRV — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=5V, GATE_NODE=TRUE, DIDT=TRUE
LCDBKLT_SW — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=53V, SWITCH_NODE=TRUE, DIDT=TRUE
PPVIN_SW_LCDBKLT_SW — MIN_LINE_WIDTH=2 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=53V, SWITCH_NODE=TRUE, DIDT=TRUE
PPVOUT_S0_LCDBKLT — MIN_LINE_WIDTH=0.5 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=53V
PPVOUT_BKLT_FB — MIN_LINE_WIDTH=0.4 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=53V

KBD BKLT LINE WIDTHS

PP5V_S0_KBDBKLT_SW — MIN_LINE_WIDTH=0.5 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=40V, SWITCH_NODE=TRUE, DIDT=TRUE
PPVOUT_S0_KBDBKLT — MIN_LINE_WIDTH=0.5 MM, MIN_NECK_WIDTH=0.2 MM, VOLTAGE=40V
PPVOUT_BKLT_FB2 — MIN_LINE_WIDTH=0.4 MM, MIN_NECK_WIDTH=0.25 MM, VOLTAGE=40V

SYNC_MASTER=SHART_J44 SYNC_DATE=11/20/2012

PAGE TITLE
LCD AND KBD BKLT DRIVER

DRAWING NUMBER <SCH_NUM> SIZE D

Apple Inc.

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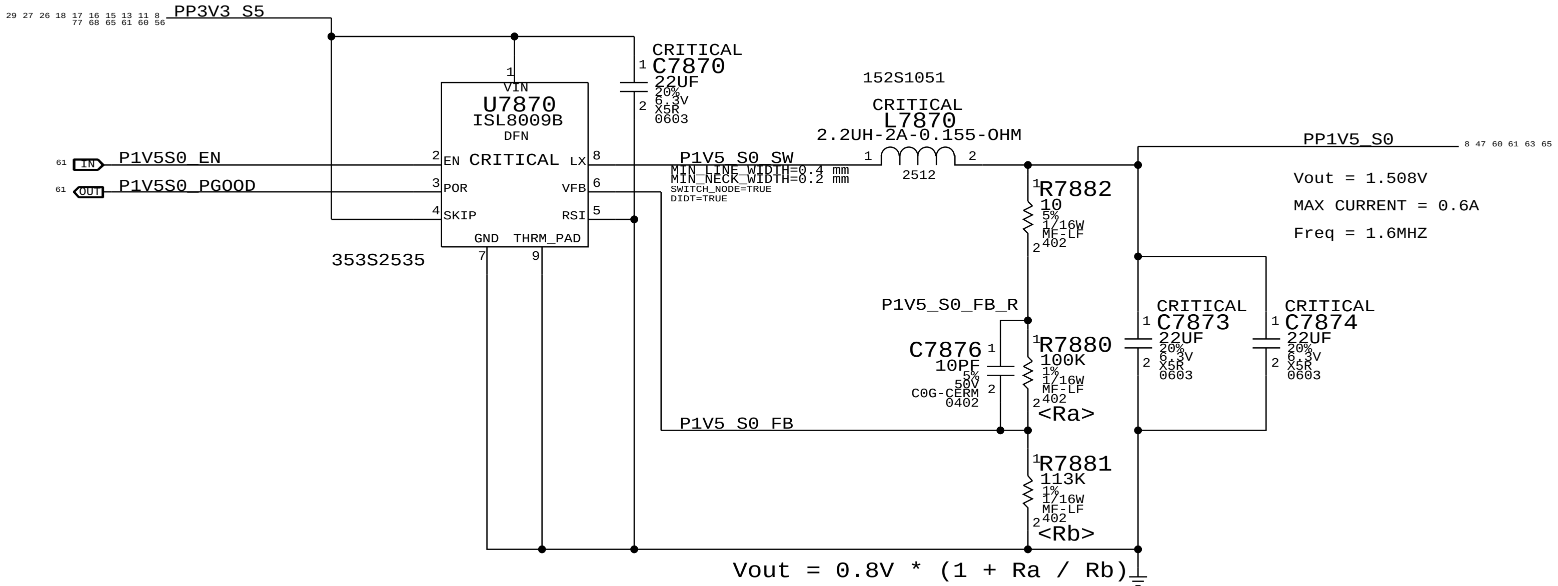
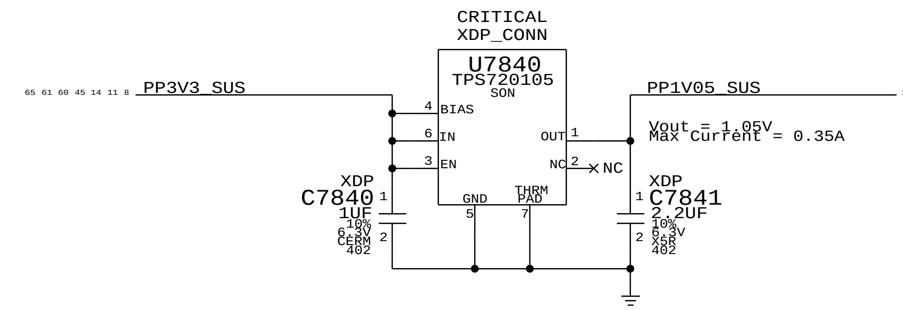
1.5V S0 Switcher



1.05V SUS LDO

Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active.
Pull-ups (3) must be 51 ohms to support XDP (not required in production).
70mA is required to support pull-ups. Alternative is strong voltage
dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



SYNC_MASTER=J41 SYNC_DATE=10/23/2012

PAGE TITLE
Misc Power Supplies

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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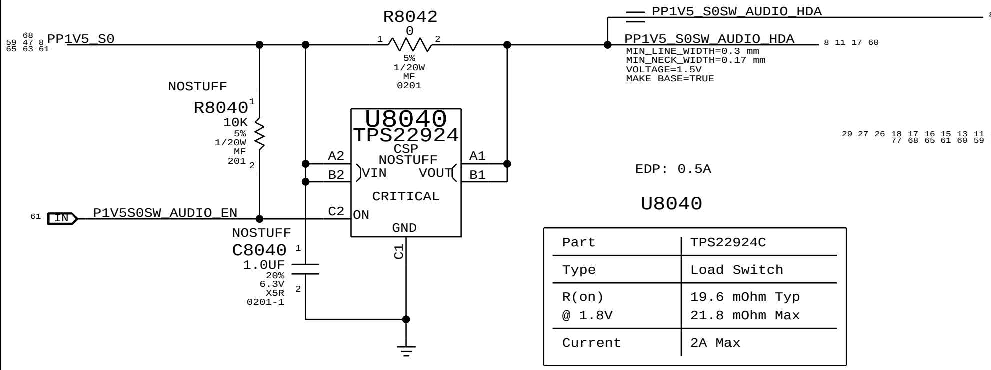
Loading specs per J41/43_PowerBudget_Riviera_rev0.99e

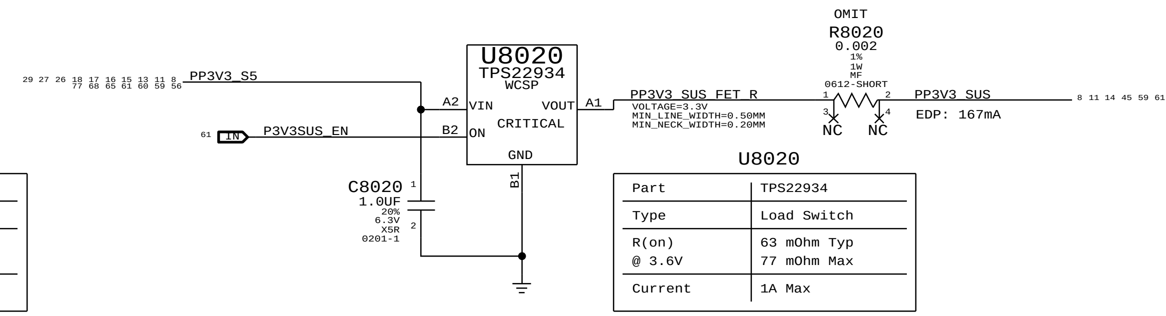
1.5V S0 Audio Switch (BYPASSED)



U8040

Part	TPS22924C
Type	Load Switch
R(on) @ 1.8V	19.6 mOhm Typ 21.8 mOhm Max
Current	2A Max

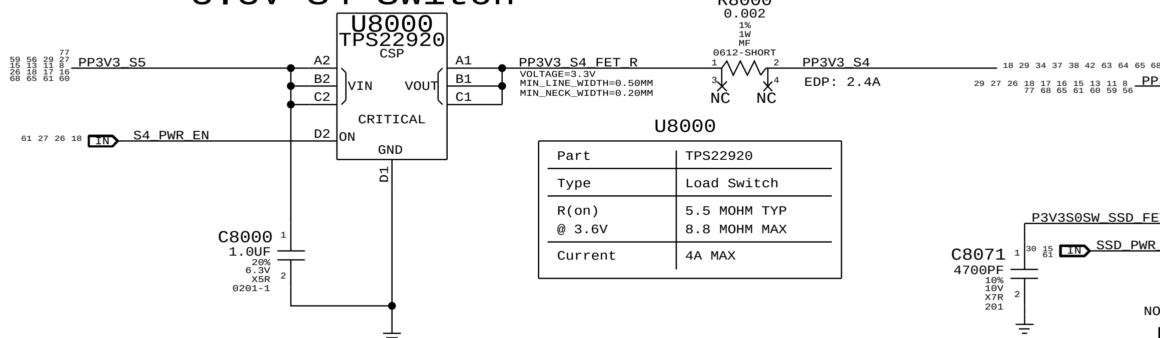
3.3V SUS Switch



U8020

Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

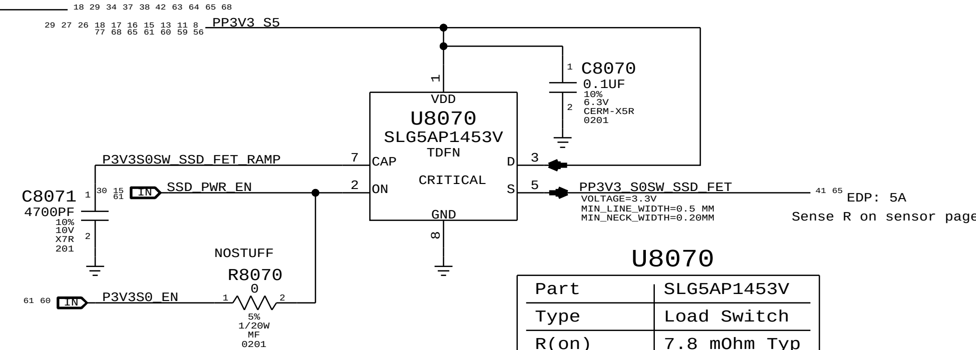
3.3V S4 Switch



U8000

Part	TPS22920
Type	Load Switch
R(on) @ 3.6V	5.5 MOHM TYP 8.8 MOHM MAX
Current	4A MAX

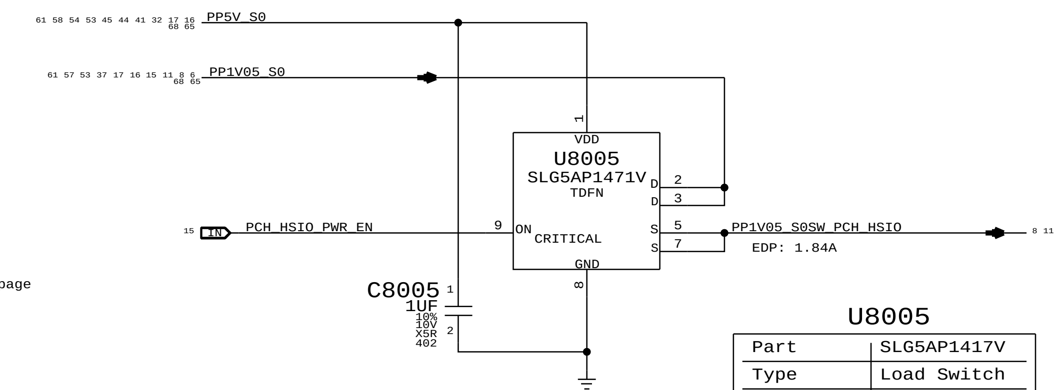
3.3V SSD Switch



U8070

Part	SLG5AP1453V
Type	Load Switch
R(on) @ 25C	7.8 mOhm Typ 8.5 mOhm Max
Current	5.3A Max

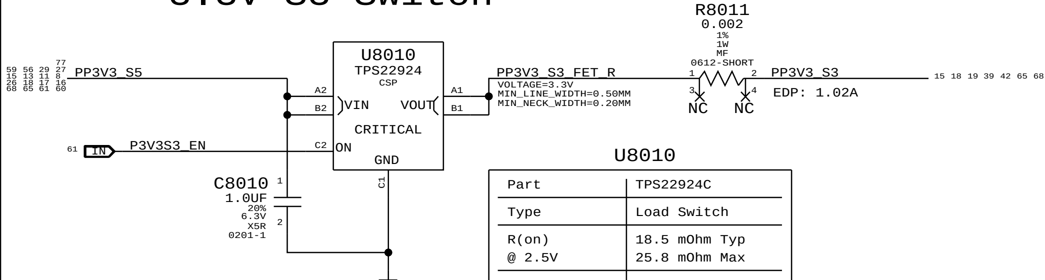
1.05V PCH HSIO Switch



U8005

Part	SLG5AP1417V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

3.3V S3 Switch

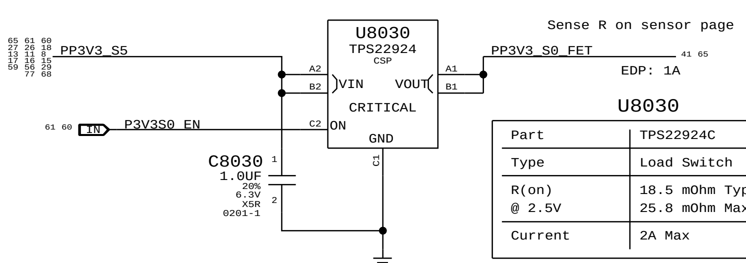


U8010

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

REMOVED THE ANALOG POWER GATE AS SLG5AP1471 SHOULD BE AVAILABLE BY THEN

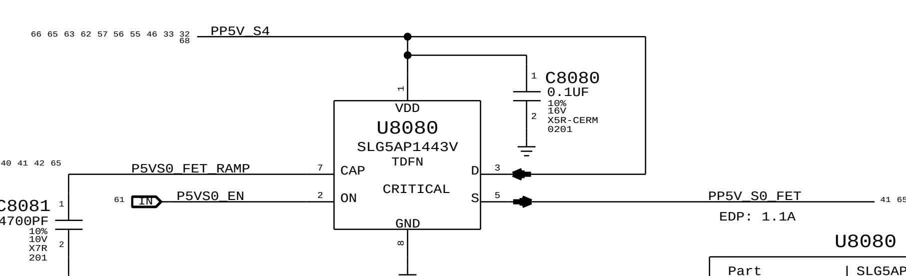
3.3V S0 Switch



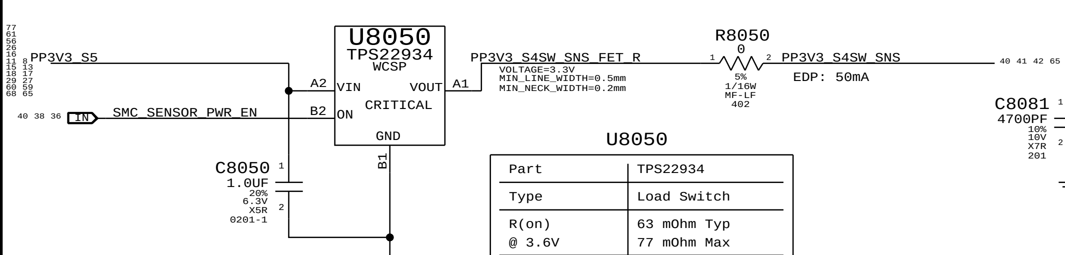
U8030

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

5V S0 Switch



3.3V Sensor Switch



U8050

Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

U8080

Part	SLG5AP1443V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

SYNC_MASTER=J41 SYNC_DATE=10/23/2012

PAGE TITLE

Power FETs

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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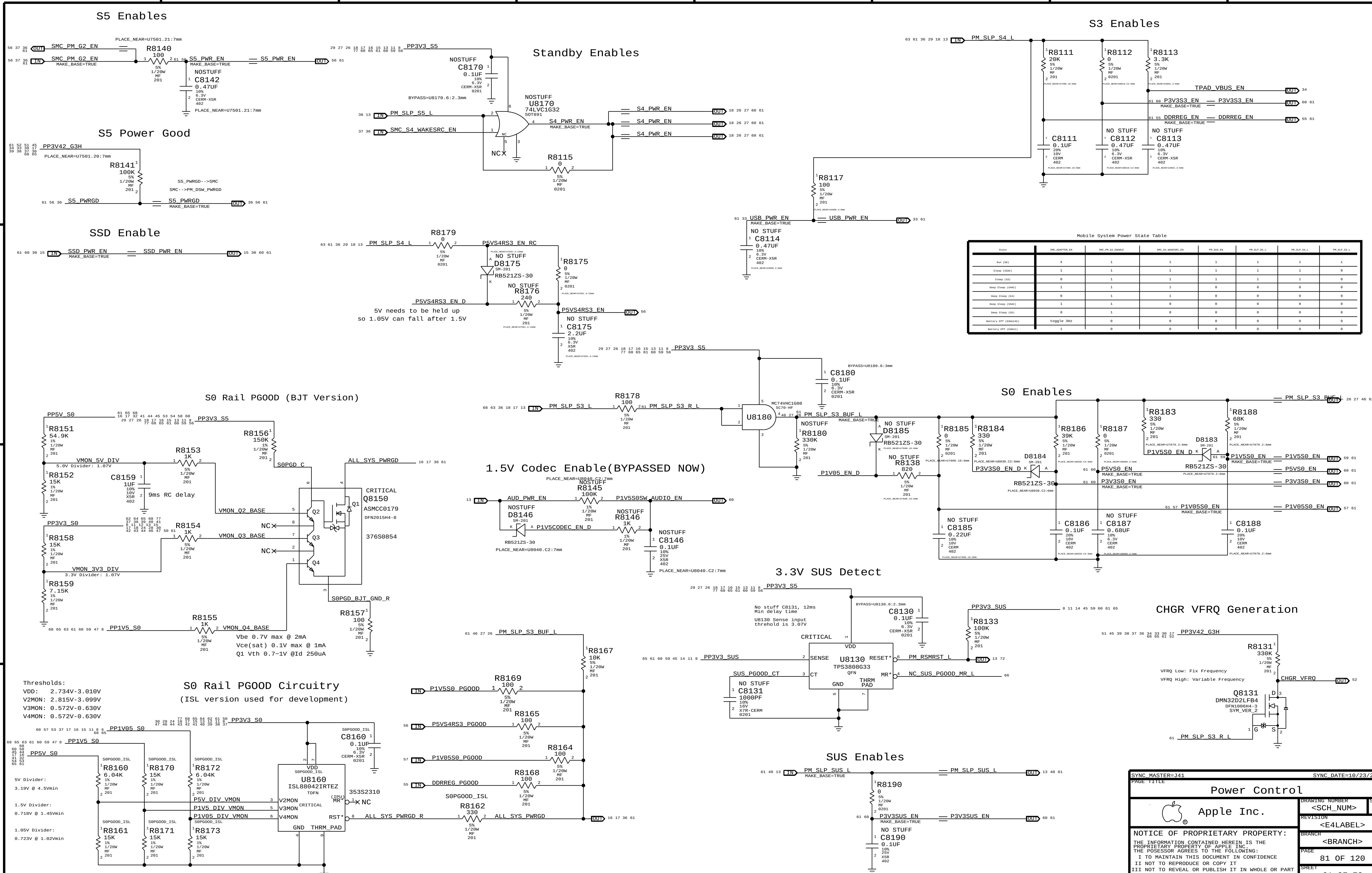

Mobile System Power State Table

State	SMC_ADAPTER_EN	SMC_PM_G2_ENABLE	SMC_S4_WAKESRC_EN	PM_SUS_EN	PM_SLP_S5_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Sleep (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (S4AC)	1	1	1	0	0	0	0
Deep Sleep (S4)	0	1	1	0	0	0	0
Deep Sleep (S5AC)	1	1	0	0	0	0	0
Deep Sleep (S5)	0	1	0	0	0	0	0
Battery Off (G3HotAC)	toggle 3Hz	0	0	0	0	0	0
Battery Off (G3Hot)	1	0	0	0	0	0	0

SYNC_MASTER=J41 SYNC_DATE=10/23/2012

PAGE TITLE

Power Control

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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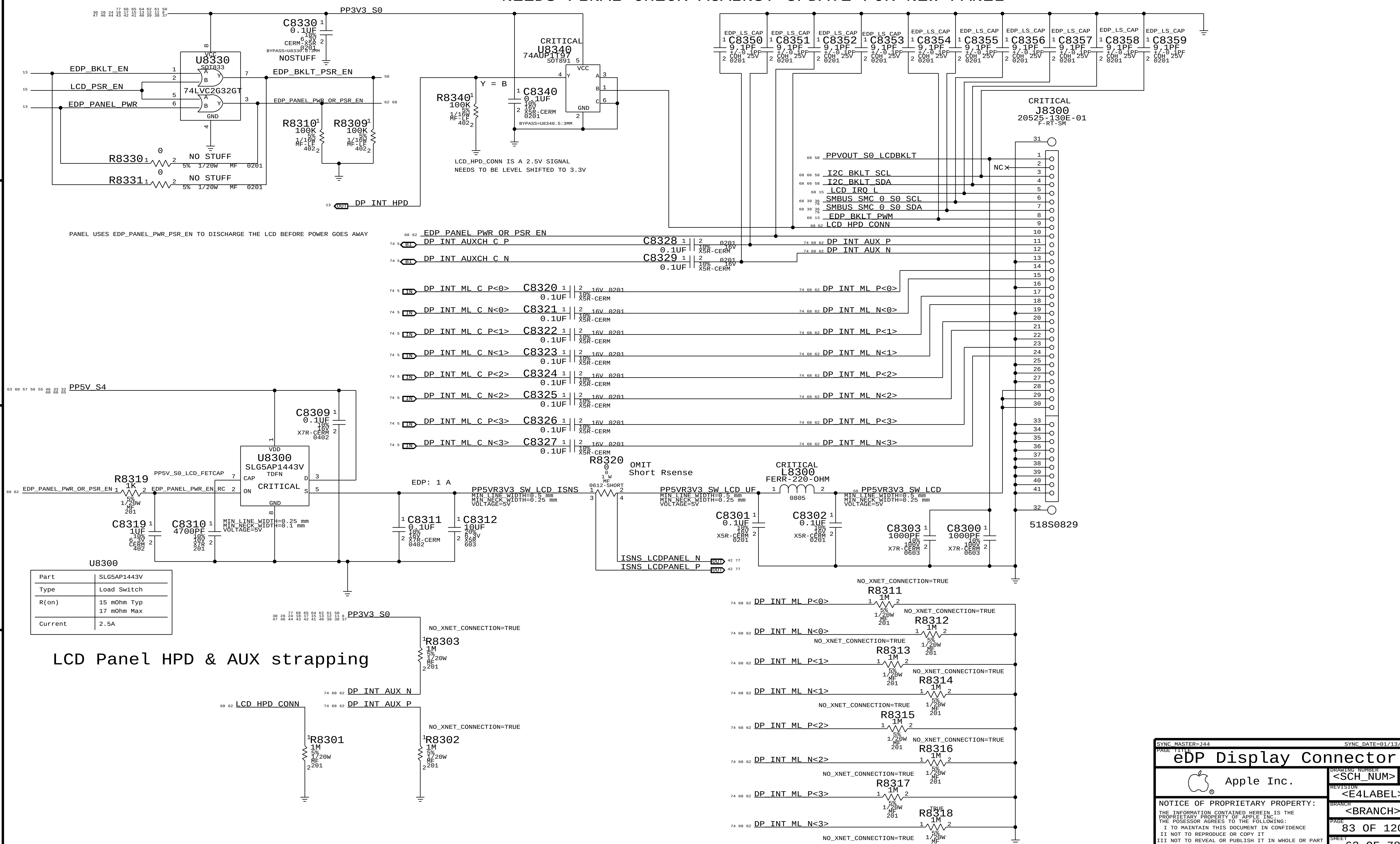
LCD PANEL INTERFACE (eDP)
NEEDS FINAL CHECK AGAINST UPDATE FOR NEW PANEL



Part	SLG5AP1443V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A

SYNC_MASTER=J44 SYNC_DATE=01/13/2013

PAGE TITLE
eDP Display Connector

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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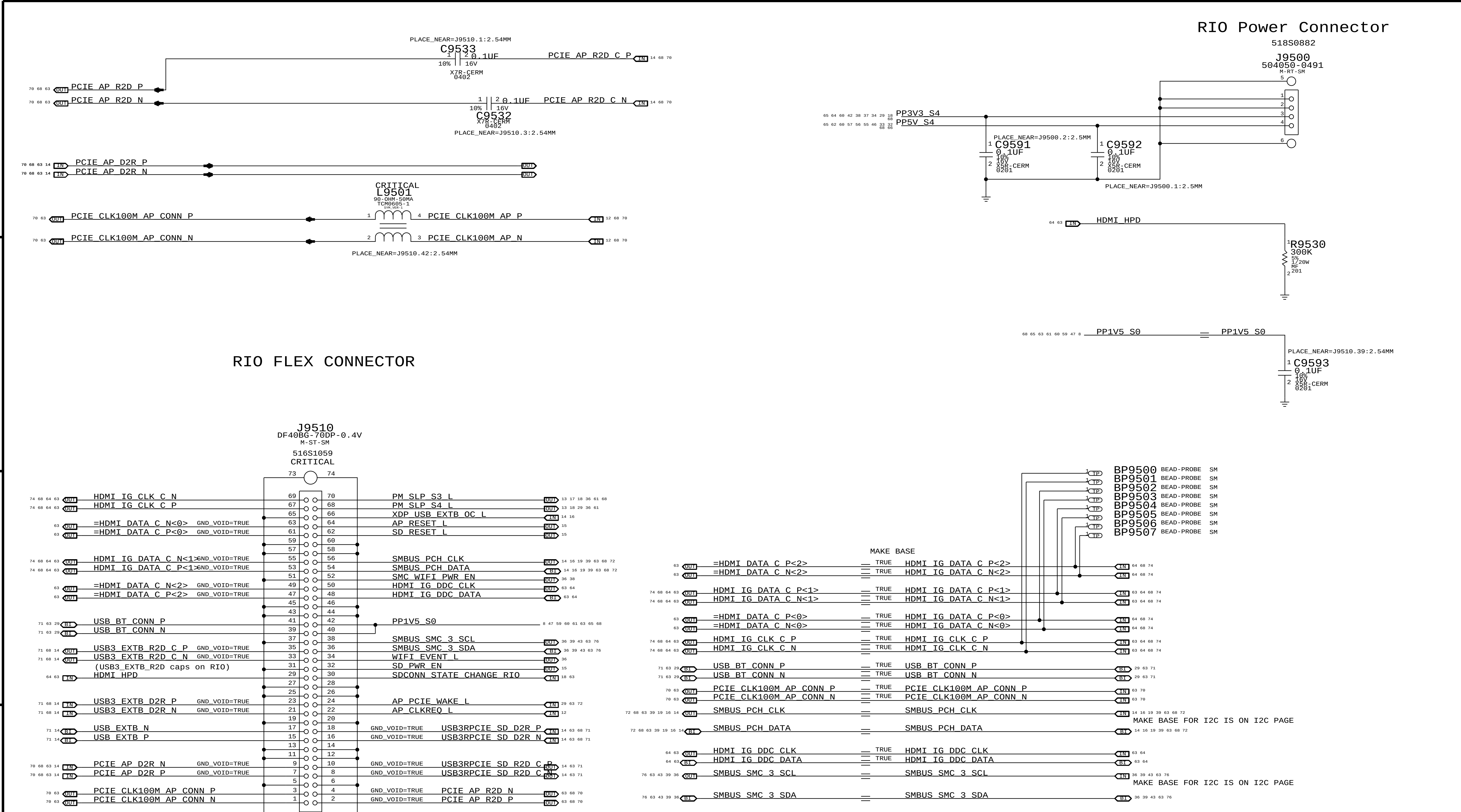

NOTE: This connector is shielded 70P Hirose Plug APN 516S1059, mates with APN 516S1058.

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=01/09/2013

PAGE TITLE: RIO Connector

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

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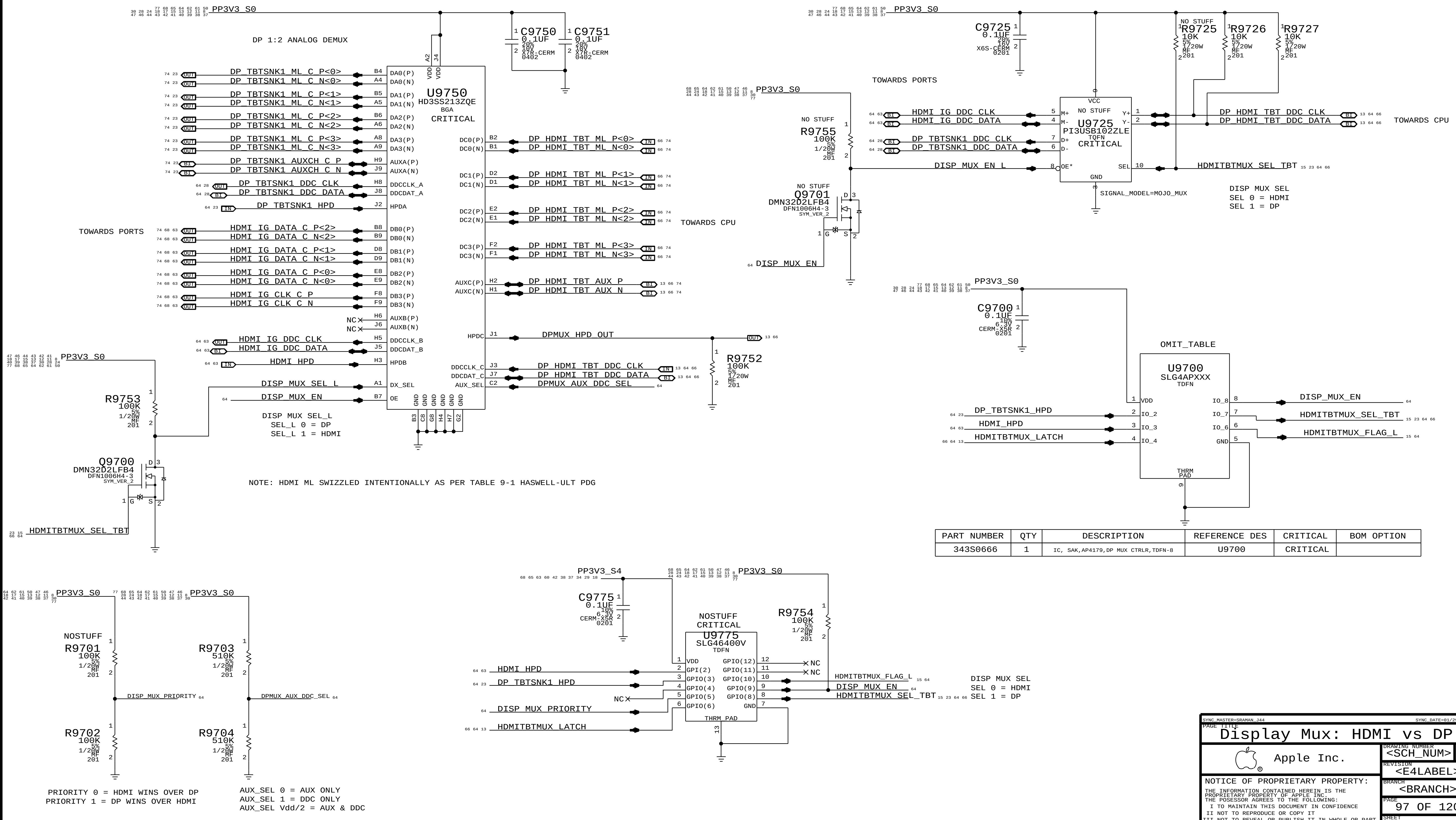
DISPLAY MUX: DP OR HDMI



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
343S0666	1	IC, SAK,AP4179,DP MUX CTRLR,TDFN-8	U9700	CRITICAL	

SYNC_MASTER=SRAMAN_J44 SYNC_DATE=01/29/2013

PAGE TITLE
Display Mux: HDMI vs DP

DRAWING NUMBER <SCH_NUM> SIZE D

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REVISION <E4LABEL>

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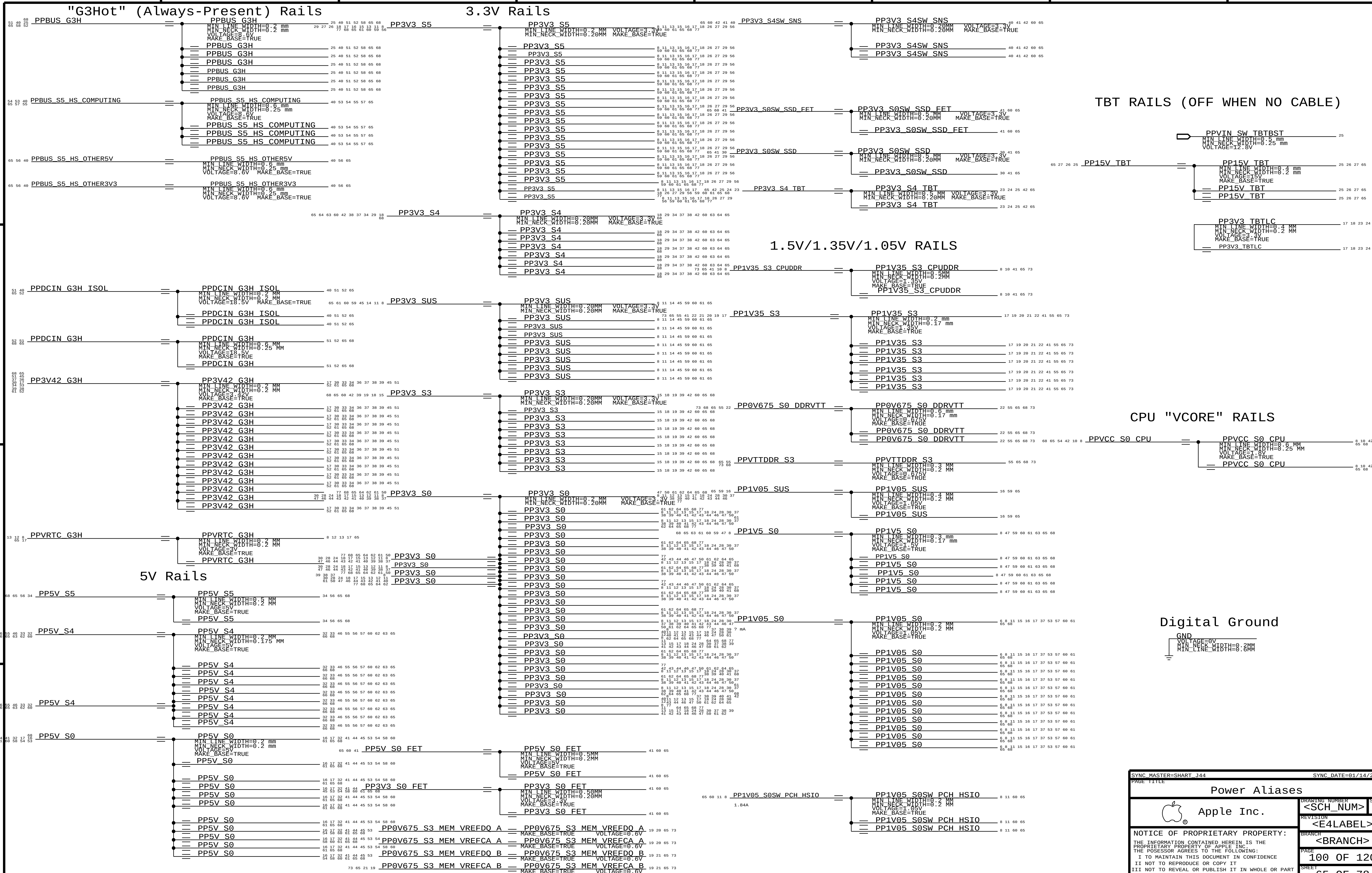

SYNC_MASTER=SHART_J44 SYNC_DATE=01/14/2013

PAGE TITLE

Power Aliases

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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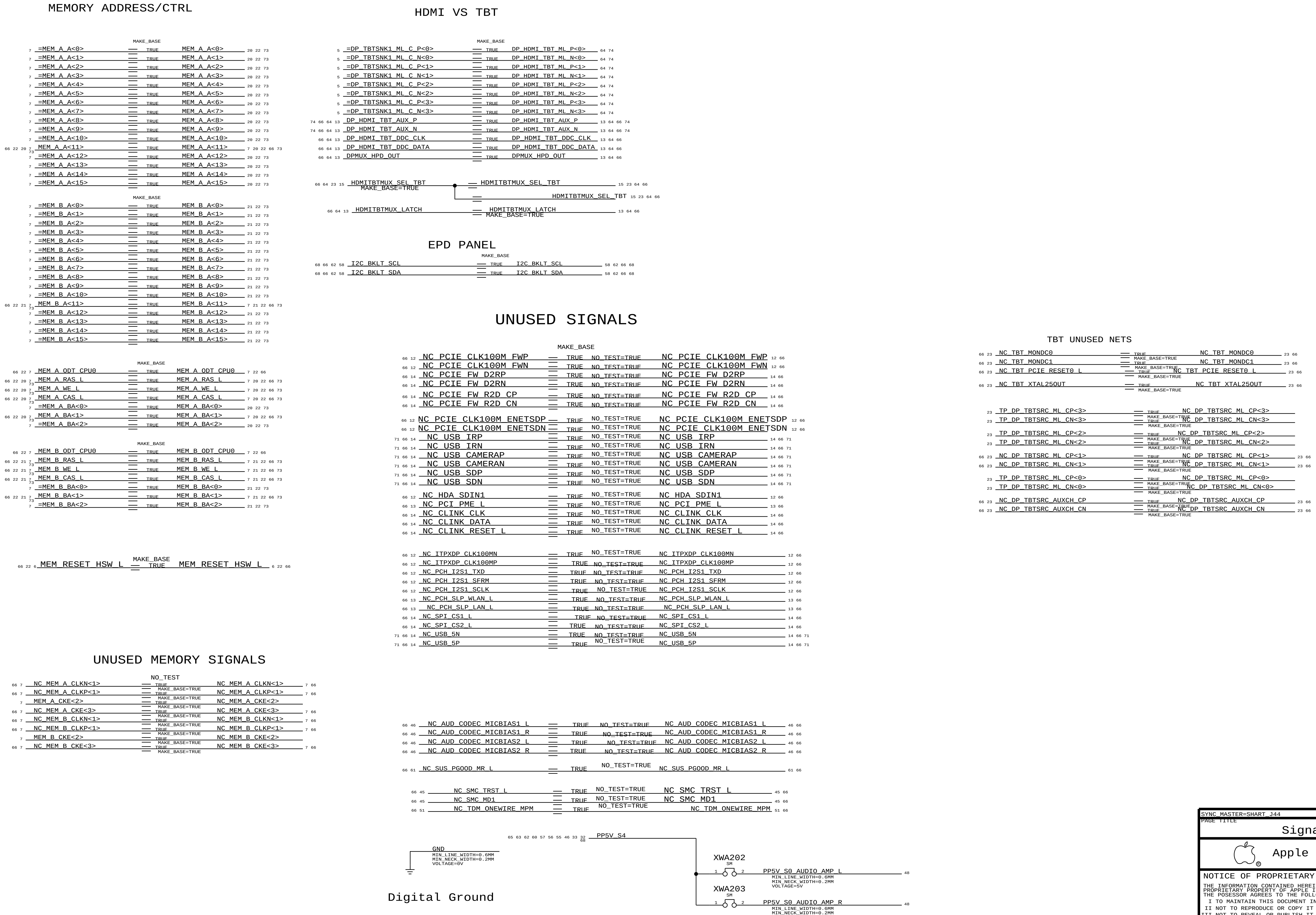
PAGE 100 OF 120

SHEET 65 OF 78



SYNC_MASTER=SHART_J44 SYNC_DATE=11/19/2012

PAGE TITLE: Signal Aliases

Apple Inc.

DRAWING NUMBER: <SCH_NUM> SIZE: D

REVISION: <E4LABEL>

BRANCH: <BRANCH>

Memory Bit/Byte Swizzle

Refs	MAKE_BASE	Net	Alias	Refs
73 68 7	TRUE	MEM_A_DQ<0>	=MEM_A_DQ<24>	20
73 68 7	TRUE	MEM_A_DQ<1>	=MEM_A_DQ<25>	20
73 68 7	TRUE	MEM_A_DQ<2>	=MEM_A_DQ<26>	20
73 68 7	TRUE	MEM_A_DQ<3>	=MEM_A_DQ<27>	20
73 68 7	TRUE	MEM_A_DQ<4>	=MEM_A_DQ<28>	20
73 68 7	TRUE	MEM_A_DQ<5>	=MEM_A_DQ<29>	20
73 68 7	TRUE	MEM_A_DQ<6>	=MEM_A_DQ<30>	20
73 68 7	TRUE	MEM_A_DQ<7>	=MEM_A_DQ<31>	20
73 68 7	TRUE	MEM_A_DQ<8>	=MEM_A_DQ<16>	20
73 68 7	TRUE	MEM_A_DQ<9>	=MEM_A_DQ<17>	20
73 68 7	TRUE	MEM_A_DQ<10>	=MEM_A_DQ<18>	20
73 68 7	TRUE	MEM_A_DQ<11>	=MEM_A_DQ<19>	20
73 68 7	TRUE	MEM_A_DQ<12>	=MEM_A_DQ<20>	20
73 68 7	TRUE	MEM_A_DQ<13>	=MEM_A_DQ<21>	20
73 68 7	TRUE	MEM_A_DQ<14>	=MEM_A_DQ<22>	20
73 68 7	TRUE	MEM_A_DQ<15>	=MEM_A_DQ<23>	20
73 68 7	TRUE	MEM_A_DQ<16>	=MEM_A_DQ<48>	20
73 68 7	TRUE	MEM_A_DQ<17>	=MEM_A_DQ<54>	20
73 68 7	TRUE	MEM_A_DQ<18>	=MEM_A_DQ<55>	20
73 68 7	TRUE	MEM_A_DQ<19>	=MEM_A_DQ<49>	20
73 68 7	TRUE	MEM_A_DQ<20>	=MEM_A_DQ<53>	20
73 68 7	TRUE	MEM_A_DQ<21>	=MEM_A_DQ<51>	20
73 68 7	TRUE	MEM_A_DQ<22>	=MEM_A_DQ<50>	20
73 68 7	TRUE	MEM_A_DQ<23>	=MEM_A_DQ<52>	20
73 68 7	TRUE	MEM_A_DQ<24>	=MEM_A_DQ<46>	20
73 68 7	TRUE	MEM_A_DQ<25>	=MEM_A_DQ<43>	20
73 68 7	TRUE	MEM_A_DQ<26>	=MEM_A_DQ<47>	20
73 68 7	TRUE	MEM_A_DQ<27>	=MEM_A_DQ<45>	20
73 68 7	TRUE	MEM_A_DQ<28>	=MEM_A_DQ<40>	20
73 68 7	TRUE	MEM_A_DQ<29>	=MEM_A_DQ<44>	20
73 68 7	TRUE	MEM_A_DQ<30>	=MEM_A_DQ<42>	20
73 68 7	TRUE	MEM_A_DQ<31>	=MEM_A_DQ<41>	20
73 68 7	TRUE	MEM_A_DQ<32>	=MEM_A_DQ<8>	20
73 68 67 20 7	TRUE	MEM_A_DQ<33>	MEM_A_DQ<33>	7 20 67 68 73
73 68 7	TRUE	MEM_A_DQ<34>	=MEM_A_DQ<10>	20
73 68 7	TRUE	MEM_A_DQ<35>	=MEM_A_DQ<11>	20
73 68 7	TRUE	MEM_A_DQ<36>	=MEM_A_DQ<12>	20
73 68 7	TRUE	MEM_A_DQ<37>	=MEM_A_DQ<13>	20
73 68 7	TRUE	MEM_A_DQ<38>	=MEM_A_DQ<14>	20
73 68 7	TRUE	MEM_A_DQ<39>	=MEM_A_DQ<15>	20
73 68 7	TRUE	MEM_A_DQ<40>	=MEM_A_DQ<0>	20
73 68 7	TRUE	MEM_A_DQ<41>	=MEM_A_DQ<1>	20
73 68 7	TRUE	MEM_A_DQ<42>	=MEM_A_DQ<2>	20
73 68 7	TRUE	MEM_A_DQ<43>	=MEM_A_DQ<3>	20
73 68 7	TRUE	MEM_A_DQ<44>	=MEM_A_DQ<4>	20
73 68 7	TRUE	MEM_A_DQ<45>	=MEM_A_DQ<5>	20
73 68 7	TRUE	MEM_A_DQ<46>	=MEM_A_DQ<6>	20
73 68 7	TRUE	MEM_A_DQ<47>	=MEM_A_DQ<7>	20
73 68 7	TRUE	MEM_A_DQ<48>	=MEM_A_DQ<34>	20
73 68 7	TRUE	MEM_A_DQ<49>	=MEM_A_DQ<39>	20
73 68 7	TRUE	MEM_A_DQ<50>	=MEM_A_DQ<37>	20
73 68 7	TRUE	MEM_A_DQ<51>	=MEM_A_DQ<35>	20
73 68 7	TRUE	MEM_A_DQ<52>	=MEM_A_DQ<36>	20
73 68 7	TRUE	MEM_A_DQ<53>	=MEM_A_DQ<38>	20
73 68 7	TRUE	MEM_A_DQ<54>	=MEM_A_DQ<32>	20
73 68 7	TRUE	MEM_A_DQ<55>	=MEM_A_DQ<33>	20
73 68 7	TRUE	MEM_A_DQ<56>	=MEM_A_DQ<56>	20
73 68 7	TRUE	MEM_A_DQ<57>	=MEM_A_DQ<59>	20
73 68 7	TRUE	MEM_A_DQ<58>	=MEM_A_DQ<62>	20
73 68 7	TRUE	MEM_A_DQ<59>	=MEM_A_DQ<61>	20
73 68 7	TRUE	MEM_A_DQ<60>	=MEM_A_DQ<58>	20
73 68 7	TRUE	MEM_A_DQ<61>	=MEM_A_DQ<57>	20
73 68 7	TRUE	MEM_A_DQ<62>	=MEM_A_DQ<60>	20
73 68 7	TRUE	MEM_A_DQ<63>	=MEM_A_DQ<63>	20

Refs		Net	Alias	Refs
73 7	TRUE	MEM_A_DQS_P<0>	=MEM_A_DQS_P<3>	20
73 7	TRUE	MEM_A_DQS_N<0>	=MEM_A_DQS_N<3>	20
73 7	TRUE	MEM_A_DQS_P<1>	=MEM_A_DQS_P<2>	20
73 7	TRUE	MEM_A_DQS_N<1>	=MEM_A_DQS_N<2>	20
73 7	TRUE	MEM_A_DQS_P<2>	=MEM_A_DQS_P<6>	20
73 7	TRUE	MEM_A_DQS_N<2>	=MEM_A_DQS_N<6>	20
73 7	TRUE	MEM_A_DQS_P<3>	=MEM_A_DQS_P<5>	20
73 7	TRUE	MEM_A_DQS_N<3>	=MEM_A_DQS_N<5>	20
73 7	TRUE	MEM_A_DQS_P<4>	=MEM_A_DQS_P<1>	20
73 7	TRUE	MEM_A_DQS_N<4>	=MEM_A_DQS_N<1>	20
73 7	TRUE	MEM_A_DQS_P<5>	=MEM_A_DQS_P<0>	20
73 7	TRUE	MEM_A_DQS_N<5>	=MEM_A_DQS_N<0>	20
73 67 20 7	TRUE	MEM_A_DQS_P<6>	MEM_A_DQS_P<6>	7 20 67 73
73 67 20 7	TRUE	MEM_A_DQS_N<6>	MEM_A_DQS_N<6>	7 20 67 73
73 7	TRUE	MEM_A_DQS_P<7>	=MEM_A_DQS_P<7>	20
73 7	TRUE	MEM_A_DQS_N<7>	=MEM_A_DQS_N<7>	20

Refs	MAKE_BASE	Net	Alias	Refs
73 68 7	TRUE	MEM_B_DQ<0>	=MEM_B_DQ<0>	21
73 68 7	TRUE	MEM_B_DQ<1>	=MEM_B_DQ<1>	21
73 68 7	TRUE	MEM_B_DQ<2>	=MEM_B_DQ<2>	21
73 68 7	TRUE	MEM_B_DQ<3>	=MEM_B_DQ<3>	21
73 68 7	TRUE	MEM_B_DQ<4>	=MEM_B_DQ<4>	21
73 68 7	TRUE	MEM_B_DQ<5>	=MEM_B_DQ<5>	21
73 68 7	TRUE	MEM_B_DQ<6>	=MEM_B_DQ<6>	21
73 68 7	TRUE	MEM_B_DQ<7>	=MEM_B_DQ<7>	21
73 68 7	TRUE	MEM_B_DQ<8>	=MEM_B_DQ<8>	21
73 68 7	TRUE	MEM_B_DQ<9>	=MEM_B_DQ<9>	21
73 68 7	TRUE	MEM_B_DQ<10>	=MEM_B_DQ<10>	21
73 68 7	TRUE	MEM_B_DQ<11>	=MEM_B_DQ<11>	21
73 68 7	TRUE	MEM_B_DQ<12>	=MEM_B_DQ<12>	21
73 68 7	TRUE	MEM_B_DQ<13>	=MEM_B_DQ<13>	21
73 68 7	TRUE	MEM_B_DQ<14>	=MEM_B_DQ<14>	21
73 68 7	TRUE	MEM_B_DQ<15>	=MEM_B_DQ<15>	21
73 68 7	TRUE	MEM_B_DQ<16>	=MEM_B_DQ<61>	21
73 68 7	TRUE	MEM_B_DQ<17>	=MEM_B_DQ<59>	21
73 68 7	TRUE	MEM_B_DQ<18>	=MEM_B_DQ<62>	21
73 68 7	TRUE	MEM_B_DQ<19>	=MEM_B_DQ<57>	21
73 68 7	TRUE	MEM_B_DQ<20>	=MEM_B_DQ<60>	21
73 68 7	TRUE	MEM_B_DQ<21>	=MEM_B_DQ<63>	21
73 68 7	TRUE	MEM_B_DQ<22>	=MEM_B_DQ<56>	21
73 68 7	TRUE	MEM_B_DQ<23>	=MEM_B_DQ<58>	21
73 68 7	TRUE	MEM_B_DQ<24>	=MEM_B_DQ<54>	21
73 68 7	TRUE	MEM_B_DQ<25>	=MEM_B_DQ<53>	21
73 68 7	TRUE	MEM_B_DQ<26>	=MEM_B_DQ<52>	21
73 68 7	TRUE	MEM_B_DQ<27>	=MEM_B_DQ<55>	21
73 68 7	TRUE	MEM_B_DQ<28>	=MEM_B_DQ<50>	21
73 68 7	TRUE	MEM_B_DQ<29>	=MEM_B_DQ<49>	21
73 68 7	TRUE	MEM_B_DQ<30>	=MEM_B_DQ<48>	21
73 68 7	TRUE	MEM_B_DQ<31>	=MEM_B_DQ<51>	21
73 68 67 21 7	TRUE	MEM_B_DQ<32>	MEM_B_DQ<32>	7 21 67 68 73
73 68 7	TRUE	MEM_B_DQ<33>	=MEM_B_DQ<17>	21
73 68 7	TRUE	MEM_B_DQ<34>	=MEM_B_DQ<18>	21
73 68 7	TRUE	MEM_B_DQ<35>	=MEM_B_DQ<19>	21
73 68 7	TRUE	MEM_B_DQ<36>	=MEM_B_DQ<20>	21
73 68 7	TRUE	MEM_B_DQ<37>	=MEM_B_DQ<21>	21
73 68 7	TRUE	MEM_B_DQ<38>	=MEM_B_DQ<22>	21
73 68 7	TRUE	MEM_B_DQ<39>	=MEM_B_DQ<23>	21
73 68 7	TRUE	MEM_B_DQ<40>	=MEM_B_DQ<24>	21
73 68 7	TRUE	MEM_B_DQ<41>	=MEM_B_DQ<25>	21
73 68 7	TRUE	MEM_B_DQ<42>	=MEM_B_DQ<26>	21
73 68 7	TRUE	MEM_B_DQ<43>	=MEM_B_DQ<27>	21
73 68 7	TRUE	MEM_B_DQ<44>	=MEM_B_DQ<28>	21
73 68 7	TRUE	MEM_B_DQ<45>	=MEM_B_DQ<29>	21
73 68 7	TRUE	MEM_B_DQ<46>	=MEM_B_DQ<30>	21
73 68 7	TRUE	MEM_B_DQ<47>	=MEM_B_DQ<31>	21
73 68 7	TRUE	MEM_B_DQ<48>	=MEM_B_DQ<46>	21
73 68 7	TRUE	MEM_B_DQ<49>	=MEM_B_DQ<45>	21
73 68 7	TRUE	MEM_B_DQ<50>	=MEM_B_DQ<42>	21
73 68 7	TRUE	MEM_B_DQ<51>	=MEM_B_DQ<41>	21
73 68 7	TRUE	MEM_B_DQ<52>	=MEM_B_DQ<43>	21
73 68 7	TRUE	MEM_B_DQ<53>	=MEM_B_DQ<44>	21
73 68 7	TRUE	MEM_B_DQ<54>	=MEM_B_DQ<40>	21
73 68 7	TRUE	MEM_B_DQ<55>	=MEM_B_DQ<47>	21
73 68 7	TRUE	MEM_B_DQ<56>	=MEM_B_DQ<35>	21
73 68 7	TRUE	MEM_B_DQ<57>	=MEM_B_DQ<37>	21
73 68 7	TRUE	MEM_B_DQ<58>	=MEM_B_DQ<32>	21
73 68 7	TRUE	MEM_B_DQ<59>	=MEM_B_DQ<39>	21
73 68 7	TRUE	MEM_B_DQ<60>	=MEM_B_DQ<34>	21
73 68 7	TRUE	MEM_B_DQ<61>	=MEM_B_DQ<33>	21
73 68 7	TRUE	MEM_B_DQ<62>	=MEM_B_DQ<38>	21
73 68 7	TRUE	MEM_B_DQ<63>	=MEM_B_DQ<36>	21

Refs		Net	Alias	Refs
73 7	TRUE	MEM_B_DQS_P<0>	=MEM_B_DQS_P<0>	21
73 7	TRUE	MEM_B_DQS_N<0>	=MEM_B_DQS_N<0>	21
73 7	TRUE	MEM_B_DQS_P<1>	=MEM_B_DQS_P<1>	21
73 7	TRUE	MEM_B_DQS_N<1>	=MEM_B_DQS_N<1>	21
73 7	TRUE	MEM_B_DQS_P<2>	=MEM_B_DQS_P<7>	21
73 7	TRUE	MEM_B_DQS_N<2>	=MEM_B_DQS_N<7>	21
73 7	TRUE	MEM_B_DQS_P<3>	=MEM_B_DQS_P<6>	21
73 7	TRUE	MEM_B_DQS_N<3>	=MEM_B_DQS_N<6>	21
73 7	TRUE	MEM_B_DQS_P<4>	=MEM_B_DQS_P<2>	21
73 7	TRUE	MEM_B_DQS_N<4>	=MEM_B_DQS_N<2>	21
73 7	TRUE	MEM_B_DQS_P<5>	=MEM_B_DQS_P<3>	21
73 7	TRUE	MEM_B_DQS_N<5>	=MEM_B_DQS_N<3>	21
73 67 21 7	TRUE	MEM_B_DQS_P<6>	MEM_B_DQS_P<6>	7 21 67 73
73 67 21 7	TRUE	MEM_B_DQS_N<6>	MEM_B_DQS_N<6>	7 21 67 73
73 7	TRUE	MEM_B_DQS_P<7>	=MEM_B_DQS_P<4>	21
73 7	TRUE	MEM_B_DQS_N<7>	=MEM_B_DQS_N<4>	21

SYNC_MASTER=SHART_J44 SYNC_DATE=11/28/2012

PAGE TITLE: Memory Bit/Byte Swizzle

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

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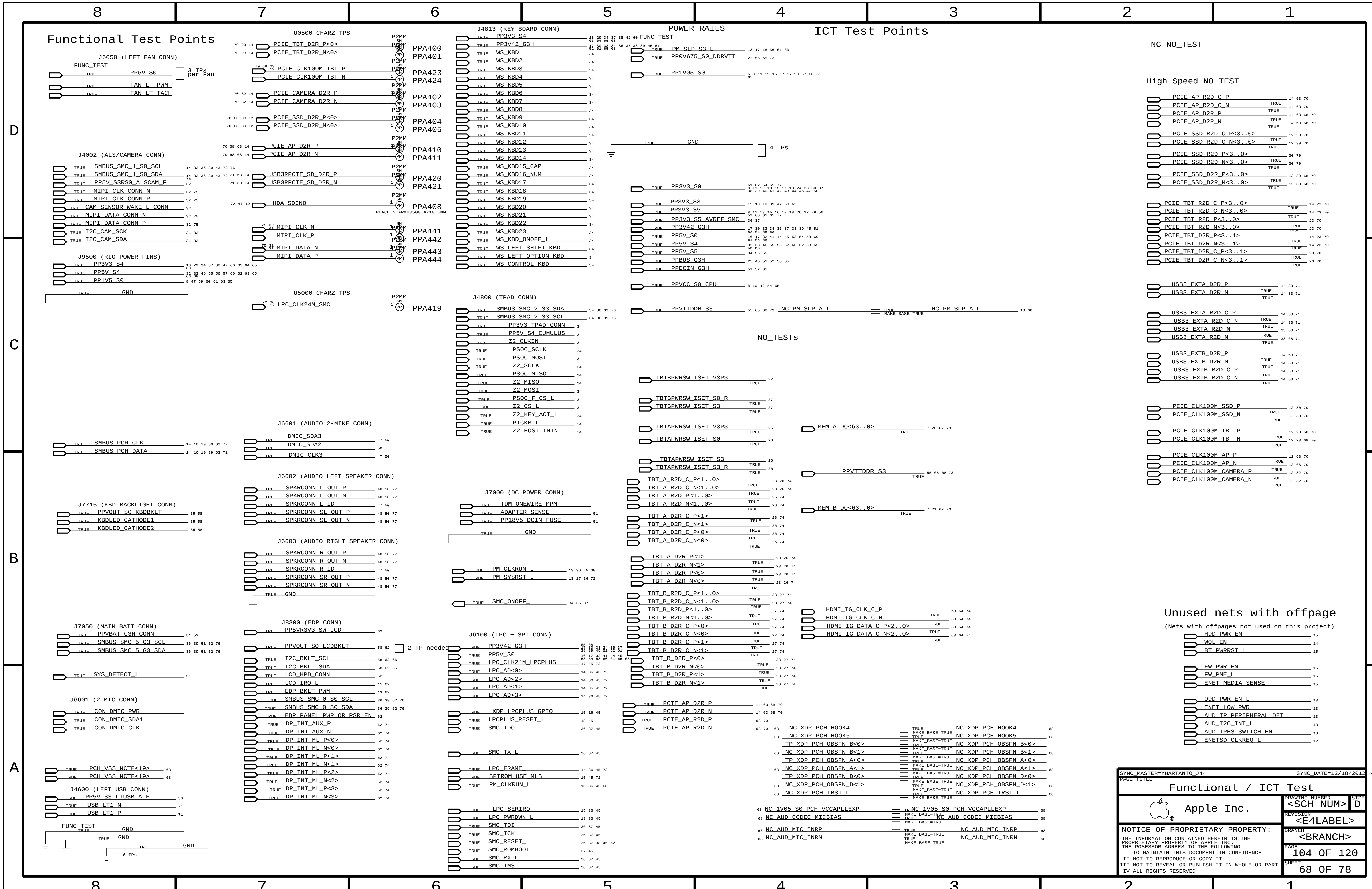

Functional Test Points
J6050 (LEFT FAN CONN)
FUNC_TEST
PP5V_S0
3 TPs per Fan
FAN_LT_PWM
FAN_LT_TACH
J4002 (ALS/CAMERA CONN)
SMBUS_SMC_1_S0_SCL
SMBUS_SMC_1_S0_SDA
PP5V_S3RS0_ALSCAM_F
MIPI_CLK_CONN_N
MIPI_CLK_CONN_P
CAM_SENSOR_WAKE_L_CONN
MIPI_DATA_CONN_N
MIPI_DATA_CONN_P
I2C_CAM_SCK
I2C_CAM_SDA
J9500 (RIO POWER PINS)
PP3V3_S4
PP5V_S4
PP1V5_S0
GND
SMBUS_PCH_CLK
SMBUS_PCH_DATA
J7715 (KBD BACKLIGHT CONN)
PPVOUT_S0_KBDBKLT
KBDLED_CATHODE1
KBDLED_CATHODE2
J7050 (MAIN BATT CONN)
PPVBAT_G3H_CONN
SMBUS_SMC_5_G3_SCL
SMBUS_SMC_5_G3_SDA
SYS_DETECT_L
J6601 (2 MIC CONN)
CON_DMIC_PWR
CON_DMIC_SDA1
CON_DMIC_CLK
PCH_VSS_NCTF<19>
J4600 (LEFT USB CONN)
PP5V_S3_LTUSB_A_F
USB_LT1_N
USB_LT1_P
FUNC_TEST
GND
6 TPs
U0500 CHARZ TPS
PCIE_TBT_D2R_P<0>
PCIE_TBT_D2R_N<0>
PCIE_CLK100M_TBT_P
PCIE_CLK100M_TBT_N
PCIE_CAMERA_D2R_P
PCIE_CAMERA_D2R_N
PCIE_SSD_D2R_P<0>
PCIE_SSD_D2R_N<0>
PCIE_AP_D2R_P
PCIE_AP_D2R_N
USB3RPCIE_SD_D2R_P
USB3RPCIE_SD_D2R_N
HDA_SDIN0
MIPI_CLK_N
MIPI_CLK_P
MIPI_DATA_N
MIPI_DATA_P
P2MM
PPA400
PPA401
PPA423
PPA424
PPA402
PPA403
PPA404
PPA405
PPA410
PPA411
PPA420
PPA421
PPA408
PLACE_NEAR=U0500.AY10:6MM
PPA441
PPA442
PPA443
PPA444
U5000 CHARZ TPS
LPC_CLK24M_SMC
PPA419
J6601 (AUDIO 2-MIKE CONN)
DMIC_SDA3
DMIC_SDA2
DMIC_CLK3
J6602 (AUDIO LEFT SPEAKER CONN)
SPKRCONN_L_OUT_P
SPKRCONN_L_OUT_N
SPKRCONN_L_ID
SPKRCONN_SL_OUT_P
SPKRCONN_SL_OUT_N
J6603 (AUDIO RIGHT SPEAKER CONN)
SPKRCONN_R_OUT_P
SPKRCONN_R_OUT_N
SPKRCONN_R_ID
SPKRCONN_SR_OUT_P
SPKRCONN_SR_OUT_N
GND
J8300 (EDP CONN)
PP5VR3V3_SW_LCD
PPVOUT_S0_LCDBKLT
2 TP needed
I2C_BKLT_SCL
I2C_BKLT_SDA
LCD_HPD_CONN
LCD_IRQ_L
EDP_BKLT_PWM
SMBUS_SMC_0_S0_SCL
SMBUS_SMC_0_S0_SDA
EDP_PANEL_PWR_OR_PSR_EN
DP_INT_AUX_P
DP_INT_AUX_N
DP_INT_ML_P<0>
DP_INT_ML_N<0>
DP_INT_ML_P<1>
DP_INT_ML_N<1>
DP_INT_ML_P<2>
DP_INT_ML_N<2>
DP_INT_ML_P<3>
DP_INT_ML_N<3>
J4813 (KEY BOARD CONN)
PP3V3_S4
PP3V42_G3H
WS_KBD1
WS_KBD2
WS_KBD3
WS_KBD4
WS_KBD5
WS_KBD6
WS_KBD7
WS_KBD8
WS_KBD9
WS_KBD10
WS_KBD11
WS_KBD12
WS_KBD13
WS_KBD14
WS_KBD15_CAP
WS_KBD16_NUM
WS_KBD17
WS_KBD18
WS_KBD19
WS_KBD20
WS_KBD21
WS_KBD22
WS_KBD23
WS_KBD_ONOFF_L
WS_LEFT_SHIFT_KBD
WS_LEFT_OPTION_KBD
WS_CONTROL_KBD
J4800 (TPAD CONN)
SMBUS_SMC_2_S3_SDA
SMBUS_SMC_2_S3_SCL
PP3V3_TPAD_CONN
PP5V_S4_CUMULUS
Z2_CLKIN
PSOC_SCLK
PSOC_MOSI
Z2_SCLK
PSOC_MISO
Z2_MISO
Z2_MOSI
PSOC_F_CS_L
Z2_CS_L
Z2_KEY_ACT_L
PICKB_L
Z2_HOST_INTN
J7000 (DC POWER CONN)
TDM_ONEWIRE_MPM
ADAPTER_SENSE
PP18V5_DCIN_FUSE
GND
PM_CLKRUN_L
PM_SYSRST_L
SMC_ONOFF_L
J6100 (LPC + SPI CONN)
PP3V42_G3H
PP5V_S0
LPC_CLK24M_LPCPLUS
LPC_AD<0>
LPC_AD<2>
LPC_AD<1>
LPC_AD<3>
XDP_LPCPLUS_GPIO
LPCPLUS_RESET_L
SMC_TDO
SMC_TX_L
LPC_FRAME_L
SPIROM_USE_MLB
PM_CLKRUN_L
LPC_SERIRQ
LPC_PWRDWN_L
SMC_TDI
SMC_TCK
SMC_RESET_L
SMC_ROMBOOT
SMC_RX_L
SMC_TMS
POWER RAILS
FUNC_TEST
PM_SLP_S3_L
PP0V675_S0_DDRVTT
PP1V05_S0
GND
4 TPs
PP3V3_S0
PP3V3_S3
PP3V3_S5
PP3V3_S5_AVREF_SMC
PP3V42_G3H
PP5V_S0
PP5V_S4
PP5V_S5
PPBUS_G3H
PPDCIN_G3H
PPVCC_S0_CPU
PPVTTDDR_S3
NC_PM_SLP_A_L
MAKE_BASE=TRUE
ICT Test Points
NO_TESTs
TBTBPWRSW_ISET_V3P3
TBTBPWRSW_ISET_S0_R
TBTBPWRSW_ISET_S3
TBTAPWRSW_ISET_V3P3
TBTAPWRSW_ISET_S0
TBTAPWRSW_ISET_S3
TBTAPWRSW_ISET_S3_R
MEM_A_DQ<63..0>
PPVTTDDR_S3
MEM_B_DQ<63..0>
TBT_A_R2D_C_P<1..0>
TBT_A_R2D_C_N<1..0>
TBT_A_R2D_P<1..0>
TBT_A_R2D_N<1..0>
TBT_A_D2R_C_P<1>
TBT_A_D2R_C_N<1>
TBT_A_D2R_C_P<0>
TBT_A_D2R_C_N<0>
TBT_A_D2R_P<1>
TBT_A_D2R_N<1>
TBT_A_D2R_P<0>
TBT_A_D2R_N<0>
TBT_B_R2D_C_P<1..0>
TBT_B_R2D_C_N<1..0>
TBT_B_R2D_P<1..0>
TBT_B_R2D_N<1..0>
TBT_B_D2R_C_P<0>
TBT_B_D2R_C_N<0>
TBT_B_D2R_C_P<1>
TBT_B_D2R_C_N<1>
TBT_B_D2R_P<0>
TBT_B_D2R_N<0>
TBT_B_D2R_P<1>
TBT_B_D2R_N<1>
HDMI_IG_CLK_C_P
HDMI_IG_CLK_C_N
HDMI_IG_DATA_C_P<2..0>
HDMI_IG_DATA_C_N<2..0>
PCIE_AP_D2R_P
PCIE_AP_D2R_N
PCIE_AP_R2D_P
PCIE_AP_R2D_N
NC_XDP_PCH_HOOK4
NC_XDP_PCH_HOOK5
TP_XDP_PCH_OBSFN_B<0>
NC_XDP_PCH_OBSFN_B<1>
TP_XDP_PCH_OBSFN_A<0>
NC_XDP_PCH_OBSFN_A<1>
TP_XDP_PCH_OBSFN_D<0>
NC_XDP_PCH_OBSFN_D<1>
NC_XDP_PCH_TRST_L
NC_XDP_PCH_OBSFN_B<0>
NC_XDP_PCH_OBSFN_A<0>
NC_XDP_PCH_OBSFN_D<0>
NC_1V05_S0_PCH_VCCAPLLEXP
NC_AUD_CODEC_MICBIAS
NC_AUD_MIC_INRP
NC_AUD_MIC_INRN
NC NO_TEST
High Speed NO_TEST
PCIE_AP_R2D_C_P
PCIE_AP_R2D_C_N
PCIE_AP_D2R_P
PCIE_AP_D2R_N
PCIE_SSD_R2D_C_P<3..0>
PCIE_SSD_R2D_C_N<3..0>
PCIE_SSD_R2D_P<3..0>
PCIE_SSD_R2D_N<3..0>
PCIE_SSD_D2R_P<3..0>
PCIE_SSD_D2R_N<3..0>
PCIE_TBT_R2D_C_P<3..0>
PCIE_TBT_R2D_C_N<3..0>
PCIE_TBT_R2D_P<3..0>
PCIE_TBT_R2D_N<3..0>
PCIE_TBT_D2R_P<3..1>
PCIE_TBT_D2R_N<3..1>
PCIE_TBT_D2R_C_P<3..1>
PCIE_TBT_D2R_C_N<3..1>
USB3_EXTA_D2R_P
USB3_EXTA_D2R_N
USB3_EXTA_R2D_C_P
USB3_EXTA_R2D_C_N
USB3_EXTA_R2D_N
USB3_EXTA_R2D_P
USB3_EXTB_D2R_P
USB3_EXTB_D2R_N
USB3_EXTB_R2D_C_P
USB3_EXTB_R2D_C_N
PCIE_CLK100M_SSD_P
PCIE_CLK100M_SSD_N
PCIE_CLK100M_TBT_P
PCIE_CLK100M_TBT_N
PCIE_CLK100M_AP_P
PCIE_CLK100M_AP_N
PCIE_CLK100M_CAMERA_P
PCIE_CLK100M_CAMERA_N
Unused nets with offpage
(Nets with offpages not used on this project)
HDD_PWR_EN
WOL_EN
BT_PWRRST_L
FW_PWR_EN
FW_PME_L
ENET_MEDIA_SENSE
ODD_PWR_EN_L
ENET_LOW_PWR
AUD_IP_PERIPHERAL_DET
AUD_I2C_INT_L
AUD_IPHS_SWITCH_EN
ENETSD_CLKREQ_L
SYNC_MASTER=YHARTANTO_J44
SYNC_DATE=12/18/2012
PAGE TITLE
Functional / ICT Test
Apple Inc.
DRAWING NUMBER
<SCH_NUM>
SIZE
D
REVISION
<E4LABEL>
BRANCH
<BRANCH>
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J44 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP,ISL2,ISL3,ISL4,ISL5,ISL6,ISL7,ISL8,ISL9,ISL10,ISL11,BOTTOM	NO_TYPE,BGA,P65BGA,BGA_MEM	MM	16.5

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=45_OHM_SE	=45_OHM_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.095 MM	0.095 MM			
50_OHM_SE	*	Y	0.066 MM	0.066 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.116 MM	0.116 MM			
45_OHM_SE	*	Y	0.083 MM	0.083 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.145 MM	0.095 MM			
40_OHM_SE	*	Y	0.102 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHM_SE	TOP,BOTTOM	Y	0.165 MM	0.095 MM			
37_OHM_SE	*	Y	0.118 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.265 MM	0.095 MM			
27P4_OHM_SE	*	Y	0.186 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	ISL2,ISL11	Y	0.105 MM	0.105 MM		0.120 MM	0.120 MM
72_OHM_DIFF	TOP,BOTTOM	Y	0.146 MM	0.146 MM		0.120 MM	0.120 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
80_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	ISL2,ISL11	Y	0.092 MM	0.092 MM		0.120 MM	0.120 MM
80_OHM_DIFF	TOP,BOTTOM	Y	0.125 MM	0.125 MM		0.155 MM	0.155 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	ISL2,ISL11	Y	0.080 MM	0.080 MM		0.120 MM	0.120 MM
85_OHM_DIFF	TOP,BOTTOM	Y	0.105 MM	0.105 MM		0.125 MM	0.125 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHM_DIFF	ISL3,ISL4,ISL9,ISL10	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	ISL2,ISL11	Y	0.078 MM	0.078 MM		0.200 MM	0.200 MM
90_OHM_DIFF	TOP,BOTTOM	Y	0.101 MM	0.101 MM		0.180 MM	0.180 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P65_BGA	*	Y	0.071MM	0.071MM		0.075MM	0.126MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1TO1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	P072_SPACE
*	*	P65BGA	P075_SPACE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
P072_SPACE	*	0.071 MM	?
P075_SPACE	*	0.075 MM	?

Stackup-Defined Spacing Rules

Note: Outer dielectric is 0.058 mm nominal,
Inner dielectric is 0.053 mm nominal.

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.1 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1x_DIELECTRIC	TOP,BOTTOM	0.058 MM	?
1x_DIELECTRIC	ISL3,ISL4,ISL9,ISL10	0.053 MM	?
1X_DIELECTRIC	ISL2,ISL5,ISL6,ISL7,ISL8,ISL11	0.101 MM	?

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	P65BGA	P65_BGA

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=12/14/2012

PAGE TITLE
PCB Rule Definitions

Apple Inc.

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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_VCCSENSE	*	25 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_08MIL	*	0.203 MM	?
CPU_12MIL	*	0.305 MM	?
CPU_18MIL	*	0.457 MM	?
CPU_25MIL	*	0.635 MM	?

PCI Express Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
CLK_PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_2SAME	*	=3X_DIELECTRIC	?
PCIE_TXRX	*	=6X_DIELECTRIC	?
PCIE_2OTHER	*	=4X_DIELECTRIC	?
PCIE_2CLK	*	=7X_DIELECTRIC	?
PCIECLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
PCIE_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIE_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
PCIE_2CLK	TOP,BOTTOM	=10X_DIELECTRIC	?
PCIECLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_*	*	*	PCIE_2OTHER
PCIE_*	=SAME	*	PCIE_2SAME
PCIE_*	CLK_*	*	PCIE_2CLK
CLK_PCIE	*	*	PCIECLK_2OTHER
PCIE_TX	*_RX	*	PCIE_TXRX
PCIE_RX	*_TX	*	PCIE_TXRX

CPU Signal Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING		
XDP_TCK0	CPU_45S	CPU_18MIL	XDP_CPU_TCK	6 16
XDP_TCK0	CPU_45S	CPU_18MIL	PCH_JTAGX	12 16
XDP_TCK1	CPU_45S	CPU_18MIL	XDP_PCH_TCK	12 16
XDP_TDO	CPU_45S		XDP_CPU_TDO	6 16
XDP_TDO	CPU_45S		XDP_PCH_TDO	12 16
XDP_TDI	CPU_45S		XDP_CPU_TDI	6 16
XDP_TDI	CPU_45S		XDP_PCH_TDI	12 16
XDP_TMS	CPU_45S		XDP_CPU_TMS	6 16
XDP_TMS	CPU_45S		XDP_PCH_TMS	12 16
XDP_TRST_L	CPU_45S		XDP_TRST_L	16
XDP_TRST_L	CPU_45S		XDP_CPUPCH_TRST_L	6 12 16
XDP_PRDY_L	CPU_45S		XDP_CPU_PRDY_L	6 16
XDP_PREQ_L	CPU_45S		XDP_CPU_PREQ_L	6 16
CPU_VCCST_PWRGD	CPU_45S	CPU_08MIL	CPU_VCCST_PWRGD	8 16 17
CPU_VCCST_PWRGD	CPU_45S	CPU_08MIL	XDP_CPU_VCCST_PWRGD	16
CPU_BPM	CPU_45S	CPU_08MIL	XDP_BPM_L<1..0>	6 16
CPU_BPM_TP	CPU_45S		XDP_BPM_L<7..2>	6 16
CPU_RCOMP_SM	CPU_27P4S	CPU_25MIL	CPU_SM_RCOMP<2..0>	6
CPU_RCOMP_EDP	CPU_27P4S	CPU_25MIL	MCP_EDP_RCOMP	5
CPU_RCOMP_OPI	CPU_27P4S	CPU_12MIL	CPU_OPI_RCOMP	6
CPU_PROCHOT	CPU_45S	CPU_08MIL	CPU_PROCHOT_L	6 36 37 53
CPU_PROCHOT	CPU_45S	CPU_08MIL	CPU_PROCHOT_R_L	6
CPU_CATERR	CPU_45S	CPU_08MIL	CPU_CATERR_L	6 36
CPU_VIDALERT	CPU_45S	CPU_18MIL	CPU_VIDALERT_L	8 53
CPU_VIDALERT	CPU_45S	CPU_18MIL	CPU_VIDALERT_R_L	8
CPU_VIDSCLK	CPU_45S	CPU_18MIL	CPU_VIDSCLK	8 53
CPU_VIDSCLK	CPU_45S	CPU_18MIL	CPU_VIDSCLK_R	8
CPU_VIDSOUT	CPU_45S	CPU_18MIL	CPU_VIDSOUT	8 53
CPU_VIDSOUT	CPU_45S	CPU_18MIL	CPU_VIDSOUT_R	8
CPU_PECI	CPU_45S	CPU_18MIL	CPU_PECI	6 37
CPU_PECI	CPU_45S	CPU_18MIL	CPU_PECI_R	36 37
CPU_PECI_SMC	CPU_45S	CPU_18MIL	SMC_PECI_L	36 37
CPU_PECI_SMC	CPU_45S	CPU_18MIL	SMC_PECI_L_R	37
CPU_CFG	CPU_45S		CPU_CFG<19..11>	6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<10..8>	6 16
CPU_CFG	CPU_45S		CPU_CFG<7..5>	6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<4>	6 16
CPU_CFG_3	CPU_45S		CPU_CFG<3>	6 16
CPU_CFG	CPU_45S		CPU_CFG<2>	6 16
CPU_CFG_PD	CPU_45S		CPU_CFG<1..0>	6 16
CPU_MEM_RESET	CPU_45S	CPU_08MIL	MEM_RESET_L	20 21 22
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_P	8 53
CPU_VCCSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCCSENSE_N	9 53

PCI Express Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING		
PCIE_SSD_D2R	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_P<3..1>	12 30 68
PCIE_SSD_D2R	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_N<3..1>	12 30 68
PCIE_SSD_D2R_PP	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_P<0>	12 30 68
PCIE_SSD_D2R_PP	PCIE_85D	PCIE_RX	PCIE_SSD_D2R_N<0>	12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_C_P<3..0>	12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_C_N<3..0>	12 30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_P<3..0>	30 68
PCIE_SSD_R2D	PCIE_85D	PCIE_TX	PCIE_SSD_R2D_N<3..0>	30 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_P<0>	14 23 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_N<0>	14 23 68
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_P<0>	23
PCIE_TBT_D2R_0	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_N<0>	23
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_P<3..1>	14 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_N<3..1>	14 23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_P<3..1>	23 68
PCIE_TBT_D2R	PCIE_85D	PCIE_RX	PCIE_TBT_D2R_C_N<3..1>	23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_P<3..0>	23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_N<3..0>	23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_C_P<3..0>	14 23 68
PCIE_TBT_R2D	PCIE_85D	PCIE_TX	PCIE_TBT_R2D_C_N<3..0>	14 23 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_P	63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_N	63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_C_P	14 63 68
PCIE_AP_R2D	PCIE_85D	PCIE_TX	PCIE_AP_R2D_C_N	14 63 68
PCIE_AP_D2R	PCIE_85D	PCIE_RX	PCIE_AP_D2R_P	14 63 68
PCIE_AP_D2R	PCIE_85D	PCIE_RX	PCIE_AP_D2R_N	14 63 68
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_P	63
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_CONN_N	63
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_P	12 63 68
PCIE_CLK100M_AP	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_AP_N	12 63 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_P	12 32 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_N	12 32 68
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P	31 32
PCIE_CLK100M_CAM	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N	31 32
PCIE_CLK100M_SSD	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_P	12 30 68
PCIE_CLK100M_SSD	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_SSD_N	12 30 68
PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_P	12 23 68
PCIE_CLK100M_TBT	CLK_PCIE_85D	CLK_PCIE	PCIE_CLK100M_TBT_N	12 23 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_P	14 32 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_N	14 32 68
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_C_P	31 32
PCIE_CAMERA_D2R	PCIE_85D	PCIE_RX	PCIE_CAMERA_D2R_C_N	31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_P	31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_N	31 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_C_P	14 32
PCIE_CAMERA_R2D	PCIE_85D	PCIE_TX	PCIE_CAMERA_R2D_C_N	14 32

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PAGE TITLE
CPU & PCIe Constraints

Apple Inc.

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USB 2 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4X_DIELECTRIC	?
USB_RBIAS	*	=6X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	TOP,BOTTOM	=6X_DIELECTRIC	?
USB_RBIAS	TOP,BOTTOM	=10X_DIELECTRIC	?

USB 3 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	*	=3X_DIELECTRIC	?
USB3_TXRX	*	=6X_DIELECTRIC	?
USB3_2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
USB3_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
USB3_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_*	*	*	USB3_2OTHER
USB3_*	=SAME	*	USB3_2SAME
USB3_TX	*_RX	*	USB3_TXRX
USB3_RX	*_TX	*	USB3_TXRX

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_25M	*	=5X_DIELECTRIC	?

SATA Interface Constraints (Not Used)

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	*	=3X_DIELECTRIC	?
SATA_TXRX	*	=6X_DIELECTRIC	?
SATA_2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
SATA_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
SATA_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA_*	*	*	SATA_2OTHER
SATA_*	=SAME	*	SATA_2SAME
SATA_TX	*_RX	*	SATA_TXRX
SATA_RX	*_TX	*	SATA_TXRX

USB Constraints

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
USB_BT	USB_85D	USB	USB_BT_P	14 29
USB_BT	USB_85D	USB	USB_BT_N	14 29
USB_BT	USB_85D	USB	USB_BT_CONN_P	29 63
USB_BT	USB_85D	USB	USB_BT_CONN_N	29 63
USB_EXTA	USB_85D	USB	USB_EXTA_P	14 33
USB_EXTA	USB_85D	USB	USB_EXTA_N	14 33
	DEFAULT	DEFAULT	SMC_DEBUGPRT_RX_L	33 36 37
	DEFAULT	DEFAULT	SMC_DEBUGPRT_TX_L	33 36 37
USB_EXTA	USB_85D	USB	USB2_EXTA_MUXED_P	33
USB_EXTA	USB_85D	USB	USB2_EXTA_MUXED_N	33
USB_EXTA	USB_85D	USB	USB2_EXTA_MUXED_F_P	33
USB_EXTA	USB_85D	USB	USB2_EXTA_MUXED_F_N	33
USB_EXTA	USB_85D	USB	USB_LT1_P	68
USB_EXTA	USB_85D	USB	USB_LT1_N	68
USB_EXTB	USB_85D	USB	USB_EXTB_P	14 63
USB_EXTB	USB_85D	USB	USB_EXTB_N	14 63
USB_TPAD	USB_85D	USB	USB_TPAD_P	14 34
USB_TPAD	USB_85D	USB	USB_TPAD_N	14 34
I330 USB_TPAD	USB_85D	USB	USB_TPAD_R_P	34
I331 USB_TPAD	USB_85D	USB	USB_TPAD_R_N	34
USB3_EXTA_D2R	USB_85D	USB3_RX	USB3_EXTA_D2R_P	14 33 68
USB3_EXTA_D2R	USB_85D	USB3_RX	USB3_EXTA_D2R_N	14 33 68
USB3_EXTA_R2D	USB_85D	USB3_TX	USB3_EXTA_R2D_P	33
USB3_EXTA_R2D	USB_85D	USB3_TX	USB3_EXTA_R2D_N	33 68
USB3_EXTA_R2D	USB_85D	USB3_TX	USB3_EXTA_R2D_C_P	14 33 68
USB3_EXTA_R2D	USB_85D	USB3_TX	USB3_EXTA_R2D_C_N	14 33 68
USB3_EXTB_D2R	USB_85D	USB3_RX	USB3_EXTB_D2R_P	14 63 68
USB3_EXTB_D2R	USB_85D	USB3_RX	USB3_EXTB_D2R_N	14 63 68
USB3_EXTB_R2D	USB_85D	USB3_TX	USB3_EXTB_R2D_C_P	14 63 68
USB3_EXTB_R2D	USB_85D	USB3_TX	USB3_EXTB_R2D_C_N	14 63 68
USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD_D2R_P	14 63 68
USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD_D2R_N	14 63 68
USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD_R2D_C_P	14 63
USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD_R2D_C_N	14 63
USB_NC	USB_85D	USB	NC_USB_IRP	14 66
USB_NC	USB_85D	USB	NC_USB_IRN	14 66
USB_NC	USB_85D	USB	NC_USB_5P	14 66
USB_NC	USB_85D	USB	NC_USB_5N	14 66
USB_NC	USB_85D	USB	NC_USB_SDP	14 66
USB_NC	USB_85D	USB	NC_USB_SDN	14 66
USB_NC	USB_85D	USB	NC_USB_CAMERAP	14 66
USB_NC	USB_85D	USB	NC_USB_CAMERAN	14 66
PCH_USB_RBIAS	PCH_USB_RBIAS	USB_RBIAS	PCH_USB_RBIAS	14
	SATA_85D	SATA_RX	DUMMY_SATA_D2R_P	
	SATA_85D	SATA_RX	DUMMY_SATA_D2R_N	
	SATA_85D	SATA_TX	DUMMY_SATA_R2D_P	
	SATA_85D	SATA_TX	DUMMY_SATA_R2D_N	
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X1	17
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2	17
SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2_R	17
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA	17 32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKP	31 32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP_R	32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP	32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALN	32
SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKN	31 32
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	17 23
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	23

Notes:
This is here to keep the SATA rules.

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=01/07/2013
PAGE TITLE
USB Constraints
Apple Inc.
DRAWING NUMBER <SCH_NUM> SIZE D
REVISION <E4LABEL>
BRANCH <BRANCH>

LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Single Net Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
PCH_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_12MIL	*	0.305 MM	?
PCH_15MIL	*	0.381 MM	?
PCH_18MIL	*	0.457 MM	?
PCH_20MIL	*	0.508 MM	?

PCH Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	
LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	14 36 45 68
LPC_AD	LPC_45S	LPC	LPC_FRAME_L	14 36 45 68
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC_R	12 17
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC	17 36 68
LPC_CLK24M_LPCPLUS	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS	17 45 68
LPC_CLK24M_LPCPLUS	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCPLUS_R	12 17
SMBUS_PCH	SMB_45S	SMB	SMBUS_PCH_CLK	14 16 19 39 63 68
SMBUS_PCH	SMB_45S	SMB	SMBUS_PCH_DATA	14 16 19 39 63 68
SML_PCH_0	SMB_45S	SMB	SML_PCH_0_CLK	14 39
SML_PCH_0	SMB_45S	SMB	SML_PCH_0_DATA	14 39
	SMB_45S	SMB	SMBUS_SMC_1_S0_SCL	14 32 36 39 43 68 76
	SMB_45S	SMB	SMBUS_SMC_1_S0_SDA	14 32 36 39 43 68 76
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	12 47
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK_R	12
HDA_SYNC	HDA_45S	HDA	HDA_SYNC	12 47
HDA_SYNC	HDA_45S	HDA	HDA_SYNC_R	12
HDA_RST	HDA_45S	HDA	HDA_RST_R_L	12
HDA_RST	HDA_45S	HDA	HDA_RST_L	12 47
HDA_SDIN	HDA_45S	HDA	HDA_SDIN0	12 47 68
HDA_SDIN	HDA_45S	HDA	CS4208_HDA_SDOUT0_R	47
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	12 47
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT_R	12 17
SPI_MLB	SPI_45S	SPI	SPI_ALT_CLK	45
SPI_MLB	SPI_45S	SPI	SPI_CLK	45
SPI_MLB	SPI_45S	SPI	SPI_CLK_R	14 45
SPI_MLB	SPI_45S	SPI	SPI_MLB_CLK	45
SPI_MLB	SPI_45S	SPI	SPI_SMC_CLK	36 45
SPI_MLB	SPI_45S	SPI	SPI_ALT_CS_L	45
SPI_MLB	SPI_45S	SPI	SPI_CS0_L	45
SPI_MLB	SPI_45S	SPI	SPI_CS0_R_L	14 45
SPI_MLB	SPI_45S	SPI	SPI_MLB_CS_L	45
SPI_MLB	SPI_45S	SPI	SPI_SMC_CS_L	36 45
SPI_MLB	SPI_45S	SPI	SPI_ALT_MISO	45
SPI_MLB	SPI_45S	SPI	SPI_MISO	14 45
SPI_MLB	SPI_45S	SPI	SPI_MISO_R	45
SPI_MLB	SPI_45S	SPI	SPI_MLB_MISO	45
SPI_MLB	SPI_45S	SPI	SPI_SMC_MISO	36 45
SPI_MLB	SPI_45S	SPI	SPI_ALT_MOSI	45
SPI_MLB	SPI_45S	SPI	SPI_MOSI	45
SPI_MLB	SPI_45S	SPI	SPI_MOSI_R	14 45
SPI_MLB	SPI_45S	SPI	SPI_MLB_MOSI	45
SPI_MLB	SPI_45S	SPI	SPI_SMC_MOSI	36 45
SPI_MLB_IO2	SPI_45S	SPI	SPI_IO<2>	14 45
SPI_MLB_IO2	SPI_45S	SPI	SPIROM_WP_L	45
SPI_MLB_IO3	SPI_45S	SPI	SPI_IO<3>	14 45
SPI_MLB_IO3	SPI_45S	SPI	SPIROM_HOLD_L	45
SPI_MLB_IO3	SPI_45S	SPI	SPIROM_USE_MLB	15 45 68
PCH_RTCX	PCH_45S	PCH_15MIL	PCH_CLK32K_RTCX1	12 17
PCH_SRTCRST	PCH_45S	PCH_15MIL	PCH_SRTCRST_L	12
PCH_RTCRST	PCH_45S	PCH_15MIL	RTC_RESET_L	12
PCH_THRMTRIP	PCH_45S	PCH_18MIL	PM_THRMTRIP_L	15 37
PCH_THRMTRIP	PCH_45S	PCH_18MIL	PM_THRMTRIP_R_L	37
	PCH_45S	PCH_15MIL	PCH_INTRUDER_L	12
	PCH_45S	PCH_15MIL	PCH_INTVRMEN	12
	PCH_45S	PCH_15MIL	PCH_DSWVRMEN	13
	PCH_45S	PCH_15MIL	PM_RSMRST_L	13 61
	PCH_45S	PCH_15MIL	PM_SYSRST_L	13 17 36 68
	PCH_45S	PCH_15MIL	XDP_DBRESET_L	16 17
	PCH_45S	PCH_15MIL	PM_PCH_SYS_PWROK	13 16 17 36
	PCH_45S	PCH_15MIL	XDP_SYS_PWROK	16
	PCH_45S	PCH_15MIL	SYS_PWROK_R	17
	PCH_45S	PCH_15MIL	PM_PCH_PWROK	13 17
	PCH_45S	PCH_15MIL	PM_S0_PGOOD	17
	PCH_45S	PCH_15MIL	SMC_DELAYED_PWRGD	17 24 25 36 37
	PCH_45S	PCH_15MIL	PM_DSW_PWRGD	13 36
	PCH_45S	PCH_15MIL	PM_PWRBTN_L	13 16 36
	PCH_45S	PCH_15MIL	XDP_CPU_PWRBTN_L	16
	PCH_45S	PCH_15MIL	PCIE_WAKE_L	13 29 31
	PCH_45S	PCH_15MIL	AP_PCIE_WAKE_L	29 63
	PCH_45S	PCH_15MIL	CAM_PCIE_WAKE_L	31
	PCH_45S	PCH_15MIL	TBT_CIO_PLUG_EVENT_L	15 18 23
PCH_CLK24M_XTAL	PCH_45S	PCH_20MIL	PCH_CLK24M_XTALIN	12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MIL	PCH_CLK24M_XTALOUT	12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MIL	PCH_CLK24M_XTALOUT_R	17
PCH_RCOMP_PCIE	PCH_27P4S	PCH_12MIL	PCH_PCIE_RCOMP	14
PCH_RCOMP_OPI	PCH_27P4S	PCH_12MIL	PCH_OPI_COMP	15
PCH_RCOMP_SATA	PCH_27P4S	PCH_12MIL	PCH_SATA_RCOMP	12

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=01/08/2013

PAGE TITLE

PCH Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

8	7	6	5	4	3	2	1

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=STANDARD	=STANDARD
MEM_72D	*	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF	=72_OHM_DIFF
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
MEM_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=2x_DIELECTRIC	?
MEM_CMD2CMD	*	=2x_DIELECTRIC	?
MEM_CMD2CTL	*	=2x_DIELECTRIC	?
MEM_CTL2CTL	*	=2x_DIELECTRIC	?
MEM_CLK2CLK	*	=4x_DIELECTRIC	?
MEM_2OTHERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	?
MEM_2GND	*	=2x_DIELECTRIC	?
MEM_2OTHER	*	=6x_DIELECTRIC	?
MEM_CMD2CMD_BM	*	=2x_DIELECTRIC	?
MEM_CMD2CTL_BM	*	=2x_DIELECTRIC	?
MEM_CTL2CTL_BM	*	=2x_DIELECTRIC	?
MEM_12MIL	*	0.305 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CTL2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2OTHERMEM	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_CMD2CMD_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CMD2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CTL2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	*	*	MEM_2OTHER
MEM_*_DQS_*	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_CTL	*	*	MEM_2OTHER
MEM_CLK	*	*	MEM_2OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTL	*	MEM_CMD2CTL
MEM_CTL	MEM_CTL	*	MEM_CTL2CTL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK
MEM_*	MEM_*	*	MEM_2OTHERMEM
MEM_CMD	MEM_CMD	BGA_MEM	MEM_CMD2CMD_BM
MEM_CMD	MEM_CTL	BGA_MEM	MEM_CMD2CTL_BM
MEM_CTL	MEM_CTL	BGA_MEM	MEM_CTL2CTL_BM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DQBYTE_0	*	MEM_DQS2OWNDATA
MEM_A_DQS_1	MEM_A_DQBYTE_1	*	MEM_DQS2OWNDATA
MEM_A_DQS_2	MEM_A_DQBYTE_2	*	MEM_DQS2OWNDATA
MEM_A_DQS_3	MEM_A_DQBYTE_3	*	MEM_DQS2OWNDATA
MEM_A_DQS_4	MEM_A_DQBYTE_4	*	MEM_DQS2OWNDATA
MEM_A_DQS_5	MEM_A_DQBYTE_5	*	MEM_DQS2OWNDATA
MEM_A_DQS_6	MEM_A_DQBYTE_6	*	MEM_DQS2OWNDATA
MEM_A_DQS_7	MEM_A_DQBYTE_7	*	MEM_DQS2OWNDATA
MEM_B_DQS_0	MEM_B_DQBYTE_0	*	MEM_DQS2OWNDATA
MEM_B_DQS_1	MEM_B_DQBYTE_1	*	MEM_DQS2OWNDATA
MEM_B_DQS_2	MEM_B_DQBYTE_2	*	MEM_DQS2OWNDATA
MEM_B_DQS_3	MEM_B_DQBYTE_3	*	MEM_DQS2OWNDATA
MEM_B_DQS_4	MEM_B_DQBYTE_4	*	MEM_DQS2OWNDATA
MEM_B_DQS_5	MEM_B_DQBYTE_5	*	MEM_DQS2OWNDATA
MEM_B_DQS_6	MEM_B_DQBYTE_6	*	MEM_DQS2OWNDATA
MEM_B_DQS_7	MEM_B_DQBYTE_7	*	MEM_DQS2OWNDATA

Haswell ULT Memory Down DDR3L 1x8 Length Matching

DDR3 Signal Group	Unit	Min Length	Max Length
CTLmax - CTLmin	mils	0	100
CTL to CLK	mils	CLK - 500	CLK + 500
CMDi to CMDj	mils	CMDj - 100	CMDj + 100
CMD to CLK	mils	CLK - 500	CLK + 500
(DQmax - DQmin) per byte	mils	0	250
(DQS - DQmax) per byte	mils	-100	150
DQS to DQS#	mils	-5	5
DQS to CLK (Rule 1)	mils	CLK - 6500	CLK + 500
Max(CLK-DQS) - Min(CLK-DQS)	mils	0	5500
CLK to CLK#	mils	-5	5

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Net Properties

ELECTRICAL CONST SET	NET TYPE: PHYSICAL	NET TYPE: SPACING	NET	PAGES
MEM_A_CLK	MEM_72D	MEM_CLK	MEM_A_CLK_P<0>	7 20 22
MEM_A_CLK	MEM_72D	MEM_CLK	MEM_A_CLK_N<0>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM_A_CKE<0>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM_A_CKE<1>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM_A_CS_L<0>	7 20 22
MEM_A_CTL	MEM_40S	MEM_CTL	MEM_A_CS_L<1>	7 20 22
MEM_A_ODT0	MEM_40S	MEM_CTL	MEM_A_ODT<0>	20 22
	MEM_40S	MEM_CTL	MEM_A_ODT<1>	20 22
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_A<15..0>	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_BA<2..0>	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_RAS_L	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_CAS_L	7 20 22 66
MEM_A_CMD	MEM_40S	MEM_CMD	MEM_A_WE_L	7 20 22 66
MEM_A_DQBYTE0	MEM_45S	MEM_A_DQBYTE_0	MEM_A_DQ<7..0>	7 67 68
MEM_A_DQBYTE1	MEM_45S	MEM_A_DQBYTE_1	MEM_A_DQ<15..8>	7 67 68
MEM_A_DQBYTE2	MEM_45S	MEM_A_DQBYTE_2	MEM_A_DQ<23..16>	7 67 68
MEM_A_DQBYTE3	MEM_45S	MEM_A_DQBYTE_3	MEM_A_DQ<31..24>	7 67 68
MEM_A_DQBYTE4	MEM_45S	MEM_A_DQBYTE_4	MEM_A_DQ<39..32>	7 20 67 68
MEM_A_DQBYTE5	MEM_45S	MEM_A_DQBYTE_5	MEM_A_DQ<47..40>	7 67 68
MEM_A_DQBYTE6	MEM_45S	MEM_A_DQBYTE_6	MEM_A_DQ<55..48>	7 67 68
MEM_A_DQBYTE7	MEM_45S	MEM_A_DQBYTE_7	MEM_A_DQ<63..56>	7 67 68
MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM_A_DQS_P<0>	7 67
MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM_A_DQS_N<0>	7 67
MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM_A_DQS_P<1>	7 67
MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM_A_DQS_N<1>	7 67
MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM_A_DQS_P<2>	7 67
MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM_A_DQS_N<2>	7 67
MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM_A_DQS_P<3>	7 67
MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM_A_DQS_N<3>	7 67
MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM_A_DQS_P<4>	7 67
MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM_A_DQS_N<4>	7 67
MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM_A_DQS_P<5>	7 67
MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM_A_DQS_N<5>	7 67
MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM_A_DQS_P<6>	7 20 67
MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM_A_DQS_N<6>	7 20 67
MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM_A_DQS_P<7>	7 67
MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM_A_DQS_N<7>	7 67
MEM_B_CLK	MEM_72D	MEM_CLK	MEM_B_CLK_P<0>	7 21 22
MEM_B_CLK	MEM_72D	MEM_CLK	MEM_B_CLK_N<0>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM_B_CKE<0>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM_B_CKE<1>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM_B_CS_L<0>	7 21 22
MEM_B_CTL	MEM_40S	MEM_CTL	MEM_B_CS_L<1>	7 21 22
MEM_B_ODT0	MEM_40S	MEM_CTL	MEM_B_ODT<0>	21 22
	MEM_40S	MEM_CTL	MEM_B_ODT<1>	21 22
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_A<15..0>	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_BA<2..0>	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_RAS_L	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_CAS_L	7 21 22 66
MEM_B_CMD	MEM_40S	MEM_CMD	MEM_B_WE_L	7 21 22 66
MEM_B_DQBYTE0	MEM_45S	MEM_B_DQBYTE_0	MEM_B_DQ<7..0>	7 67 68
MEM_B_DQBYTE1	MEM_45S	MEM_B_DQBYTE_1	MEM_B_DQ<15..8>	7 67 68
MEM_B_DQBYTE2	MEM_45S	MEM_B_DQBYTE_2	MEM_B_DQ<23..16>	7 67 68
MEM_B_DQBYTE3	MEM_45S	MEM_B_DQBYTE_3	MEM_B_DQ<31..24>	7 67 68
MEM_B_DQBYTE4	MEM_45S	MEM_B_DQBYTE_4	MEM_B_DQ<39..32>	7 21 67 68
MEM_B_DQBYTE5	MEM_45S	MEM_B_DQBYTE_5	MEM_B_DQ<47..40>	7 67 68
MEM_B_DQBYTE6	MEM_45S	MEM_B_DQBYTE_6	MEM_B_DQ<55..48>	7 67 68
MEM_B_DQBYTE7	MEM_45S	MEM_B_DQBYTE_7	MEM_B_DQ<63..56>	7 67 68
MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM_B_DQS_P<0>	7 67
MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM_B_DQS_N<0>	7 67
MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM_B_DQS_P<1>	7 67
MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM_B_DQS_N<1>	7 67
MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM_B_DQS_P<2>	7 67
MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM_B_DQS_N<2>	7 67
MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM_B_DQS_P<3>	7 67
MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM_B_DQS_N<3>	7 67
MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM_B_DQS_P<4>	7 67
MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM_B_DQS_N<4>	7 67
MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM_B_DQS_P<5>	7 67
MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM_B_DQS_N<5>	7 67
MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM_B_DQS_P<6>	7 21 67
MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM_B_DQS_N<6>	7 21 67
MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM_B_DQS_P<7>	7 67
MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM_B_DQS_N<7>	7 67
		MEM_PWR	PP1V35_S3	17 19 20 21 22 41 55 65
		MEM_PWR	PP1V35_S3_CPUDDR	8 10 41 65
		MEM_PWR	PP0V675_S0_DDRVTT	22 55 65 68
		MEM_PWR	PPVTTDDR_S3	55 65 68
		MEM_12MIL	CPU_DIMMA_VREFDQ	7 19
		MEM_12MIL	CPU_DIMMA_VREFDQ_A_ISOL	19
		MEM_12MIL	CPU_DIMMB_VREFDQ	7 19
		MEM_12MIL	CPU_DIMMB_VREFDQ_B_ISOL	19
		MEM_12MIL	CPU_DIMM_VREFCA	7 19
		MEM_12MIL	CPU_DIMM_VREFCA_A_ISOL	19
I146		MEM_12MIL	CPU_DIMM_VREFCA_B_ISOL	19
		MEM_12MIL	PP0V675_S3_MEM_VREFDQ_A	19 20 65
		MEM_12MIL	PP0V675_S3_MEM_VREFDQ_B	19 21 65
		MEM_12MIL	PP0V675_S3_MEM_VREFCA_A	19 20 65
		MEM_12MIL	PP0V675_S3_MEM_VREFCA_B	19 21 65

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=01/02/2013

PAGE TITLE
Memory Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

REVISION <E4LABEL>

BRANCH <BRANCH>

Thunderbolt, DP, HDMI Constraints

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt & DisplayPort Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTDP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_2SAME	*	=3X_DIELECTRIC	?
TBTDP_TXRX	*	=6X_DIELECTRIC	?
TBTDP_2OTHER	*	=4X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
TBTDP_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
TBTDP_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBTDP_*	*	*	TBTDP_2OTHER
TBTDP_*	=SAME	*	TBTDP_2SAME
TBTDP_TX	*_RX	*	TBTDP_TXRX
TBTDP_RX	*_TX	*	TBTDP_TXRX

DisplayPort & HDMI Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
HDMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	*	=3x_DIELECTRIC	?
DP_2OTHER	*	=4x_DIELECTRIC	?
HDMICLK_2OTHER	*	=7x_DIELECTRIC	?
HDMICLK_2DPHDMI	*	=4x_DIELECTRIC	?
HDMIDATA_2SAME	*	=3x_DIELECTRIC	?
HDMIDATA_2OTHER	*	=4x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMICLK_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?
HDMICLK_2DPHDMI	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMIDATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
HDMIDATA_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDMI_DATA	*	*	HDMIDATA_2OTHER
HDMI_DATA	=SAME	*	HDMIDATA_2SAME
HDMI_DATA	TBTDP_TX	*	HDMIDATA_2SAME
HDMI_DATA	TBTDP_RX	*	TBTDP_TXRX
HDMI_CLK	*	*	HDMICLK_2OTHER
HDMI_CLK	HDMI_DATA	*	HDMICLK_2DPHDMI
HDMI_CLK	DISPLAYPORT	*	HDMICLK_2DPHDMI
HDMI_CLK	TBTDP_TX	*	HDMICLK_2DPHDMI

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	*	*	DP_2OTHER
DISPLAYPORT	=SAME	*	DP_2SAME
DISPLAYPORT	HDMI_DATA	*	DP_2SAME
DISPLAYPORT	TBTDP_TX	*	DP_2SAME
DISPLAYPORT	TBTDP_RX	*	TBTDP_TXRX

DisplayPort/TMDS intra-pair matching should be 0.127mm. Inter-pair matching should be within 2.54cm. Max Length 241.3mm.
DisplayPort AUX CH intra-pair matching should be 0.127mm. Max length 330.2mm.
SOURCE: Calpella SFF DG Rev 1.5 (407364) and Family GPU DG-04202-001-v04.
MAX LENGTH OF DISPLAYPORT/TMDS TRACES: 13 INCHES.

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_P<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_ML_C_N<3..0>	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_P	
	DP_85D	DISPLAYPORT	DP_TBTSRC_AUXCH_C_N	
SPI_TBT_CLK	TBT_SPI_45S	TBT_SPI	TBT_SPI_CLK	23
SPI_TBT_MOSI	TBT_SPI_45S	TBT_SPI	TBT_SPI_MOSI	23
SPI_TBT_MISO	TBT_SPI_45S	TBT_SPI	TBT_SPI_MISO	23
SPI_TBT_CS_L	TBT_SPI_45S	TBT_SPI	TBT_SPI_CS_L	23
DP_HDMI_TBT_ML	DP_85D	DISPLAYPORT	DP_HDMI_TBT_ML_P<3..0>	64 66
DP_HDMI_TBT_ML	DP_85D	DISPLAYPORT	DP_HDMI_TBT_ML_N<3..0>	64 66
DP_HDMI_TBT_AUX	DP_85D		DP_HDMI_TBT_AUX_P	13 64 66
DP_HDMI_TBT_AUX	DP_85D		DP_HDMI_TBT_AUX_N	13 64 66
HDMI_CLOCK	HDMI_85D	HDMI_CLK	HDMI_IG_CLK_C_P	63 64 68
HDMI_CLOCK	HDMI_85D	HDMI_CLK	HDMI_IG_CLK_C_N	63 64 68
HDMI_DATA	HDMI_85D	HDMI_DATA	HDMI_IG_DATA_C_P<2..0>	63 64 68
HDMI_DATA	HDMI_85D	HDMI_DATA	HDMI_IG_DATA_C_N<2..0>	63 64 68

Only used on hosts supporting Thunderbolt video-in

Thunderbolt, DP, HDMI Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	
TBT_A_R2D	TBTDP_85D	TBTDP_TX	TBT_A_R2D_C_P<1..0>	23 26 68
TBT_A_R2D	TBTDP_85D	TBTDP_TX	TBT_A_R2D_C_N<1..0>	23 26 68
TBT_A_R2D	TBTDP_85D	TBTDP_TX	TBT_A_R2D_P<1..0>	26 68
TBT_A_R2D	TBTDP_85D	TBTDP_TX	TBT_A_R2D_N<1..0>	26 68
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<1>	23 26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<1>	23 26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<1>	26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<1>	26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_A_LSX_ML_P<1>	26
DP_A_LSX_ML	DP_85D	DISPLAYPORT	DP_A_LSX_ML_N<1>	26
DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_P<3>	23 26
DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_C_N<3>	23 26
DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_P<3>	26
DP_TBTPA_ML	DP_85D	DISPLAYPORT	DP_TBTPA_ML_N<3>	26
TBT_A_D2R_0	TBTDP_85D	TBTDP_RX	TBT_A_D2R_C_P<0>	26 68
TBT_A_D2R_0	TBTDP_85D	TBTDP_RX	TBT_A_D2R_C_N<0>	26 68
TBT_A_D2R_0	TBTDP_85D	TBTDP_RX	TBT_A_D2R_P<0>	23 26 68
TBT_A_D2R_0	TBTDP_85D	TBTDP_RX	TBT_A_D2R_N<0>	23 26 68
TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R_C_P<1>	26 68
TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R_C_N<1>	26 68
TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R_P<1>	23 26 68
TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R_N<1>	23 26 68
TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R1_AUXDDC_P	26
TBT_A_D2R_1	TBTDP_85D	TBTDP_RX	TBT_A_D2R1_AUXDDC_N	26
DP_TBTPA_AUXCH	DP_85D		DP_TBTPA_AUXCH_C_P	23 26
DP_TBTPA_AUXCH	DP_85D		DP_TBTPA_AUXCH_C_N	23 26
DP_TBTPA_AUXCH	DP_85D		DP_TBTPA_AUXCH_P	26
DP_TBTPA_AUXCH	DP_85D		DP_TBTPA_AUXCH_N	26
TBT_B_R2D	TBTDP_85D	TBTDP_TX	TBT_B_R2D_C_P<1..0>	23 27 68
TBT_B_R2D	TBTDP_85D	TBTDP_TX	TBT_B_R2D_C_N<1..0>	23 27 68
TBT_B_R2D	TBTDP_85D	TBTDP_TX	TBT_B_R2D_P<1..0>	27 68
TBT_B_R2D	TBTDP_85D	TBTDP_TX	TBT_B_R2D_N<1..0>	27 68
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<1>	23 27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<1>	23 27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<1>	27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<1>	27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_B_LSX_ML_P<1>	27
DP_B_LSX_ML	DP_85D	DISPLAYPORT	DP_B_LSX_ML_N<1>	27
DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_P<3>	23 27
DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_C_N<3>	23 27
DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_P<3>	27
DP_TBTPB_ML	DP_85D	DISPLAYPORT	DP_TBTPB_ML_N<3>	27
TBT_B_D2R_0	TBTDP_85D	TBTDP_RX	TBT_B_D2R_C_P<0>	27 68
TBT_B_D2R_0	TBTDP_85D	TBTDP_RX	TBT_B_D2R_C_N<0>	27 68
TBT_B_D2R_0	TBTDP_85D	TBTDP_RX	TBT_B_D2R_P<0>	23 27 68
TBT_B_D2R_0	TBTDP_85D	TBTDP_RX	TBT_B_D2R_N<0>	23 27 68
TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R_C_P<1>	27 68
TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R_C_N<1>	27 68
TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R_P<1>	23 27 68
TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R_N<1>	23 27 68
TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R1_AUXDDC_P	27
TBT_B_D2R_1	TBTDP_85D	TBTDP_RX	TBT_B_D2R1_AUXDDC_N	27
DP_TBTPB_AUXCH	DP_85D		DP_TBTPB_AUXCH_C_P	23 27
DP_TBTPB_AUXCH	DP_85D		DP_TBTPB_AUXCH_C_N	23 27
DP_TBTPB_AUXCH	DP_85D		DP_TBTPB_AUXCH_P	27
DP_TBTPB_AUXCH	DP_85D		DP_TBTPB_AUXCH_N	27
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_P<3..0>	5 23
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_C_N<3..0>	5 23
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_P<3..0>	23
DP_TBTSNK0_ML	DP_85D	DISPLAYPORT	DP_TBTSNK0_ML_N<3..0>	23
DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_C_P	13 23
DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_C_N	13 23
DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_P	23
DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK0_AUXCH_N	23
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_P<3..0>	23 64
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_C_N<3..0>	23 64
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_P<3..0>	23
DP_TBTSNK1_ML	DP_85D	DISPLAYPORT	DP_TBTSNK1_ML_N<3..0>	23
DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_C_P	23 64
DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_C_N	23 64
DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_P	23
DP_TBTSNK_AUXCH	DP_85D		DP_TBTSNK1_AUXCH_N	23
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_C_P<3..0>	5 62
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_C_N<3..0>	5 62
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_P<3..0>	62 68
DP_INT_ML	DP_85D	DISPLAYPORT	DP_INT_ML_N<3..0>	62 68
DP_INT_AUXCH	DP_85D		DP_INT_AUXCH_C_P	5 62
DP_INT_AUXCH	DP_85D		DP_INT_AUXCH_C_N	5 62
DP_INT_AUXCH	DP_85D		DP_INT_AUX_P	62 68
DP_INT_AUXCH	DP_85D		DP_INT_AUX_N	62 68

Notes:
AUX and DDC was removed from DISPLAYPORT or TBTDP_RX/TX because it's not high speed, and to save routing space.

Only used on dual-port hosts.

SYNC_MASTER=YHARTANTO_J44 SYNC_DATE=01/08/2013

PAGE TITLE
TBT,DP,HDMI Constraints

Apple Inc.

DRAWING NUMBER <SCH_NUM> SIZE D

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BRANCH <BRANCH>

MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICLK_2OTHER	*	=7X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIPI_2OTHER	TOP,BOTTOM	=6X_DIELECTRIC	?
MIPI_2CLK	TOP,BOTTOM	=8X_DIELECTRIC	?
MIPICLK_2OTHER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_2OTHER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICLK_2OTHER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	*	=2x_DIELECTRIC	?
S2_DQS2OWNDATA	*	=2x_DIELECTRIC	?
S2_CMD2CMD	*	=2x_DIELECTRIC	?
S2_CMD2CTRL	*	=2x_DIELECTRIC	?
S2_CTRL2CTRL	*	=2x_DIELECTRIC	?
S2_2OTHERMEM	*	=4x_DIELECTRIC	?
S2MEM_2PWR	*	=2x_DIELECTRIC	?
S2MEM_2GND	*	=2x_DIELECTRIC	?
S2MEM_2OTHER	*	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_DQS2OWNDATA	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CMD	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CMD2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_CTRL2CTRL	TOP,BOTTOM	=4x_DIELECTRIC	?
S2_2OTHERMEM	TOP,BOTTOM	=6x_DIELECTRIC	?
S2MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
S2MEM_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_2OTHER
S2_MEM_DQS*	*	*	S2MEM_2OTHER
S2_MEM_CMD	*	*	S2MEM_2OTHER
S2_MEM_CTRL	*	*	S2MEM_2OTHER
S2_MEM_CLK	*	*	S2MEM_2OTHER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS2OWNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS2OWNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P	31 32
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N	31 32
S2_MEM_CKE	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE	31 32
S2_MEM_CS	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CS_L	31 32
	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT	32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	31 32
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	31 32
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	31 32
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	31 32
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	31 32
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>	31 32
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>	31 32
S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>	31 32
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>	31 32
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>	31 32
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P	31 32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N	31 32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P	32 68
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N	32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P	31 32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N	31 32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P	32 68
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N	32 68
		S2_MEM_PWR	PP1V35_CAM	31 32
		S2_MEM_PWR	PP0V675_CAM_VREF	31 32
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA	32
I149		S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ	32

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SMC SMBus & Charger Net Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
SMBUS_SMC_2	SMB_45S	SMB	SMBUS_SMC_2_S3_SCL	34 36 39 68
SMBUS_SMC_2	SMB_45S	SMB	SMBUS_SMC_2_S3_SDA	34 36 39 68
SMBUS_SMC_1	SMB_45S	SMB	SMBUS_SMC_1_S0_SCL	14 32 36 39 43 68 72
SMBUS_SMC_1	SMB_45S	SMB	SMBUS_SMC_1_S0_SDA	14 32 36 39 43 68 72
SMBUS_SMC_0	SMB_45S	SMB	SMBUS_SMC_0_S0_SCL	36 39 62 68
SMBUS_SMC_0	SMB_45S	SMB	SMBUS_SMC_0_S0_SDA	36 39 62 68
SMBUS_SMC_5	SMB_45S	SMB	SMBUS_SMC_5_G3_SCL	36 39 51 52 68
SMBUS_SMC_5	SMB_45S	SMB	SMBUS_SMC_5_G3_SDA	36 39 51 52 68
SMBUS_SMC_3	SMB_45S	SMB	SMBUS_SMC_3_SCL	36 39 43 63
SMBUS_SMC_3	SMB_45S	SMB	SMBUS_SMC_3_SDA	36 39 43 63

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SMC Constraints

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_45S	*	=1TO1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.1 MM	0.1 MM
THERM_45S	*	=1TO1_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.1 MM	0.1 MM
DIG_AUDIO	*	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	=1TO1_DIFFPAIR	0.1 MM	0.1 MM
ANL_AUDIO	*	=1TO1_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
ANL_AUDIO_WIDE	*	=1TO1_DIFFPAIR	0.3 MM	0.3 MM	10 MM	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2X_DIELECTRIC	?
THERM	*	=2X_DIELECTRIC	?
AUDIO	*	=2X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
GND	PCIE_*	*	GND_P2MM
GND	SATA_*	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SB_POWER	SATA_*	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_45S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.070 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D	TOP			0.100 MM	500 MIL		
CPU_27P4S	BOTTOM			0.230 MM	100 MIL		
USB3_85D	TOP			0.100 MM	500 MIL		
USB3_85D	ISL10			0.075 MM			0.090 MM
DP_85D	ISL9			0.075 MM			0.090 MM
PCIE_85D	ISL10			0.075 MM			0.090 MM

DDR3 Loaded Segment Constraint Relaxations

Alternate single ended and differential impedances between devices.

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_40S	BGA_MEM	MEM_45S
MEM_72D	BGA_MEM	MEM_85D

Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_72D	BOTTOM			0.127 MM	6.35 MM		
MEM_85D	TOP			0.100 MM	6.35 MM		

DP, SATA, HDMI, PCIE CONSTRAINT RELAXATIONS

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
DP_85D	BGA	P65_BGA
PCIE_85D	BGA	P65_BGA
CLK_PCIE_85D	BGA	P65_BGA
HDMI_85D	BGA	P65_BGA

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SENSE_45S	*	SENSE_45S
THERM_45S	*	THERM_45S
DIG_AUDIO	*	DIG_AUDIO
ANL_AUDIO	*	ANL_AUDIO
1TO1_DIFFPAIR	*	1TO1_DIFFPAIR

J44 Specific Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
THERM_DP_TBT_D1	THERM_45S	THERM	TBTTHMSNS_D1_P	23 43
THERM_DP_TBT_D1	THERM_45S	THERM	TBTTHMSNS_D1_N	43
THERM_DP_CPU_D1	THERM_45S	THERM	CPUTHMSNS_D1_P	43
THERM_DP_CPU_D1	THERM_45S	THERM	CPUTHMSNS_D1_N	43
THERM_DP_CPU_D2	THERM_45S	THERM	CPUTHMSNS_D2_P	43
THERM_DP_CPU_D2	THERM_45S	THERM	CPUTHMSNS_D2_N	43
	SENSE_45S	SENSE	NC_ISNS_CAMERAP	42
	SENSE_45S	SENSE	NC_ISNS_CAMERAN	42
SENSE_DP	SENSE_45S	SENSE	ISNS_CPUDDR_P	41
SENSE_DP	SENSE_45S	SENSE	ISNS_CPUDDR_N	41
SENSE_DP_LCDBKLT	SENSE_45S	SENSE	ISNS_LCDBKLT_P	40 58
SENSE_DP_LCDBKLT	SENSE_45S	SENSE	ISNS_LCDBKLT_N	40 58
SENSE_DP_TBT	SENSE_45S	SENSE	ISNS_TBT_P	42
SENSE_DP_TBT	SENSE_45S	SENSE	ISNS_TBT_N	42
SENSE_DP	SENSE_45S	SENSE	ISNS_LCDPANEL_P	42 62
SENSE_DP	SENSE_45S	SENSE	ISNS_LCDPANEL_N	42 62
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_COMPUTING_P	40 42
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_COMPUTING_N	40 42
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_OTHER5V_P	40
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_OTHER5V_N	40
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_OTHER3V3_P	40
SENSE_DP	SENSE_45S	SENSE	ISNS_HS_OTHER3V3_N	40
SENSE_DP_CPUVR	SENSE_45S	SENSE	CPUVR_ISNS_P	41
SENSE_DP_CPUVR	SENSE_45S	SENSE	CPUVR_ISNS_N	41
SENSE_DP_CPUVR	SENSE_45S	SENSE	CPUVR_ISNS_R_P	41
SENSE_DP_CPUVR	SENSE_45S	SENSE	CPUVR_ISNS_R_N	41
	SENSE_45S	SENSE	CPUVR_ISNS1_P	41 54
	SENSE_45S	SENSE	CPUVR_ISNS1_N	41 54
	SENSE_45S	SENSE	CPUVR_ISNS2_P	41 54
	SENSE_45S	SENSE	CPUVR_ISNS2_N	41 54
SENSE_DP	SENSE_45S	SENSE	ISNS_1V05_S0_P	41 57
SENSE_DP	SENSE_45S	SENSE	ISNS_1V05_S0_N	41 57
SENSE_DP	SENSE_45S	SENSE	ISNS_SSD_P	41
SENSE_DP	SENSE_45S	SENSE	ISNS_SSD_N	41
SENSE_DP	SENSE_45S	SENSE	ISNS_TPAD_P	
SENSE_DP	SENSE_45S	SENSE	ISNS_TPAD_N	
	SENSE_45S	SENSE	NC_ISNS_DDR_S3P	41 55
	SENSE_45S	SENSE	NC_ISNS_DDR_S3N	41 55
SENSE_DP	SENSE_45S	SENSE	ISNS_PP3V3S0_P	41
SENSE_DP	SENSE_45S	SENSE	ISNS_PP3V3S0_N	41
SENSE_DP	SENSE_45S	SENSE	ISNS_PP5VS0_P	41
SENSE_DP	SENSE_45S	SENSE	ISNS_PP5VS0_N	41
SENSE_DP_CPUHIGAIN	SENSE_45S	SENSE	ISNS_CPUHIGAIN_P	42 43
SENSE_DP_CPUHIGAIN	SENSE_45S	SENSE	ISNS_CPUHIGAIN_N	42 43
SENSE_DP_CPUHIGAIN	SENSE_45S	SENSE	ISNS_CPUHIGAIN_R_P	42
SENSE_DP_CPUHIGAIN	SENSE_45S	SENSE	ISNS_CPUHIGAIN_R_N	42
SENSE_DP_CHGR_CSI	SENSE_45S	SENSE	CHGR_CSI_P	52
SENSE_DP_CHGR_CSI	SENSE_45S	SENSE	CHGR_CSI_N	52
SENSE_DP_CHGR_CSI	SENSE_45S	SENSE	CHGR_CSI_R_P	52
SENSE_DP_CHGR_CSI	SENSE_45S	SENSE	CHGR_CSI_R_N	52
SENSE_DP_CHGR_CSO	SENSE_45S	SENSE	CHGR_CSO_P	52
SENSE_DP_CHGR_CSO	SENSE_45S	SENSE	CHGR_CSO_N	52
SENSE_DP_CHGR_CSO	SENSE_45S	SENSE	CHGR_CSO_R_P	42 52
SENSE_DP_CHGR_CSO	SENSE_45S	SENSE	CHGR_CSO_R_N	42 52

J44 Specific Net Properties

ELECTRICAL CONST SET	NET TYPE PHYSICAL	NET TYPE SPACING	NET	PAGES
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_LO2_L_P	46 48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_LO2_L_N	46 48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_SPKRAMP_LIN_P	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_SPKRAMP_LIN_N	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	SPKRAMP_LIN_P	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	SPKRAMP_LIN_N	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_LO2_R_P	46 48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_LO2_R_N	46 48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_SPKRAMP_RIN_P	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	AUD_SPKRAMP_RIN_N	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	SPKRAMP_RIN_P	48
AUDIO_DP_AMPTWT	ANL_AUDIO	AUDIO	SPKRAMP_RIN_N	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_LO3_L_P	46 48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_LO3_L_N	46 48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_SPKRAMP_LSUBIN_P	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_SPKRAMP_LSUBIN_N	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	LSUBIN_P	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	LSUBIN_N	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_LO3_R_P	46 48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_LO3_R_N	46 48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_SPKRAMP_RSUBIN_P	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	AUD_SPKRAMP_RSUBIN_N	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	RSUBIN_P	48
AUDIO_DP_AMPSUB	ANL_AUDIO	AUDIO	RSUBIN_N	48
AUDIO_DP_SPKSUB	DIG_AUDIO	AUDIO	SPKRCONN_SL_OUT_P	48 50 68
AUDIO_DP_SPKSUB	DIG_AUDIO	AUDIO	SPKRCONN_SL_OUT_N	48 50 68
AUDIO_DP_SPKSUB	DIG_AUDIO	AUDIO	SPKRCONN_SR_OUT_P	48 50 68
AUDIO_DP_SPKSUB	DIG_AUDIO	AUDIO	SPKRCONN_SR_OUT_N	48 50 68
AUDIO_DP_SPKTWT	DIG_AUDIO	AUDIO	SPKRCONN_L_OUT_P	48 50 68
AUDIO_DP_SPKTWT	DIG_AUDIO	AUDIO	SPKRCONN_L_OUT_N	48 50 68
AUDIO_DP_SPKTWT	DIG_AUDIO	AUDIO	SPKRCONN_R_OUT_P	48 50 68
AUDIO_DP_SPKTWT	DIG_AUDIO	AUDIO	SPKRCONN_R_OUT_N	48 50 68
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CH_HS_GND	46 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_HS_MIC_P	
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_SLEEVE	50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_SLEEVE_XW	49 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_HP_PORT_REFCH	46 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_HS_MIC_P	49 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	CODEC_HS_MIC_P	46
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	HS_MIC_P	46 49
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_HS_MIC_N	
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_RING2	50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_CONN_RING2_XW	49 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_HP_PORT_REFUS	46 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_HS_MIC_N	49 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	AUD_US_HS_GND	46 50
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	HS_MIC_N	46 49
AUDIO_DP_MIC	ANL_AUDIO_WIDE	AUDIO	CODEC_HS_MIC_N	46
		SB_POWER	PP3V3_S5	8 11 13 15 16 17 18 26 27 29 56 59 60 61 65 68 61 62 64 65 68
		SB_POWER	PP3V3_S0	8 11 12 13 15 17 18 24 28 30 37 38 39 40 41 42 43 44 46 47 50
		GND	GND	

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Kismet:

AFP://KISMET.APPLE.COM/KISMET-PROJECTS/J44

Useful Wiki Links:

Schematic Conventions - https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions
Schematic Design Wiki - https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design

MobileMac HW Radar:

<rdar://component/497591> MobileMac HW | Task
<rdar://component/497587> MobileMac HW | Schematic
<rdar://component/497585> MobileMac HW | New Bugs
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<rdar://component/497589> MobileMac HW | Architecture

Other Info:

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